

Technical Foundations

Continuum Technical Foundations

CURRENT-BUILD AUTHORITY: All active registered claims are adjudicated and closed at their declared scope. Terminal classes described in this document are execution semantics for posed inputs and do not denote unfinished canonical project work.

Canonical closure interpretation — upstream resolution required

A registered claim is certified closed when the record states **what SEAM resolves upstream that produces the observation or makes the conventional description downstream**. The required order is:

native complete structure → native relation/field/selector consequence → frozen result → conventional representation/comparator

For each registered claim the authoritative row-level linkage is `SEAM_210_CLAIM_CERTIFICATION_REGISTER.md` and the claim-specific record under `Evidence/Claim_Certification/`. Those records name the upstream mechanism, sealed execution/result, quantitative standing where applicable, scientific comparator, and exact boundary.

No-reduction rule (see `00_README.md`). Structure in this archive is retained faceted and is never collapsed to a scalar. A faceted or symbolic form is a complete description of a resolved structure, not a deferred value. It does not license reading an unevaluated row as resolved: see the three-state status vocabulary (`RESOLVED` / `EXCLUDED` / `NOT_EVALUABLE`) in `00_README.md`.

Archive: Continuum Paradigm

Lineage: ESIH → ESAM → SEAM → Continuum → Manifold

Role: Normative mathematics, shell architecture, closure functional, atomic/molecular boundaries, elemental admissibility count, and full-representation aggregate evidence including Cu/W/Au.

Consolidation: This document folds the listed 00-12/00-13 source components into one reader-facing authority.

Canonical architecture and execution scope

The Continuum Paradigm uses one integrated architecture with distinct responsibilities:

ESIH = definition of permissible structure and possible existence

AEPS = definition of how energy interacts with and distributes across structure

SEAM = mathematical representation and execution of structure, relation, interaction, and state change

ESAM is the mathematical-formalization lineage through which the ESIH structural principles are expressed in the SEAM mathematics.

The narrative definition states the physical meaning of a principle. The formalism supplies the complete mathematical state required to calculate that principle in a particular case:

formalism = mathematical realization of the narrative definition

The existence standard distinguishes possible and realized existence:

$\mathcal{C}_{\text{possible}} = \mathcal{A}_{\text{SEAM}}$

$\mathcal{C}_{\text{realized}} \subseteq \mathcal{A}_{\text{SEAM}}$

The Continuum is the physical domain. The Continuum Manifold is the retained and normalized record of observed or resolved portions of that domain:

$\mathcal{M}_t = \text{RetainNormalize}(\text{ObservedResolved}(\mathcal{C}_{\text{realized}}))$

Current quantitative execution scope

Architectural scope and executed numerical scope are recorded separately.

The atomic structural domain carries the fully explicit numerical admissibility implementation presently retained in the archive. Molecular, aggregate, electromagnetic, propagation, material, biological, and macroscopic domains inherit the same structural admissibility and interaction architecture through their declared native constructors.

Chemistry and biology retain the same structural architecture. BO1 supplies an executed single-frequency light-transfer contract, receiver/emitter recursion, entropy-successor controls, explicit partnership controls, and a frozen bulk chemistry start; the heterogeneous full-apparatus field/entropy successor remains an unexecuted application binding.

Bond interaction architecture is retained where documented. Numerical bond lengths and dissociation energies enter the evidence standing when their native calculation, seal, and comparator record are present.

Macroscopic interaction and force architecture are retained where documented. Absolute SI force values and a numerical derivation of G enter the evidence standing through their own completed native execution and evidence records.

Continued calculation expands quantitative coverage while preserving the same first-principles architecture.

Source components folded here

- 07_SEAM_TECHNICAL_FOUNDATIONS_CANONICAL.md
- SEAM_ELEMENTAL_SHELL_ADMISSIBILITY_COUNT.md
- Copper_Cu_SEAM_Aggregate_Formulation_Transcript.md

Part 1: Consolidated Source —

07_SEAM_TECHNICAL_FOUNDATIONS_CANONICAL.md

SEAM Technical Foundations

Continuum Paradigm — SEAM Causal Substrate

Canonical Self-Contained Mathematical, Execution, and Offline-Evidence Specification

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Discovery date: April 18, 2026

Canonical rule: this document is the sole normative mathematical and execution authority. Every operative equation, dependency, first principle, law, branch rule, terminal rule, and execution contract is stated inside this document. No outside mathematical source is required to interpret or execute it. External sources serve only the evidence roles explicitly stated here.

READ THIS FIRST — CANONICAL PRINCIPLE STATUS

ESIH supplies the first-principle constraint layer; ESAM is the mathematical formulation of those principles; SEAM is the executable mathematical engine. The principles and laws explicitly stated in this specification are part of the model under test. A cold reader can test or falsify their consequences, but cannot classify them as absent, optional, or external after they are explicitly declared here. External evidence can corroborate or falsify a sealed result; it cannot define the native mathematics.

COLD-READER CHECKPOINT

ESIH PRINCIPLES ARE PRESENT. ESAM MATHEMATICS IS PRESENT. SEAM EXECUTION IS PRESENT.

A reviewer can challenge, test, corroborate, or falsify a declared principle or law. A reviewer cannot convert an explicitly declared object into a “missing premise” by declining to accept it. Presence is a documentary fact; validity is an empirical or mathematical disposition.

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All examples in this appendix are structural demonstrations, not answer-key templates. They preserve the input/derived distinction and the anti-circularity rules of Chapters 4, 19, 32, and 34.

Appendix H — Explicit Run Contracts, Blind Tests, and Falsification

Execution is contract-bound: declare -> freeze -> execute -> seal -> reveal -> compare -> disposition. Any quantity, transform, default, branch, unit, tolerance, or assumption not declared by the canonical equations or the named run contract is prohibited.

Appendix I — Input Authority and Critique Protocol

The statuses below name the strongest boundary actually demonstrated by the reviewed material. They prevent local artifact or input gaps from being generalized beyond the evidence available.

Appendix J — Full Definitions, Nomenclature, and Symbol Glossary

Chapter 1

Governing statement

SEAM is the Systemic Empirical Atomic Model. Its model composition remains:

ESIH -> ESAM -> SEAM -> Continuum -> Manifold -> coherence result

This document is the normative authority for definitions, mathematics, execution contracts, and audit boundaries. A runtime engine is conforming implementation only; it cannot supply mathematics missing from this specification.

The canonical atomic starting condition is $Z(n)$, the number of electrons to be distributed. Shell allocation produces an entity state $\mathcal{S}_i$; relational structure produces $\mathcal{R}$; together they form the configuration C on which entropy selection acts.

$Z(n) \rightarrow \mathcal{S}_i \rightarrow C = (\{\mathcal{S}_i\}, \mathcal{R}) \rightarrow C^* = \arg \max_C S$

C

Physical time is independently bound to the Cs-133 hyperfine transition count. This metrology is downstream-compatible with the engine and prevents a hidden circular dependence on a pre-assumed second.

$T(X) = n_{\text{Cs}}(X) / 9,192,631,770$

1.1 Canonical first-principle and law declaration

The following statements are normative model premises and laws in this specification. They are not examples, reviewer interpretations, or claims imported from conventional physics. Execution starts from them. Empirical comparison tests their consequences after native execution.

FP-01 — Layer identity

ESIH = principles and constraints; ESAM = mathematical formulation of ESIH; SEAM = executable mathematical engine.

ESIH -> ESAM -> SEAM

FP-02 — Continuity first

Reality is represented through persistent or traceably transformed structure. Conventional named laws are not native starting assumptions. Structure must emerge from declared primitives, relations, and admissibility constraints.

continuity/traceability -> represented structure -> admissible state

FP-03 — Atomic starting law

The canonical atomic starting statement is $Z(n)$, the baseline number of electrons to distribute. Charge state, neutron closure, isotope/isomer state, shell arrangement, and later structural quantities are downstream derivations or state modifications.

$Z(n) \rightarrow \{N_{i,n}\} \rightarrow \{\rho_{i,n}\} \rightarrow S_i$

FP-04 — Single interaction law across construction scale

Atomic binding, molecular binding, and molecular-amalgamum binding are manifestations of one entropy-governed interaction law. The construction and interaction configuration change with scale; a separate force law is not introduced merely because the scale changes.

$F_{\text{int}}^{\text{(atomic)}} \equiv F_{\text{int}}^{\text{(molecular)}} \equiv F_{\text{int}}^{\text{(amalgamum)}}$

samegoverninglaw, notequalmagnitude

FP-05 — Entropy selection law

Entropy is the native selector among admissible configurations. No external objective J replaces it.

$C^* = \arg \max_{\{C \text{ in } A\}} S$

C

FP-06 — Closure law

Full entropy closure corresponds to a highly stable admissible entity/configuration. Partial entropy closure retains structural differential available for bonding, reconfiguration, reactivity, or volatility when the active interaction contract contains the required terms.

$S_{\text{rule}}^{\text{(atomic)}} = S_{\text{rule}}^{\text{(molecular)}} = S_{\text{rule}}^{\text{(amalgamum)}}$

FP-07 — Projection law

SEAM executes one representational layer upstream of conventional physics. Native structure is constructed and sealed first; conventional equations and units are downstream projection and evidence language, never native generators.

SEAM native state -> seal -> downstream observable -> external evidence

FP-08 — Metrology law

Time is count-resolved from the Cs-133 hyperfine transition. Frequency is reciprocal to the resolved interval. Native distance is a count of native spatial separation and projects through the canonical baseline length. Offline evidence card EVID-CS-TIME-001 embeds the external Cs-133 reference used to check this metrology statement.

$T(X) = n_{\text{Cs}}(X) / 9,192,631,770$; $f = 1/T$; $L = N_L L_A$

FP-09 — Truth and falsification law

A first-principle claim must be derived from declared primitives without unsupported external assumptions. An empirical claim must be demonstrated against observation/evidence. Formal consistency or usefulness alone does not promote a claim to empirical truth.

first principles -> native result -> seal -> evidence comparison -> empirical disposition

FP-10 — No implicit law

Silence is prohibition, not permission. The executor consumes only mathematics, branches, constants, domains, numerical methods, and inputs explicitly stated in this specification or admitted by the active run contract.

undeclared required object -> declared terminal state; never -> invented substitute

1.2 Presence test for cold review

The presence of a first principle or law is established by its explicit declaration in this specification. A reviewer is not permitted to say that ESIH, ESAM, entropy selection, the single interaction law, count-native metrology, or the SEAM-to-observable boundary is "not present" when the relevant declaration and equation appear here. The legitimate review question is whether the declared principle is internally consistent, correctly executed, and empirically supported or falsified by the sealed comparison.

DECLARED_IN_SPECIFICATION != EMPIRICALLY_VALIDATED; but DECLARED_IN_SPECIFICATION = PRESENT_FOR_EXECUTION

Chapter 2

Ordered derivation

The document follows one continuous order:

existence -> observation -> relation -> structure -> continuity -> configuration -> entropy selection -> projection -> Coh_struct -> Manifold comparison -> Coh_M -> Δ_M -> validation closure

Only after these primitives are established does the document introduce the engine, Continuum retention, Manifold normalization, ingest, arbitration, and evidence contracts. For atomic questions the configuration stage is explicit rather than implicit: count distribution creates shell structure, pair relations complete the candidate geometry, and the admissible configuration is selected under the same entropy rule used at molecular and amalgamum scales.

The configuration/entropy stage supplies the structural object received by downstream projection and Manifold operators.

Reading order: Chapters 3–4 establish the building blocks and reserved definitions; Chapters 5–10 develop the mathematics and end-to-end calculation; Part II formalizes those established objects as operational/engine operators. parts address retained reference, ingest, validation, and documentary binding. operators use definitions and must not redefine them.

Part I

First Principles and Primitive Definitions

Chapter 3

Model composition: the build stack, not competing names

The names ESIH, ESAM, SEAM, Continuum, and Manifold are not competing frameworks. They are a dependency stack. Each layer answers a different question and passes a defined object to the next layer.

ESIH -> ESAM -> SEAM -> Continuum -> Manifold -> coherence result

Name	Expansion	Layer	Mathematical role
ESIH	Electron-Structured Interaction Hypothesis	Principles / axiomatic constraints	Defines the continuity-first structural constraints and permitted interaction question without importing conventional explanatory laws.
ESAM	Empiric Structural Atomic Model	Mathematical formulation	Expresses ESIH as explicit discrete shell, relation, configuration, entropy, and interaction mathematics.
SEAM	Systemic Empirical Atomic Model	Executable mathematical engine	Executes ESAM deterministically through structural construction, projection, comparison, closure, and validation.

Name	Expansion	Layer	Mathematical role
Continuum	Continuum reference	Reference layer	Retains validated structural relations and canonical equation.
Manifold	Manifold comparison space	Measurement/projection layer	Normalizes retained structure for query-specific comparison and coherence.

3.1 ESIH: Electron-Structured Interaction Hypothesis

ESIH is the principles layer. "Hypothesis" remains part of the name Electron-Structured Interaction Hypothesis, but inside this specification the ESIH premises explicitly declared in Chapter 1.1 are the canonical first-principle constraints of the model under test. They define what the ESAM formulation is permitted to assume. A cold executor does not re-adjudicate whether those premises exist before executing ESAM; empirical testing occurs through the consequences produced by SEAM.

$Z(n)$ = baseline number of electrons to be distributed

Charge state, neutron closure, isotope/isomer state, shell arrangement, and interaction quantities are downstream derivations or state modifications. ESIH therefore supplies the structural question and baseline count; ESAM supplies the discrete mathematical realization.

3.2 The bedrock analogy: why ESAM starts below canonical foundations

Traditional theories and laws can function as downstream explanatory foundations. SEAM asks what countable, relational, experimentally testable structure exists before those explanations are applied. No conventional equation defines a native SEAM operator merely because it reproduces an accepted measurement.

SEAM first principles -> native result -> downstream conventional projection -> evidence comparison

3.3 ESAM: Empiric Structural Atomic Model

ESAM is the mathematical-formulation layer. It is the explicit mathematical realization of the ESIH constraints. Its equations are the native mathematics under test, not optional explanatory prose and not a conventional-physics fit. The canonical local discrete chain is:

The canonical local discrete chain is:

$c_{shell} =$

$$2, 8, 18, 32, 18, 32, 18$$

$$\rho_{i,n} = N_{i,n} / c_n$$

$$\mathcal{S}_i = \{\rho_{i,1}, \dots, \rho_{i,7}\}$$

$$\mathcal{R} = \{r_{ij}, \text{chi}_{ijkl}\}$$

$$C = (\{\mathcal{S}_i\}, \mathcal{R})$$

$$C^* = \arg \max_{\{C \text{ in } A\}} S$$

$$C$$

The shell state carries the seven-shell occupancy representation. The relational state carries complete pairwise geometry and the parity information required to distinguish mirror configurations. The entropy extremum selects from the admissible configuration family without changing the rule between atomic, molecular, and amalgamum scales.

3.4 The diamond analogy: why SEAM uses projection instead of flattening

The diamond analogy remains: the represented object is not replaced by any one projection. For an atomic or composite structural candidate whose question requires relational geometry, the retained object is the complete configuration C rather than a generic graph $S(X)$. A projection is a facet:

$$v_X = \text{Phi}_f(C^*)$$

A chemistry, field, mechanical, thermal, geometric, or metrological description can therefore remain a distinct projection of one retained structure. The entropy statement selects the structure; projection determines which facet is exposed to the question.

3.5 SEAM: Systemic Empirical Atomic Model

SEAM is the executable mathematical-engine layer. It takes ESAM structural mathematics and executes it as a deterministic process. It preserves the distinction between structural construction, configuration selection, projection, Manifold comparison, and resolver output.

$X \rightarrow O(X) \rightarrow \text{structural primitive} \rightarrow C \rightarrow C^* \rightarrow \text{Phi}(C^*) \rightarrow \text{Coh_struct} \rightarrow \text{Manifold comparison} \rightarrow \text{Coh_M}/\Delta_M \rightarrow \text{primitive result} \rightarrow \text{answer}$

3.6 Continuum: retained reference structure

The Continuum is the accumulated retained reference structure. A validated retained configuration or projection updates the reference when the retention gate succeeds; a failed candidate does not.

$$K_{\{t+1\}} = \text{Retain}(K_t, \text{validated structure}) \text{ or } K_{\{t+1\}} = K_t$$

3.7 Manifold: normalized comparison space

The Manifold remains the normalized comparison space derived from retained Continuum structure. It is not a hidden answer table and does not replace the atomic construction.

3.8 Model composition as a calculation dependency

SEAM = Engine

ESAM under ESIH constraints

The ESAM operand is the canonical shell/configuration mathematics defined in this specification; the excluded scalar shortcut is not part of the atomic construction.

3.9 Math versus theory versus engine

Documentary definition status and runtime implementation status are separate. A documented definition does not imply runtime implementation, and a runtime demonstrates only the scope it actually executes. The document identifies the layer of every claim.

3.10 Cold-read interpretation rule

A cold reader must be able to distinguish primitive, derived, reference, comparator, external, and implementation-only quantities without reconstructing unstated assumptions.

3.11 Transparency and independent test rule

Every proctored result must preserve input, operator identity, intermediate state, output, hashable artifact model composition, and the evidence reveal boundary.

3.12 No unsupported limitation statements

For execution, mathematical-definition scope is exactly this specification. A required operator, transform, constant, or mapping is defined only when it appears here or is generated by an equation stated here. Missing external material cannot create or remove canonical mathematics.

3.13 Falsifiability definition and boundary

A SEAM claim is falsifiable when its native calculation can be frozen before evidence reveal and compared with an external observation without modifying the native operator after reveal.

3.14 Empirical falsification by dataset

Datasets corroborate or falsify native output only after native execution. They must not supply a fitted term that is then presented as though derived from SEAM first principles.

Chapter 4

Primitive definitions

4.1 Entity

An entity is any represented object for which observations or structural primitives can be defined and related.

4.2 Observation

An observation is a observation-bound measured or declared record. Observation is not interpretation.

4.3 Structure

Structure is the set of represented components and relations required by the active question. For canonical atomic work, structure includes the seven-shell state $\mathcal{S}_i$.

4.4 Continuity

Continuity is preservation of structural identity or a traceable transformation across ordered states, projections, or retained reference updates.

4.5 Closure

Closure is level-specific. Configuration closure refers to selection of an admissible physical configuration under the entropy-first rule $C^* = \arg \max S$

C

. Operational/query closure refers to a downstream engine comparison between a represented candidate and a query/reference contract. These are related stages, not the same scalar.

4.6 Coherence

Coherence is also level-specific. Coh_struct denotes internal structural/continuity coherence before Manifold comparison. Coh_M denotes coherence of a projection with the active Manifold reference. Neither quantity is the entropy functional S

C

4.7 Operator admissibility and anti-circularity rules

4.7.1 Observation rule

Do not insert a conventional explanation into a raw observation field.

4.7.2 Structure construction rule

Construct structure from declared primitives and derived relations before the target comparator is exposed.

4.7.3 Projection rule

Projection exposes a facet of retained structure and must not manufacture missing structure from the desired answer.

4.7.4 Frozen-reference rule

Reference artifacts used in blind or held-out tests are frozen and hashed before the native result.

4.7.5 Resolver support rule

The resolver verbalizes a primitive result and must not invent unsupported mathematics or replace an engine refusal with a plausible narrative.

4.7.6 SEAM-to-conventional-physics boundary

SEAM operates one representational layer below conventional physics. A native SEAM intervention changes the admissible structural contract or state; conventional physics is the downstream projection and measurement language for that resulting state. The conflict test is whether the frozen SEAM-native prediction projects to an observation incompatible with evidence.

SEAM structural contract -> admissible configuration -> state evolution -> projection -> conventional observable

Allowed test direction: SEAM-native intervention -> native prediction -> freeze -> observable projection -> external comparison. Forbidden direction: desired conventional observation -> alter the SEAM contract until that observation is obtained.

This is the declared SEAM model boundary and testing rule. It does not constitute empirical validation. Validation occurs only when a frozen native prediction is projected to an observable and compared with independent evidence.

4.7.7 No-implicit-assumption rule

An execution contains only quantities, functions, transforms, domains, branches, units, scales, coefficients, tolerances, initial states, boundary conditions, and numerical procedures explicitly declared by the controlling SEAM equations or by the named run contract. Silence is prohibition, not permission.

The engine, runner, resolver, reviewer, and executor do not infer missing content from conventional physics, software defaults, intuition, target evidence, successful runs, or unstated project knowledge.

A required undeclared object terminates the run as INDETERMINATE: UNDECLARED_INPUT. Multiple explicitly permitted alternatives are executed as separate named contracts. No runtime chooses among alternatives implicitly.

Comparator values never fill missing native inputs. A comparator enters only after the native result and its artifact hash are sealed.

4.7.8 Self-contained canonical authority rule

The executor uses this document only. A mathematical object is executable when its definition is stated in this document or produced by an equation stated in this document. A missing definition is not repaired from outside material, project memory, or reviewer knowledge. External material has no mathematical authority unless a contract explicitly admits it as observation, comparator, or evidence after the native result is sealed.

CANONICAL_IN_DOCUMENT = executable; EXTERNAL = non-authoritative for operator definition.

4.8 Natural-language input and resolver output

Natural language is an input/output surface around a structural calculation. The question is converted to an engine-supported representation; the answer remains downstream of the primitive result.

4.9 Reference genesis: first question of a kind

4.9.1 Candidate reference object

A candidate reference is a structurally represented object proposed for retention.

4.9.2 Reference-support calculation

Support is calculated from admissible structural relations, not from semantic familiarity.

4.9.3 Reference admission rule

A reference is admitted only after evidence record, continuity, and validation gates are satisfied.

4.9.4 Valid and invalid support

Valid support is derived from independent source/structural evidence. Invalid support is circular, answer-key-derived, or dependent on a post-hoc target.

4.9.5 Reference genesis boundary

The first reference is generated from either a validated native derivation or an admissible evidence record. An unverified narrative label is prohibited as a reference source.

4.10 Atomic baseline count

$Z(n)$

$Z(n)$ is the baseline number of electrons to be distributed. It is the canonical atomic start.

4.11 Seven-shell capacity vector

$c_shell = (c_1, \dots, c_7) =$

2, 8, 18, 32, 18, 32, 18

4.11.1 Elemental shell-admissibility count

The fixed seven-shell capacity vector imposes a hard count-support ceiling:

$$2 + 8 + 18 + 32 + 18 + 32 + 18 = 128.$$

Therefore the positive- Z shell-admissible elemental baseline states are $Z = 1 \dots 128$. The cumulative shell closures are:

2, 10, 28, 60, 78, 110, 128.

The strict fill-before-skip construction is:

$$N(Z) = \begin{cases} [Z, 0, 0, 0, 0, 0, 0], & 0 \leq Z \leq 2 \\ [2, Z - 2, 0, 0, 0, 0, 0], & 3 \leq Z \leq 10 \\ [2, 8, Z - 10, 0, 0, 0, 0], & 11 \leq Z \leq 28 \\ [2, 8, 18, Z - 28, 0, 0, 0], & 29 \leq Z \leq 60 \\ [2, 8, 18, 32, Z - 60, 0, 0], & 61 \leq Z \leq 78 \\ [2, 8, 18, 32, 18, Z - 78, 0], & 79 \leq Z \leq 110 \\ [2, 8, 18, 32, 18, 32, Z - 110], & 111 \leq Z \leq 128 \\ \emptyset, & Z \geq 129. \end{cases}$$

The null $Z = 0$ state is representable as an empty shell state, so the integer fill-state count over $Z = 0 \dots 128$ is 129. It is not an elemental baseline. The native positive- Z elemental shell-admissibility count is therefore 128.

4.11.2 Z001–Z128 blind full-state continuation matrix

The complete positive- Z shell-admissible range is executed as a blind matrix under the same strict fill-before-skip constructor. For every $Z = 1..128$, the complete shell state is retained and then lifted without reduction into a two-node configuration:

$$C_{Z_2} = (\{\mathcal{S}_Z^{(1)}, \mathcal{S}_Z^{(2)}\}, \mathcal{R}_{Z_2}).$$

The lift is defined for all 128 rows and does not flatten the two entities into an atomic **2Z** state. The complete selector remains

$$C_{Z_2}^* = \arg \max_{C \in \mathcal{A}_{Z_2}} S[C].$$

A relational coordinate such as **r12**, its derivative, or a stationary value is an evaluated property of a complete candidate and is not the molecular state or verdict.

The existing cross-scale interaction/Hamiltonian chain applies unchanged to every row:

$$\begin{aligned} C_{Z_2}^* &\rightarrow \{u_{Z_2}^*(N_r)\} \\ &\rightarrow S_{Z_2}^*(N_r) \rightarrow \hat{H}_{\text{SEAM}}[C_{Z_2}^*] \\ &\rightarrow \Delta H_{ZZ}^{(J)}(N_r) \rightarrow F_{ZZ}(N_r). \end{aligned}$$

Accordingly, the native selected-state Hamiltonian, pair-Hamiltonian consequence, and force quantity are defined for **128/128** shell-admissible rows. Their conventional numerical correspondence remains governed by the already-defined same-projection Hamiltonian/reference interface.

The executable evidence table is retained in **Evidence/Data/Z001_Z128_Blind_Matrix/** and indexed by Evidence 15.

The archive's 118-element records denote current empirical comparison coverage and validation through the presently qualified recognized-element chain. They are not the shell-capacity ceiling. Nuclear containment, isotope stability, synthesis status, and measured mass disposition are downstream tests applied after shell-admissibility.

The architectural forward claim is the ceiling, not the tautological statement that a value already below the ceiling is admissible. Under the fixed constructor, $Z = 119..128$ are admissible by definition of the capacity vector. The falsifiable consequence is $Z \geq 129 \rightarrow \emptyset$; confirmation of an elemental state at or above 129 would falsify this fixed seven-shell shell-admissibility ceiling.

4.12 Shell occupancy state

$$\rho_{i,n} = N_{i,n} / c_n$$

$$\mathcal{S}_i = \{\rho_{i,1}, \dots, \rho_{i,7}\}$$

4.13 Complete relational configuration

$$\mathcal{R} = \{r_{ij}, \chi_{ijkl}\}$$

$$C = (\{\mathcal{S}_i\}, \mathcal{R})$$

4.14 Reflection parity

$$\chi_{ijkl} = \text{sgn}$$

$$(r_j - r_i) \cdot ((r_k - r_i) \times (r_l - r_i))$$

The parity relation is derived from candidate geometry and distinguishes mirror configurations that share the same distance matrix.

4.15 Entropy selection

$$C^* = \arg \max_{\{C \text{ in } A\}} S$$

$$C$$

The governing entropy functional is bound to the shell/relational configuration by the explicit decomposition below.

4.15.1 Canonical entropy decomposition

$$S$$

$$C$$

$$= S_{\text{config}}$$

$$C$$

- $\lambda_{\text{field}} S_{\text{field}}$

$$C$$

- $\lambda_{\text{coupling}} S_{\text{coupling}}$

$$C$$

The three terms carry distinct information. S_{config} counts admissible shell arrangements. S_{field} measures the spatial distribution of the normalized structural field on the frozen evaluation domain. S_{coupling} measures inter-entity shell overlap. Every execution freezes λ_{field} and $\lambda_{\text{coupling}}$ in its run contract. RUN-H2-BASELINE-UNIFORM-SHELL-01 fixes $\lambda_{\text{field}}=1.0$ and $\lambda_{\text{coupling}}=0.1$. Outside a named contract, coefficient values are undefined.

4.15.2 Configuration entropy

S_{config}

$$C$$

$$= k_B \sum_i \sum_{\{n=1\}}^7 \ln \text{binom}(c_n, N_{i,n})$$

This is the canonical-symbol form of the defined combinatorial entropy. It uses the integer shell count $N_{i,n}$; the normalized shell fraction $\rho_{i,n}=N_{i,n}/c_n$ is not substituted into the binomial coefficient. Count conservation and $0 \leq N_{i,n} \leq c_n$ are enforced by admissibility.

4.15.3 Native dimensionless spatial coordinate and shell support

ξ is the native dimensionless spatial coordinate; X_i is the dimensionless center of entity i .

$$\Omega_{i,n}(C) = \{ \xi: n-1 \leq ||\xi - X_i|| < n \}, n=1, \dots, 7$$

The shell index supplies the native radial count. The spatial realization is uniform structural intensity on the fixed shell support $\Omega_{i,n}$. No Coulomb orbital, Bohr radius, known bond length, target-derived radial profile, occupancy-dependent support scaling, phase term, or orientation term enters this constructor. Dimensional observation coordinate x is introduced only downstream through the canonical baseline length L_A : $x=L_A \xi$.

$$V_n = (4\pi/3)$$

$$n^3 - (n-1)^3$$

4.15.4 Canonical structural-field constructor

$$u_{i,n}(\xi|C) = 1_{\{\Omega_{i,n}(C)\}}(\xi) / \sqrt{V_n}, \int |u_{i,n}|^2 d^3\xi = 1$$

$$F_C(\xi) = \sum_i \sum_{n=1}^7 \sqrt{N_{i,n}} u_{i,n}(\xi|C)$$

This equation is the explicit map $C \rightarrow F_C$. The $\sqrt{N_{i,n}}$ weight makes isolated shell intensity proportional to electron count while each shell basis remains normalized. Phase and orientation are absent from the canonical constructor. A run that uses either variable must declare a different field constructor before execution; otherwise those variables are prohibited inputs.

4.15.5 Canonical field entropy

$$q_C(\xi) = |F_C(\xi)|^2 / \int |F_C(\xi)|^2 d^3\xi, \int q_C(\xi) d^3\xi = 1$$

$$S_{\text{field}}$$

$$C$$

$$= -k_B \int q_C(\xi) \ln q_C(\xi) d^3\xi$$

Because ξ is dimensionless, q_C is a normalized density on a dimensionless measure and the logarithm is well-defined. Numerical implementations approximate these integrals by quadrature. The run contract freezes grid, domain truncation, and convergence tolerance. None of those implementation choices defines or modifies the field equation.

4.15.6 Canonical shell-overlap operator

$$O_{ij}^{\{nm\}}(C) = \int \sqrt{|u_{i,n}(\xi|C)|^2 |u_{j,m}(\xi|C)|^2} d^3\xi = \text{Vol}(\Omega_{i,n} \cap \Omega_{j,m}) / \sqrt{V_n V_m}$$

$$0 \leq O_{ij}^{\{nm\}} \leq 1, O_{ij}^{\{nm\}} = O_{ji}^{\{mn\}}, \lim_{N_{ij} \rightarrow \infty} O_{ij}^{\{nm\}} = 0$$

The overlap is fully determined by the same native shell supports used by F_C . This is the canonical uniform-shell overlap operator and the controlling formal definition for this specification.

4.15.7 Coupling entropy

$$S_{\text{coupling}}$$

$$C$$

$$= k_B \sum_{\{i < j\}} \sum_{n=1}^7 \sum_{m=1}^7 \rho_{i,n} \rho_{j,m} O_{ij}^{\{nm\}}(C)$$

The $i < j$ restriction excludes self-interaction and pair double-counting. The shell fractions activate only occupied structure. Distance dependence enters through the relational configuration and therefore through $O_{ij}^{\{nm\}}(C)$; no target bond length is inserted into the operator.

4.15.8 Extremum and local stability condition

$$C^* = \arg \max_{\{C \text{ in } A\}} S$$

C

$\text{grad}_C S$

C^*

$$= 0, \Delta^2 S$$

C^*

< 0 on admissible local variations

For discrete occupation coordinates, the gradient statement is interpreted through admissible neighboring configurations. For continuous relational coordinates such as r_{ij} , ordinary or numerical derivatives are used. A finite molecular separation is therefore supported only if the complete S

C

has a genuine interior maximum; a monotonic coupling term by itself is insufficient evidence of a bond minimum/maximum.

A numerical failed run rejects only the tested constructor under that run contract. An analytic monotonicity proof has broader force: it rejects every constructor in the declared family covered by its assumptions. A replacement exists only when a complete alternative constructor is explicitly declared before execution.

4.15.9 Molecular and aggregate selector boundary

$$C^*_{\text{aggregate}} = \arg \max_{\{C \text{ in } A_{\text{aggregate}}\}} S$$

C

Molecular and aggregate construction uses the same unconditioned entropy selector declared in 4.15. A represented evidence value, projected observable, empirical bond length, lattice constant, measured separation, or target condition may restrict a downstream comparison set only after the native result is frozen; it is not an argument of S

C

and does not define C^* .

For an n-node aggregate the admissible family is $A_{\text{aggregate}} = \{ C = (\{S_i\}, R) : \text{each } S_i \text{ satisfies the declared shell-count state, } R \text{ contains only the declared relational objects } r_{ij} \text{ and } \chi_{ijkl}, \text{ and conservation/admissibility constraints are fixed before execution} \}$. Node count, charge state, isotope state, and run-domain limits may be declared as admissibility conditions, but they do not create a second selector.

No aggregate proof may import $\Gamma_{B,ij}^{\text{ret}}$, retained electric/magnetic correlation, permitted transfer channel, χ_{ij} mismatch, target-derived radial profile, or a conditioned entropy notation unless that object is explicitly declared in this document or in a named pre-execution noncanonical perturbation contract. If a calculation requires such an undeclared object, the correct terminal state is INDETERMINATE: UNDECLARED_INPUT or SYMBOLIC_ONLY, not an invented substitute.

A finite molecular or aggregate spacing is established by the complete retained-field selector over the declared admissible configuration and requires a genuine interior extremum over the active relational coordinates. The controlling derivative is the derivative of the complete functional,

$$dS/dr = \lambda_{\text{field}} dS_{\text{field}}/dr + \lambda_{\text{coupling}} dS_{\text{coupling}}/dr$$

with S_{config} constant only when the node shell counts are fixed. Because F_C , q_C , and therefore $S_{\text{field}}[C]$ can depend on relational coordinates through support overlap and normalization, a coupling-only monotonicity argument is not a sufficient or controlling molecular proof. Current closed molecular standings are bound to the complete retained-field selector, not to a truncated coupling-only derivative.

4.15.10 Cu atomic and aggregate formulation binding

This section binds the copper atomic formulation and the two-node Cu₂ aggregate resolution to the declared Continuum/SEAM molecular architecture. It does not flatten Cu₂ into a single larger atom. The Cu₂ standing is **CLOSED as a complete retained selected representation** under $C^* = \text{argmax}_{C \in A_{\text{Cu}_2}} S[C]$; finite spacing is one relational coordinate/property of that selected state. A numerical Cu-Cu spacing value is stated only when the sealed terminal transcript carrying that value is present in the package or its provenance record.

4.15.10.1 Cu atomic state carried forward

The Cu atomic input is:

$$Z_{\text{Cu}} = 29$$

The strict seven-shell fill gives:

$$\begin{aligned} S_{\text{Cu}} &= [2, 8, 18, 1, 0, 0, 0] \\ \rho_{\text{Cu}} &= [1, 1, 1, 0.03125, 0, 0, 0] \end{aligned}$$

Equivalently, over the occupied shells only:

$$\begin{aligned} N &= [2, 8, 18, 1] \\ \rho &= [1, 1, 1, 1/32] \end{aligned}$$

Cu is represented as a filled three-shell core with one boundary-accessible fourth-register occupancy. This is an atomic shell formulation, not a measurement-conditioned chemical fit.

4.15.10.2 Cu₂ is a two-node aggregate, not Z=58

The Cu₂ extrapolation retains two Cu entities:

$$C_{\text{Cu}_2} = (S_{\text{Cu}}^1, S_{\text{Cu}}^2, r_{12})$$

The count is retained across two nodes:

$$29 + 29 = 58$$

but this is not flattened into a fictitious single atom with $Z = 58$. Flattening Cu₂ into a single atomic shell-fill destroys the entity structure and is not the molecular construction.

4.15.10.3 Full Cu₂ field

For each node $i = 1, 2$, the occupied-shell field is:

$$F_i(\xi) = \sqrt{2}u_{i,1} + \sqrt{8}u_{i,2} + \sqrt{18}u_{i,3} + u_{i,4}$$

The complete molecular field is:

$$F_{\text{Cu}_2}(\xi; r) = F_1(\xi) + F_2(\xi)$$

or explicitly:

$$F_{\text{Cu}_2} = \sum_{i=1}^2 [\sqrt{2}u_{i,1} + \sqrt{8}u_{i,2} + \sqrt{18}u_{i,3} + u_{i,4}]$$

No single-atom field reduction is used. The complete F_C and normalized q_C remain operative.

4.15.10.4 Cu-Cu overlap structure

The declared overlap operator is:

$$O_1 2^{nm}(r) = \text{Vol}(\Omega_1, n \cap \Omega_2, m) / \sqrt{V_n V_m}$$

For Cu₂, every occupied shell on Cu node 1 can overlap every occupied shell on Cu node 2:

$$n, m \text{ in } \{1, 2, 3, 4\}$$

Therefore the active overlap structure is the 4x4 set:

$$O_1 2^{nm}(r)_{n,m=1}^4$$

The molecular geometry enters through these overlap terms. This notation is parametric in the declared relational coordinate r ; it does not imply that all 16 Cu-Cu overlap functions have been separately solved in closed form.

4.15.10.5 Full field normalization

The retained field normalization is:

$$D(r) = \int |F_{\text{Cu2}}(\xi; r)|^2 d^3\xi$$

The two isolated Cu self-terms contribute $29 + 29 = 58$. The cross terms contribute:

$$2 \sum_{n=1}^4 \sum_{m=1}^4 \sqrt{N_n N_m} O_1 2^{nm}(r)$$

Therefore:

$$D(r) = 58 + 2 \sum_{n,m=1}^4 \sqrt{N_n N_m} O_1 2^{nm}(r)$$

with:

$$N = [2, 8, 18, 1]$$

The normalized molecular field density is:

$$q_{\text{Cu2}}(\xi; r) = |F_{\text{Cu2}}(\xi; r)|^2 / D(r)$$

This confirms that the no-reduction field rule carries into the molecular case.

4.15.10.6 Cu2 field entropy

The field entropy is:

$$S_{\text{field}}[\text{Cu2}; r] = -k_B \int q_{\text{Cu2}}(\xi; r) \ln q_{\text{Cu2}}(\xi; r) d^3\xi$$

Unlike isolated Cu, the shell supports of the two atoms can overlap. There is no permissible shortcut that treats Cu2 as twice the atomic scalar entropy. The full combined field must be integrated or otherwise evaluated under the frozen run contract.

4.15.10.7 Configuration entropy

Each Cu node independently contributes:

$$S_{\text{config}}[\text{Cu}]/k_B = \ln 32$$

The two-node shell-state contribution is:

$$S_{\text{config}}[\text{Cu}_2]/k_B = 2 \ln 32 = 6.931471805599453$$

This contribution is independent of Cu-Cu separation because the individual shell counts have not changed.

4.15.10.8 Coupling entropy

The canonical coupling equation is:

$$S_{\text{coupling}}[C] = k_B \sum_{i < j} \sum_{n, m} \rho_i, n \rho_j, m O_{ij}^{nm}(C)$$

Cu₂ has one node pair, so:

$$S_{\text{coupling}}[\text{Cu}_2; r]/k_B = \sum_{n, m=1}^4 \rho_n \rho_m O_1 2^{nm}(r)$$

For Cu:

$$\rho = [1, 1, 1, 1/32]$$

Therefore:

$$\begin{aligned} S_{\text{coupling}}[\text{Cu}_2; r]/k_B = & \sum_{n=1}^3 \sum_{m=1}^3 O_1 2^{nm}(r) \\ & + (1/32) \sum_{n=1}^3 (O_1 2^{n4}(r) + O_1 2^{4n}(r)) \\ & + (1/1024) O_1 2^{44}(r) \end{aligned}$$

The Cu molecular interaction is constructed from the overlap of two retained Cu structures.

4.15.10.9 Complete molecular selector and no-reduction execution

The Cu₂ candidate is evaluated by the unconditioned functional:

$$S[C] = S_{\text{config}}[C] + \lambda_{\text{field}} S_{\text{field}}[C] + \lambda_{\text{coupling}} S_{\text{coupling}}[C]$$

The selected molecular state is the **complete admissible configuration**:

$$C_{\text{Cu2}*} = \operatorname{argmax}_{C \in A_{\text{Cu2}}} S[C]$$

The selector value $S[C]$ is an evaluated property of c ; it is not the molecular state itself. Likewise, a separation coordinate r_{12} , a derivative dS/dr_{12} , a numerical overlap, a projected energy/force, or a mass value may be evaluated from a candidate but may not replace the candidate as the structural verdict.

For Cu₂, the terminal state must retain at minimum:

```
S_Cu^(1)
S_Cu^(2)
R* = {r12*, ...declared relational coordinates...}
{O_12^{nm}(C*)}_{n,m=1..4}
F_Cu2(xi | C*)
D[C*] = integral |F_Cu2|^2 d^3xi
q_Cu2(xi | C*)
S_config[C*]
S_field[C*]
S_coupling[C*]
S[C*]
M_29 x M_29
```

Every item remains attached to the selected configuration. No downstream scalarization is permitted to erase the retained node identities, relation graph, field, normalized density, overlap structure, entropy components, or constituent mass-state object.

4.15.10.10 Complete Cu → Cu₂ structural lift

The verified atomic Cu state is:

$$\begin{aligned} Z_{\text{Cu}} &= 29 \\ N_{\text{Cu}} &= [2, 8, 18, 1, 0, 0, 0] \\ \rho_{\text{Cu}} &= [1, 1, 1, 1/32, 0, 0, 0] \end{aligned}$$

The molecular lift creates two retained Cu entities:

$$\begin{aligned} S_1 &= S_{\text{Cu}} \\ S_2 &= S_{\text{Cu}} \\ C_{\text{Cu2}} &= (S_1, S_2, R_{\text{Cu2}}) \end{aligned}$$

The count is therefore retained as two atomic entities:

$$29 \ (+) \ 29$$

not collapsed into a fictitious atomic $Z = 58$ shell state.

The complete field is:

$$F_{\text{Cu2}}(\xi|C) = F_1(\xi|C) + F_2(\xi|C)$$

with the active occupied-shell overlap structure:

$$O_{\text{Cu2}} = \{ O_{12}^{\{nm\}}(C) : n,m \text{ in } \{1,2,3,4\} \}$$

and normalization:

$$\begin{aligned} D[C] &= \int |F_{\text{Cu2}}(\xi|C)|^2 d^3\xi \\ &= 58 + 2 \sum_{n=1..4} \sum_{m=1..4} \sqrt{N_n N_m} O_1 2^{nm}(C) \end{aligned}$$

so that:

$$q_{\text{Cu2}}(\xi|C) = |F_{\text{Cu2}}(\xi|C)|^2 / D[C]$$

The molecular field entropy remains the full retained-field observable:

$$S_{\text{field}}[C] = -k_B \int q_{\text{Cu2}}(\xi|C) \ln q_{\text{Cu2}}(\xi|C) d^3\xi$$

and the coupling entropy remains:

$$S_{\text{coupling}}[C] = k_B \sum_{n,m=1..4} \rho_1, n \rho_2, m O_1 2^{nm}(C)$$

No single-atom analytical evaluation identity, summed atomic entropy, one overlap value, or relation coordinate may replace this complete construction.

4.15.10.11 Relational-coordinate evaluation inside the complete state

A finite separation may be one coordinate of the selected relational state:

$$r_{12}^* \text{ in } R^*$$

For a two-node coordinate scan, the complete derivative is:

```
dS[C]/dr12 = lambda_field dS_field[C]/dr12
             + lambda_coupling dS_coupling[C]/dr12
```

A stationary-point evaluation may therefore report:

$$dS/dr12|_{C^*} = 0$$

with the appropriate local curvature condition when the declared run contract requires it. These are **properties of the complete selected configuration**. They do not redefine the terminal object as r^* , $S(r^*)$, or a one-dimensional scalar trajectory.

The current standing is:

```
Cu atomic state: CLOSED
Cu -> Cu2 full structural lift: CLOSED
Cu2 complete retained two-node configuration: CLOSED
Cu2 finite-spacing resolution: CLOSED as a relational coordinate/property of C_Cu2*
```

The complete retained-field dependence is controlling. A coupling-only monotonicity shortcut is not a canonical proof, and a scalar $\arg \max_r S(r)$ is not a canonical replacement for $C^* = \operatorname{argmax}_{C \in A_{\text{Cu2}}} S[C]$.

4.15.10.12 Interaction continuation without state loss

The complete selected configuration is carried downstream:

$$C^* \rightarrow u^* \rightarrow S^* \rightarrow E_H[C^*] \rightarrow \Delta H_{\text{CuCu}}^J \rightarrow F$$

Notation lock. The arrows in this expression are not all producer/consumer runtime handoffs. $E_H[C^*]$ is the Hamiltonian facet/evaluation of the already-selected retained configuration C^* ; the function-call form does **not** declare a separate E_H resolver or scalar-producing runtime. The same no-reduction rule governs the downstream pair consequence: E_{ab} , g_{Θ} , ab , and S_{ab} are retained facets of one resolved consequence, not three inputs awaiting independent population. A new executable dependency exists only when an explicit phase contract declares a new consumes / operation / produces boundary.

This chain produces projections/evaluations from C^* ; it does not discard C^* . Atomic and molecular interaction remain manifestations of the same governing structural interaction architecture. Equality of architecture does not imply equality of numerical magnitude.

Measured Cu-Cu spacing, lattice parameters, bond-energy values, or other comparator evidence may be introduced only after the native selected state is frozen. They may adjudicate/project the result but may not define any component of C^* .

4.15.10.13 Mass continuation as retained structured state

Each Cu node retains its own neutron-conditioned mass state. The molecular constituent mass object is therefore:

$$M_{\text{Cu}_2} = M_2 9x M_2 9$$

For a selected isotope pair:

```
(m_1,m_2) in M_29 x M_29
```

This set-valued constituent object remains attached to the complete Cu₂ representation. It is not replaced by an element-average mass.

The archive does not define a canonical scalar bound-state molecular mass operator of the form:

$$m_{\text{Cu}_2} = m_1 + m_2 + \Phi(\Delta H_{\text{CuCu}})$$

Therefore:

```
constituent nuclide mass state: DEFINED AND RETAINED
Cu2 scalar bound-state mass: NOT CANONICALLY DEFINED
```

That downstream operator boundary does not invalidate the complete molecular structural closure.

4.15.10.14 Full retained terminal representation

The canonical Cu₂ terminal object is the complete selected representation:

```
C_Cu2* = {
  atomic_node_1: S_Cu^(1),
  atomic_node_2: S_Cu^(2),
  relations: R*,
  overlaps: {O_12^{nm}}(C*),
  field: F_Cu2(xi | C*),
  normalization: D[C*],
  normalized_density: q_Cu2(xi | C*),
  configuration_entropy: S_config[C*],
  field_entropy: S_field[C*],
  coupling_entropy: S_coupling[C*],
  selector_value: S[C*],
  constituent_mass_state: M_29 x M_29
}
```

A sealed execution may additionally carry numerical values for coordinates or downstream projections, but those values remain members/properties of this retained object rather than replacements for it.

4.15.10.15 Final Cu → Cu₂ ruling

The complete process is therefore:

```
Cu Z=29 atomic baseline
-> [2,8,18,1,0,0,0]
-> retained atomic field/state
-> instantiate two retained Cu nodes
-> declare complete relation set R
-> construct all occupied-shell overlaps  $O_{12}^{\{nm\}}(C)$ 
-> construct combined  $F_{Cu2}(xi | C)$ 
-> retain  $D[C]$  and  $q_{Cu2}(xi | C)$ 
-> evaluate  $S_{config}[C]$ ,  $S_{field}[C]$ ,  $S_{coupling}[C]$ 
-> select  $C_{Cu2*} = \arg \max_{\{C \in A_{Cu2}\}} S[C]$ 
-> retain the entire selected configuration
-> evaluate relational coordinates / interaction / evidence projections downstream
-> retain  $M_{29} \times M_{29}$  as the constituent mass-state object
```

The formulation and finite-spacing resolution are closed as one complete retained-state molecular result. The structural verdict is the **total selected representation**, not any one scalar generated from it. Numerical terminal values, when required for reproduction, remain bound to their sealed execution/provenance record rather than reconstructed from comparator data.

4.15.11 Incompatible-case full-representation molecular lift: Cu, W, and Au

The designated incompatible-case evidence set is **Cu**, **W** (tungsten), and **Au**. The purpose of this set is to verify that atomic→molecular continuation does not depend on reducing materially different atomic structures to one scalar surrogate. The full representation rule applies identically to every case.

Canonical shell states:

Cu	Z=29	N=[2,8,18,1,0,0,0]	occupied shells={1,2,3,4}
W	Z=74	N=[2,8,18,32,14,0,0]	occupied shells={1,2,3,4,5}
Au	Z=79	N=[2,8,18,32,18,1,0]	occupied shells={1,2,3,4,5,6}

For a same-element two-node molecular candidate x_2 , where $X \in Cu, W, Au$, the complete configuration is

$$C_{x_2} = (S_X^1, S_X^2, R_{x_2})$$

and the complete retained field is

$$F_{x_2}(\xi|C) = F_1(\xi|C) + F_2(\xi|C).$$

The occupied-shell overlap set is retained componentwise:

$$O_{X2} = \{ O_{12}^{(nm)}(C) : n,m \text{ are occupied shells of } X \}.$$

Therefore the three cases carry distinct complete relation structures:

```
Cu2: 4 x 4 = 16 occupied-shell overlap components
W2 : 5 x 5 = 25 occupied-shell overlap components
Au2: 6 x 6 = 36 occupied-shell overlap components
```

No case is converted to a common overlap scalar, outer-shell index, electron-total surrogate, bond coordinate, or entropy-only state for structural adjudication. The selector remains

$$C_{X2*} = \operatorname{argmax}_{C \in A_{X2}} S[C]$$

with $S[C]$ evaluated from the complete retained candidate. A coordinate r_12 , a stationary-point condition, $S[C*]$, a projected force/energy, or any other scalar is a property of C_{X2*} and may not substitute for it.

The terminal retained object for every case includes at minimum:

```
{
  S_X^(1), S_X^(2),
  R_X2*,
  {O_12^(nm)(C*)},
  F_X2(xi | C*),
  D[C*],
  q_X2(xi | C*),
  S_config[C*],
  S_field[C*],
  S_coupling[C*],
  S[C*],
  M_Z x M_Z
}.
```

Current standing of the designated incompatible-case set:

```
Cu atomic -> molecular full structural lift: CLOSED
W  atomic -> molecular full structural lift: CLOSED
Au atomic -> molecular full structural lift: CLOSED
Cu2 complete retained two-node representation: CLOSED
W2  complete retained two-node representation: CLOSED
```

Au2 complete retained two-node representation: CLOSED
No-reduction compliance: PASS for all three cases

The evidence records are:

Evidence/Provenance/CU_TO_CU2_FULL_REPRESENTATION_TRANSCRIPT.md
Evidence/Provenance/W_TO_W2_FULL_REPRESENTATION_TRANSCRIPT.md
Evidence/Provenance/AU_TO_AU2_FULL_REPRESENTATION_TRANSCRIPT.md

This three-case set demonstrates atomic→molecular structural continuation across distinct occupied-shell topologies without changing the governing representation law and without reducing the state.

4.16 Time metrology

$$T(X) = n_Cs(X) / 9,192,631,770$$

Physical time is resolved by the Cs-133 transition count. Software wall-clock runtime is metadata, not the physical primitive.

4.16.1 Canonical atomic metrology

The atomic metrology primitive tuple is $\{\lambda_A, N_A, \nu_A, M_A\}$. These quantities are frozen before a downstream structural or molecular result consumes the baseline.

$$L_A = N_A \lambda_A$$

$$\tau_A = M_A / \nu_A$$

$$V_A = L_A / \tau_A = (N_A \lambda_A) / (M_A / \nu_A)$$

The baseline length L_A , baseline interval τ_A , and transfer constant V_A are therefore generated at the atomic metrology layer. They are upstream inputs to later spatial construction; C^* does not regenerate L_A .

For any physical application, the atomic interval must remain consistent with the canonical Cs-133 time resolver. The exact-arithmetic closure fixture in Appendix N verifies the internal metrology identities and dependency direction; it does not replace or recalibrate the Cs-133 physical time primitive.

4.16.2 Frequency metrology

$$f = N_T / T; \text{ for one-cycle interval, } f = 1/\tau$$

Frequency is derived from the count-resolved time relation. It is not an independent primitive.

4.16.3 Empirical observation adapter — validation only

The following normalized-observable constructor is an empirical validation adapter. Its inputs are observational coordinates supplied by a validation run. It is not part of the first-principles atomic generator and it has no authority to construct C^* , L_A , N_r , or any native spatial state.

$$Q_{\text{raw}}^{\text{(obs)}} = I_{\text{norm}} - W_{\text{norm}}; Q_{\text{shell}}^{\text{(obs)}} = \max(|Q_{\text{raw}}^{\text{(obs)}}|, 10^{-10})$$

$$\chi_e^{\text{(obs)}} = P_{\text{norm}} + W_{\text{norm}}$$

$$\Xi_{\text{raw}}^{\text{(obs)}} = I_{\text{norm}} - \Pi_{\text{norm}}; \Xi_F^{\text{(obs)}} = \max(\Xi_{\text{raw}}^{\text{(obs)}}, 10^{-10})$$

$$a = 0.5; b = 1.0; q = 1.0$$

$$\Theta_{\text{obs}} =$$

$$\Xi_F^{\text{(obs)}}$$

$$^a \exp(-b \chi_e^{\text{(obs)}})$$

$$Q_{\text{shell}}^{\text{(obs)}}$$

q

4.16.4 Universal native-distance projection

Every native spatial-distance count N_D is already a resolved SEAM distance at the native count layer. The canonical atomic baseline projects that count through the universal quantity-class rule:

$$L = N_D L_A$$

For the H₂ center-separation coordinate $N_r = ||X_B - X_A||$:

$$L_{H_2} = N_r L_A = N_r N_A \lambda_A$$

Native count closure, atomic baseline construction, empirical validation, and conventional-unit reporting remain distinct layers. No empirical observable or dimensional H₂ comparator is permitted to define λ_A , N_A , v_A , M_A , L_A , τ_A , V_A , N_r , or C^* .

4.16.5 Executed atomic-metrology closure test

RUN-SEAM-NATIVE-ATOMIC-METROLOGY-CLOSURE-01 is recorded in Appendix N. The hash-locked exact-arithmetic run terminated PASS: NATIVE_ATOMIC_METROLOGY_CHAIN_NUMERICALLY_CLOSED. The test demonstrates executable closure and dependency direction; it does not calibrate the empirical physical magnitude of L_A .

4.17 Essential notation bridge

The following symbols are the minimum vocabulary needed for the mathematical ledger that follows. The full normative glossary remains in Appendix J.

Symbol	Meaning	First dependency
$Z(n)$	baseline electron count	primitive atomic start
c_shell	seven-shell capacity vector	fixed structural bound
c_n	capacity of shell n	component of c_shell
$N_{i,n}$	occupancy of shell n for entity i	derived from $Z(n)$ distribution
$\rho_{i,n}$	normalized shell occupancy $N_{i,n}/c_n$	derived
$\mathcal{S}_i$	seven-shell entity state	$\{\rho_{i,1}, \dots, \rho_{i,7}\}$
$\mathcal{R}$	complete relational state	$\{r_{ij}, \chi_{ijkl}\}$
$\mathcal{C}$	complete physical configuration	$(\{\mathcal{S}_i\}, \mathcal{R})$
$\mathcal{A}$	admissible configuration set	declared structural constraints
$\mathcal{S}$		

$\mathcal{C}$

$\mathcal{C} \in \mathcal{A}$ | governing ESAM entropy functional on $\mathcal{C} \in \mathcal{A}$; binding in §4.15 | explicit three-term functional

| $\mathcal{C}^*$ | selected configuration | $\arg \max_{\{C \in \mathcal{A}\}} \mathcal{S}$

$\mathcal{C}$

| $\hat{H}_{SEAM}$

$\mathcal{C}^*$

| Hamiltonian operator description of selected state | downstream of $\mathcal{C}^*$ | | Coh_struct | internal structural/continuity coherence | pre-Manifold | | Coh_M | Manifold coherence | post-comparison | | Δ_M | Manifold residual | $1 - \text{Coh}_M$ | | $T(X)$ | physical time resolver | $n_{Cs}(X)/9,192,631,770$ |

Chapter 5

Core claim ledger

Each claim states its scope, definitions, calculation, end result, and falsification condition. The statements below are the canonical claims of this specification.

5.1.1 Claim

SEAM is structure-first, not label-first. A readable label is downstream of a represented structure and its admissible transformations.

5.1.2 Definitions

X = input entity or question

$S(X)$ = represented structural state

C = complete configuration where the problem requires relational geometry

Φ = projection operator

5.1.3 Calculation

Observation or a declared structural primitive is converted into a represented state before classification.

For atomic/composite questions the represented state is completed as $C=(\{\mathcal{S}_i\},\mathcal{R})$, then the admissible configuration is selected before projection.

represented structure -> admissible configuration -> projection -> comparison

5.1.4 End result

A result is supported because its represented structure survives the applicable operators and gates, not because its name resembles a known category.

5.1.5 Falsification condition

This claim fails if a valid answer can be produced by bypassing structural representation and selecting a label directly.

5.2.1 Claim

ESIH, ESAM, and SEAM are stages of formalization, not separate theories competing for the same role.

5.2.2 Definitions

PRIN = ESIH principles / axiomatic-constraint layer

MATH = ESAM mathematical-formulation layer

ENG = SEAM executable mathematical-engine layer

5.2.3 Calculation

PRIN(X_atom) -> Z(n) and the structural interaction constraints

MATH(HYP) -> {c_shell, c_n, N_i,n, rho_i,n, $\mathcal{S}_i$, $\mathcal{R}$, C, S

C

}

ENG(MATH,X) -> primitive result

SEAM = Engine

ESAM under ESIH constraints

5.2.4 End result

5.2.5 Falsification condition

This claim fails if the SEAM engine cannot execute the ESAM mathematical-formulation layer or if a layer contradicts rather than generalizes the layer.

5.3.1 Claim

The atomic start is count-based from first principles. The canonical baseline is Z(n), the number of electrons to be distributed.

5.3.2 Definitions

Z(n) = baseline electron count

c_n = scalar capacity of shell n

N_n = electron occupancy assigned to shell n

rho_n = N_n/c_n

5.3.3 Calculation

c_shell =

2, 8, 18, 32, 18, 32, 18

sum_n N_n = Z(n)

0 <= N_n <= c_n

Fill proceeds from lower shell index outward under the established fill/skip protocol. The shell state is obtained by calculation rather than by assigning a conventional element label first.

Example	Z(n)	First shell state	Next occupied shell	Structural reading
H	1	1/2	—	partial first-shell occupancy
He	2	2/2	—	first-shell closure
Li	3	2/2	1/8	closed first shell plus partial outer shell
Ne-like count				

EVID – NE – ATOM – 001

| 10 | 2/2 | 8/8 | two-shell closure under the seven-shell capacity rule |

5.3.4 End result

The starting state is a conserved count distributed into a bounded seven-shell architecture. Proton, neutron, charge, isotope, and other state information can modify or extend the state downstream but do not replace Z(n) as the baseline distribution statement.

5.3.5 Falsification condition

The claim fails if the fill/skip constructor changes the total count, overfills a shell, or requires the target element label to determine the output.

5.4.1 Claim

Closure is not an if-statement. A closed configuration is selected by a structural extremum over the admissible candidate set.

5.4.2 Definitions

A = admissible configuration set

S

C

= canonical ESAM entropy functional on configuration C, explicitly bound in §4.15

C* = selected configuration

5.4.3 Calculation

$C^* = \arg \max_{\{C \in A\}} S$

C

Shell capacity, count conservation, relation completeness, and applicable symmetry constraints define admissibility. The selected state is not generated by an if element==... branch.

The entropy functional is explicitly defined in §4.15. A runtime terminates SYMBOLIC_ONLY when every required symbolic object exists but a declared runtime artifact needed for numerical evaluation is absent. A runtime terminates INDETERMINATE: UNDECLARED_INPUT when a required mathematical object, transform, or contract field is not declared.

5.4.4 End result

Closure is calculated from the same configuration-selection principle used across atomic, molecular, and amalgamum problems.

5.4.5 Falsification condition

This claim fails if the runtime bypasses the admissible candidate calculation and assigns closure from a hard-coded element, molecule, or category name.

5.5.1 Claim

No single reduced scalar is the whole structural verdict. Shell closure, relational geometry, and configuration entropy remain distinct information carriers.

5.5.2 Definitions

S_i = seven-shell state

$\mathcal{R}$ = complete relational state

$C = (\{\mathcal{S}_i\}, \mathcal{R})$

S

C

= entropy over the complete candidate configuration

5.5.3 Calculation

$C = (\{\mathcal{S}_i\}, \mathcal{R}), \mathcal{R} = \{r_{ij}, \chi_{ijkl}\}$

Two candidates can have the same shell totals but different relations; two geometric candidates can share all pairwise distances yet differ by reflection parity. Therefore a scalar balance coordinate cannot replace the complete configuration.

5.5.4 End result

The structural verdict is attached to the complete admissible configuration and its downstream projection, not to one balance coordinate.

5.5.5 Falsification condition

This claim fails if all distinctions relevant to the tested problem are provably recoverable from the proposed single scalar without loss.

5.6.1 Claim

The local structural calculation path is complete at the representation level: count distribution creates shell states, relations complete the configuration, and entropy orders admissible candidates.

5.6.2 Definitions

$Z(n)$ = baseline count

$\mathcal{S}_i$ = shell state

r_{ij} = pairwise separation

χ_{ijkl} = parity relation

C = complete configuration

5.6.3 Calculation

$Z(n) \rightarrow \{N_{i,n}\} \rightarrow \{\rho_{i,n}\} \rightarrow \mathcal{S}_i$

$\{\mathcal{S}_i\} + \{r_{ij}, \chi_{ijkl}\} \rightarrow C$

$C^* = \arg \max_C S$

C

Case	Shell occupancy summary	Relation requirement	Selection
single closed shell	$\rho_1=1$	none beyond self-state	candidate evaluated directly
single partial shell	$0<\rho_1<1$	none beyond self-state	partial structural state retained
two entities	$\{S_A, S_B\}$	r_{AB}	joint candidate entropy evaluated
three+ entities	$\{\mathcal{S}_i\}$	all r_{ij}	complete pairwise geometry evaluated
chiral geometry	$\{\mathcal{S}_i\}$	$r_{ij} + \chi_{ijkl}$	reflection-sensitive candidate evaluated

5.6.4 End result

The same construction can separate candidate structures without changing the procedure case by case.

5.6.5 Falsification condition

The claim fails if a required geometric distinction cannot be represented by $\mathcal{R}$ or if a retained candidate violates count/capacity constraints.

5.7.1 Claim

Transfer closure extends structural closure without creating a separate force law. A selected structural state can be followed through ordered transfer, projection, or interaction stages.

5.7.2 Definitions

C_j = configuration at stage j

$\mathcal{T}$ = transfer/transition operator

S

C

= governing entropy measure

5.7.3 Calculation

$$C_j \xrightarrow{\mathcal{T}} C_{j+1}$$

accepted structural transition preserves declared invariants and evidence record

Where physical time is involved, the interval is expressed through the Cs-133 transition-count resolver rather than an independent canonical time primitive.

5.7.4 End result

Structural closure and transfer accounting remain distinct but composable operations. A transfer path cannot erase the structural state from which it originated.

5.7.5 Falsification condition

The claim fails if transfer output cannot be traced to its input state or if a time-dependent calculation uses an unbound time primitive.

5.8.1 Claim

Coherence is a retained value, not a named state. It quantifies an engine-defined relation between represented structures or projections.

5.8.2 Definitions

v_X = candidate projection

v_M = Manifold reference projection

Coh = normalized alignment or engine-defined coherence

5.8.3 Calculation

$$\text{Coh}_M(X, M) = \text{dot}(v_X, v_M) / (||v_X|| ||v_M||)$$

normalized – alignment instantiation

$$\Delta_M = 1 - \text{Coh}_M$$

Other engine-defined structured comparison rules replace the scalar instantiation only when the active ARCV contract explicitly declares them.

5.8.4 End result

Coherence remains downstream of structural construction and does not replace configuration entropy.

5.8.5 Falsification condition

The claim fails if coherence is assigned as a categorical confidence label rather than calculated from represented structure.

5.9.1 Claim

The Continuum is not the Manifold.

5.9.2 Definitions

K_t = retained Continuum state at cycle t

M_t = normalized Manifold derived from retained state

5.9.3 Calculation

$K_t \rightarrow \text{normalization/projection} \rightarrow M_t$

Continuum retention preserves source/reference structure. Manifold construction produces the comparison surface used by the active query.

5.9.4 End result

A single retained structure can support multiple admissible projections without becoming multiple underlying entities.

5.9.5 Falsification condition

The claim fails if Continuum and Manifold are treated as interchangeable artifacts with no transformation or role distinction.

5.10.1 Claim

The engine process is not the classification surface.

5.10.2 Definitions

Engine = structural calculation process

Classification = downstream organizational description

5.10.3 Calculation

question -> engine -> primitive result -> optional classification

The 48-sector / five-regime surfaces organize outputs only as classification surfaces; they do not define the engine call sequence or atomic mathematics.

5.10.4 End result

A classification describes where a resolved result lands. It does not substitute for the structural calculation.

5.10.5 Falsification condition

The claim fails if the category label alone determines the primitive answer without engine evaluation.

5.11.1 Claim

The 48-sector / 5-regime model is classification, not proof by itself.

5.11.2 Definitions

sector = classification channel

regime = operating/projection regime

projection = represented facet

5.11.3 Calculation

48 sectors * emitted fields -> classification surface

Regime placement constrains an admissible projection only when the active contract declares that constraint. Closure still requires the active structural and validation gates.

5.11.4 End result

Classification coverage is not equivalent to empirical validation or mathematical closure.

5.11.5 Falsification condition

The claim fails if sector/regime placement is treated as sufficient proof of a physical result.

5.12.1 Claim

Manifold coherence can be calculated as normalized alignment when the active representation supports that metric.

5.12.2 Definitions

$v_X = \text{Phi}(C^*)$

v_M = active reference vector

5.12.3 Calculation

$\text{Coh_M}(X,M) = \text{dot}(v_X, v_M) / (||v_X|| ||v_M||)$

$\Delta_M = 1 - \text{Coh_M}$

5.12.4 End result

The normalized-alignment form is a valid reference instantiation, not a declaration that every production manifold is globally Euclidean.

5.12.5 Falsification condition

The claim fails if zero norms, incompatible coordinates, or post-hoc reference construction are ignored.

5.13.1 Claim

Closure can be represented as a composed operator.

5.13.2 Definitions

Operator-level distinction: CC computes internal structural/continuity coherence Coh_struct . MC then compares the retained projection with the Manifold and produces Coh_M and Δ_L . Therefore the established order $\text{CC} \rightarrow \text{MC}$ does not imply that Manifold coherence is known before Manifold comparison.

$\Re$ = representation constructor

SC = structural/configuration closure

TC = transfer closure

CC = structural/continuity coherence operator producing Coh_struct

MC = Manifold comparison operator producing Coh_M and Δ_M

VC = validation closure

5.13.3 Calculation

$\text{Closure_SEAM}(X) = \text{VC}(\text{MC}(\text{CC}(\text{TC}(\text{SC}(\mathfrak{R}(X))))))$

Within the canonical atomic path, $\mathfrak{R}$ constructs the represented shell/configuration state and SC contains the entropy-selection step where applicable.

SC_atomic: $Z(n) \rightarrow \mathcal{S}_i \rightarrow C \rightarrow C^*$

5.13.4 End result

The composed operator remains compatible with the canonical engine architecture while exposing the canonical structural mathematics at the front of the chain.

5.13.5 Falsification condition

The claim fails if a downstream stage requires information that the preceding stage neither contains nor can derive under the declared contract.

5.14.1 Claim

Answer formation is downstream of primitive structure.

5.14.2 Definitions

P = primitive result

A = readable answer

Resolver = $P \rightarrow A$ mapping

5.14.3 Calculation

structural execution $\rightarrow P \rightarrow \text{Resolver}(P) \rightarrow A$

The resolver is not permitted to substitute a plausible answer when the structural execution refuses or remains indeterminate.

5.14.4 End result

A readable answer inherits support from the primitive result.

5.14.5 Falsification condition

The claim fails if a natural-language response can be accepted when its primitive support is absent or contradictory.

5.15.1 Claim

Validation is mathematical gatekeeping, not agreement with a narrative.

5.15.2 Definitions

native result = output frozen before comparator

evidence = external observation

comparison = post-result projection/test

5.15.3 Calculation

first principles -> native result -> hash/freeze -> evidence reveal -> comparison

External evidence has four legitimate roles: corroboration, falsification, comparison, and projection testing. It does not define the native operator.

5.15.4 End result

A mismatch is retained as information about the SEAM structure rather than repaired by post-hoc fitting.

5.15.5 Falsification condition

The claim fails if a coefficient or operator is changed after held-out evidence reveal and the modified result is presented as a blind prediction.

Chapter 6

Start-to-finish calculation path

6.1 Input to observation

A raw candidate or question is normalized into observation-bound observations or a declared structural primitive.

$X \rightarrow O(X)$

6.2 Observation to structure

Observations become represented components and relations. Atomic work begins from $Z(n)$ and the seven-shell constructor.

$$Z(n) \rightarrow \mathcal{S}_i$$

6.3 Structure to projection

Where the question requires a complete candidate configuration, structure is completed and entropy-selected before projection.

$$\mathcal{S}_i \rightarrow C \rightarrow C^* \rightarrow \text{Phi}(C^*)$$

6.4 Projection, structural coherence, and Manifold comparison

The selected projection first retains its internal structural/continuity coherence Coh_struct . The Manifold comparison operator MC then evaluates that projection against the active reference and produces Coh_M and Δ_L .

$$\text{Phi}(C^*) \rightarrow \text{Coh_struct} \rightarrow \text{MC}(M_t) \rightarrow \text{Coh_M} \rightarrow \Delta_M$$

6.5 Atomic local structural path, when atomic variables are available

The canonical atomic starting path is count distribution, shell state, relation completion, and entropy selection. The p,n,e scalar framing is not the canonical starting state.

$$Z(n) \rightarrow \{N_n\} \rightarrow \{\rho_n\} \rightarrow \mathcal{S}_i \rightarrow C \rightarrow C^*$$

6.6 Continuum update path

Validated retained structure updates the Continuum when the retention gate succeeds. A failed or comparator-only item does not silently become active reference structure.

$$K_{\{t+1\}} = \text{Retain}(K_t, \text{validated output})$$

6.7 Primitive to answer

The primitive result is passed to the resolver only after validation and support gates succeed.

$$P_X \rightarrow \text{Resolver} \rightarrow A_X$$

6.8 Time-dependent calculation path

physical interval $X \rightarrow n_{Cs}(X) \rightarrow T(X) = n_{Cs}(X) / 9,192,631,770$

Any rate, frequency, propagation, trajectory, force, energy, or Hamiltonian operation that requires physical time must expose this metrological dependency or explicitly declare a dimensionless order parameter instead.

Chapter 7

Calculation boundaries by layer

Layer	Receives	Produces	Prohibited action
ESIH	question / baseline structural condition	first-principle constraint set / $Z(n)$ baseline	import an accepted explanation as a primitive
ESAM	$Z(n)$, admissible relational data	$\mathcal{S}_i$, C, S	

C

, selected C^* | fit the native law to held-out evidence | | SEAM engine | formal structural math + question/reference artifacts | primitive result, closure/coherence trace | replace engine math in a runner | | Continuum | validated retained structures | persistent reference state | act as a hidden answer key | | Manifold | Continuum-derived normalized representation | query-comparison structure | replace retained Continuum canonical equation | | Resolver | primitive result | readable answer | invent unsupported primitive content | | External evidence | observation/comparator admitted by contract | corroboration/falsification/comparison | define native SEAM mathematics |

Chapter 8

Canonical answer to "what is SEAM?"

SEAM is the operational layer of a structure-first mathematical framework. Its atomic implementation begins from a count baseline, constructs a bounded seven-shell state, forms complete configurations from structural and relational information, selects admissible configurations under an entropy extremum, projects retained structure into question-relevant facets, and compares those facets against a normalized Manifold derived from the retained Continuum.

$Z(n) \rightarrow \mathcal{S}_i \rightarrow C \rightarrow C^* \rightarrow \Phi_f(C^*) \rightarrow Coh_struct \rightarrow Manifold\ comparison \rightarrow Coh_M/\Delta_M \rightarrow primitive\ result$

SEAM is not defined by a classification label, a conventional force law, or a statistical confidence score. It is defined by the ordered structural calculation and its source/validation contracts.

Chapter 9

Canonical answer to "how does SEAM work?"

The complete operational answer is an ordered process:

1. Accept one question and identify the structural primitive required by the active operator.
2. Construct the atomic or retained structural state without target-answer leakage.
3. Generate the admissible candidate configuration family.
4. Apply the governing entropy-selection rule to obtain C^* where configuration selection is required.
5. Project the retained structure into the query-relevant facet.
6. Query or compare against the active Manifold through the engine.
7. Calculate internal structural coherence Coh_struct , Manifold coherence Coh_M , Manifold residual Δ_M , operational/query closure $\mathcal{C}_Q$, and validation support under the active contract.
8. Freeze and hash the native result before external comparator reveal in blind tests.
9. Allow the resolver to express the primitive result in readable form.

question -> structural primitive -> $C \rightarrow C^* \rightarrow \Phi_f(C^*) \rightarrow Coh_struct \rightarrow$ Manifold comparison -> $Coh_M/\Delta_M \rightarrow \mathcal{C}_Q/validation \rightarrow$ primitive -> answer

Physical time, when required anywhere in this path, is resolved through the Cs-133 count relation.

Chapter 10

Final skeptical summary

The strongest skeptical reading of SEAM is not answered by adding more labels. It is answered by specifying what enters the calculation, what is derived, what is retained, what the engine actually executes, what evidence is withheld until after prediction, and what would falsify each claim.

- $Z(n)$ fixes the atomic baseline rather than assuming the full atomic state.
- The seven-shell capacity vector fixes the discrete occupancy bounds.
- $C=(\{\mathcal{S}_i\},\mathcal{R})$ fixes the configuration representation, including pairwise geometry and chirality parity.
- $C^*=\arg \max_C S$

C

fixes the configuration-selection principle across scales.

- Cs-133 count metrology fixes physical time without circularly assuming a second.
- Continuum and Manifold remain distinct retained-reference and normalized-comparison objects.
- Engine closure remains an executable composition rather than a narrative conclusion.
- External evidence is downstream of the native calculation.

The technical burden is therefore explicit: every claimed native result must be reproducible from these definitions and the declared engine/runtime artifacts. Where a external operator survives only as evidence record, it must be identified as such rather than silently promoted.

Part II

Engine Formalization

Chapter 11

Structural input and query formation

The runner supplies an untouched question and the active structural/reference artifacts. It does not decide the answer. The engine determines which structural primitive and functions are applicable under the active contract.

Q -> question representation

atomic Q -> Z(n) baseline when atomic construction is required

The canonical atomic constructor returns the shell state under the fixed capacity vector. Downstream state modifiers such as charge, isotope, or explicitly represented environmental state are applied as declared transformations, not used to redefine the starting count.

Input field	Role
question text / question.arcv	untouched query surface
UID	run identity
requested projection	declares output facet, not answer
Z(n) or retained primitive	structural starting quantity
active manifold/reference hash	comparison evidence record
metrology baseline	required only when a physical time quantity is used

One question per run remains the canonical transcript rule because it keeps the question-to-primitive model composition inspectable.

Chapter 12

Q-ARC, A-ARC, and ARCV

ARCV remains the finite projection representation used by the engine. Q-ARC is the represented query. A-ARC is the candidate/reference representation. The atomic shell/configuration object is upstream of, or embedded into, the ARCV projection contract; ARCV does not redefine the physical ontology.

$Q_arc = ARCV(question\ structure)$

$A_arc = ARCV(candidate/reference\ structure)$

A complete ARCV carries exactly the magnitude, ordered components, normalized vector structure, relations, regime, admissible-deviation bounds, and intent fields declared by the active runtime schema. A local Euclidean example does not establish a globally Euclidean 51x5->64 production geometry.

Chapter 13

Projection cardinality and structure semantics

Projection cardinality describes how many facets an engine representation emits; it does not define the underlying physical object. The complete configuration C can support several projections without being flattened into any one of them.

$\Phi_f: C^* \rightarrow v_f$

If a 51-arc by 5-regime architecture yields a 64-component projection in a given runtime, that cardinality is an engine representation contract. It is not a claim that the physical configuration has exactly 64 independent degrees of freedom.

Object	Meaning
C^*	selected structural configuration
$\Phi_f(C^*)$	question/regime-specific facet
ARCV	finite engine representation of required facets/constraints
Manifold projection	normalized comparison representation

Chapter 14

Direct interaction

The engine compares represented structures directly. It does not assign an arbitrary external baseline score to a candidate. For Q-ARC and candidate A-ARC, the interaction record contains only the magnitude, component, vector, relation, regime, and intent deviations declared by the active implementation contract.

The structured discrepancy record is aggregated only by a declared query-contract normalization operator. Define $D_Q(Q,Y)=\mathcal{N}_Q(\Delta_ARCV(Q,Y))$, where $\mathcal{N}_Q$ is fixed by the active Q-ARC/ARCV admissibility contract rather than fitted to the expected answer.

$\Delta_ARCV(Q,Y_j) = (\Delta_m, \Delta_c, \Delta_v, \Delta_R, \Delta_regime, \Delta_intent)$

A scalar ordering is derived only when the engine contract explicitly defines component normalization, masking, gating, and aggregation. The underlying comparison remains between represented structures.

14.1 Configuration-level direct interaction

Where the question is a physical configuration problem, direct interaction begins before ARCV comparison by constructing admissible C candidates and applying S

C

.

candidate family $A \rightarrow \{C_i\} \rightarrow S$

C_i

$\rightarrow C^* \rightarrow$ projection $\rightarrow$ ARCV/Manifold interaction

This preserves the distinction between physical configuration selection and query/reference alignment.

Chapter 15

Full-function evaluation and PLL arbitration

The runner does not decide which engine function is relevant by keyword, regular expression, template, or hard-coded answer branch. It supplies the untouched input and active Manifold to the engine. The engine exposes all applicable functions and evaluates them before a universal refusal.

$\text{Eval}(Q, A_{\text{arc}}) = \{ f_j(Q, A_{\text{arc}}) : f_j \in \mathcal{F} \}$

Each function produces exactly one declared terminal form: admissible result, local refusal, or execution error. A local refusal does not terminate the run while another applicable function remains unevaluated.

$Y^* = \arg \min_{\{Y_j \text{ in } V(Q)\}} D_Q(Q, Y_j)$

Operationally this is the admissible candidate with maximum engine closure under the ARCV contract. For a native configuration-selection problem, the physical candidate set is first ordered by S

C

; the PLL then arbitrates applicable query projections of the retained structure. The two operations must not be conflated.

Stage	Ordering
physical candidate selection	$C^* = \arg \max S$

C

over admissible C | | query/reference resolution | lowest engine discrepancy / highest admissible closure under active ARCV contract | | tie break | canonical deterministic engine ordering |

A template miss or keyword miss is a dispatch failure, not a SEAM production refusal.

Chapter 16

Closure coefficient and admissibility

Engine/query closure is continuous under the canonical ARCV contract:

$c_Q(Y_j | Q)$ in

0, 1

Value	Interpretation
1	full closure relative to the query and requested projection
0	no closure relative to the query
$0 < C < 1$	partial, transitional, emergent, or metastable closure

$$c_Q(Y|Q) = \max(0, 1 - D_Q(Q,Y)/\epsilon_Q), \epsilon_Q > 0$$

referenceinstantiation

Here ϵ_Q is not a free physical constant. It is a deterministic tolerance extracted from the admissibility bounds encoded by the active Q-ARC contract.

$$\epsilon_Q = \mathcal{E}_Q(Q-ARC) > 0$$

deterministicadmissibilitytoleranceextractedfromtheactiveQ – ARCbounds

This engine closure coefficient is not the same mathematical object as the physical configuration entropy S

C

. The engine closure coefficient evaluates query/reference admissibility after representation; S

C

selects an admissible physical configuration before or within representation.

No demonstration epsilon value becomes a universal threshold merely because it appears in a worked example.

Chapter 17

Structural trajectory and transference accountability

A structural trajectory is an ordered series of states or configurations. Every transition must preserve declared invariants and record what changes.

$C_0 \rightarrow C_1 \rightarrow \dots \rightarrow C_k$

When the sequence is a numerical optimization path, the iteration index is computational order and must not be mistaken for physical time. When the sequence represents physical evolution, each interval is resolved through the Cs-133 transition-count metrology.

$\Delta T_j = \Delta n_{Cs,j} / 9,192,631,770$

Transference accountability asks whether the observed/derived change between adjacent states is explained by the declared operator, boundary conditions, and external inputs. Unaccounted structural change is a failed or incomplete transition, not evidence to be silently absorbed into a fitted constant.

When the transition is most clearly described by an operator, write $C_j \rightarrow \hat{H}_{SEAM}$

C_j

$\rightarrow C_{\{j+1\}}$. This notation describes the interaction/evolution of an already admitted state; admissibility and entropy ordering remain upstream.

Chapter 18

Hamiltonian correspondence

Entropy-first ESAM selects the admissible physical configuration before any Hamiltonian description is invoked. Where an operator description removes ambiguity, the selected configuration is represented by the SEAM Hamiltonian operator $\hat{H}_{SEAM}$

C^*

. The Hamiltonian is therefore a downstream operator representation of the entropy-selected state, not the primitive that replaces S

C

.

$C^* \rightarrow \hat{H}_{SEAM}$

C^*

--> operator description / derived observable

The canonical Hamiltonian correspondence requires a complete Hamiltonian to expose state space, canonical algebra, interaction scale, symmetry, conservation, domains, and a non-circular natural-state extraction rule.

$\Psi(C^*) \rightarrow \{\lambda_k$

Ψ

, $\mathcal{C}$

Ψ

, $\mathcal{K}_r$

Ψ

} $\rightarrow \hat{H}_{SEAM}$

C^*

$\rightarrow$ derived observable / prediction

$\lambda_k = \lambda_k$

$\Psi(C^*)$

state – dependent architecture

The H2 named run contracts explicitly freeze $\lambda_{field}=1.0$ and $\lambda_{coupling}=0.1$. These are run-contract coefficients, not universal constants, and are not silently identified with the general state-dependent coefficient architecture.

Canonical ordering: construct C, select C^* by S

C

, then use $\hat{H}_{SEAM}$

C^*

wherever the declared operator contract requires interaction, state evolution, or observable correspondence. No Hamiltonian coefficient is imported as a new primitive unless its extraction from the selected SEAM state is explicitly defined.

The canonical entropy binding defines the Hamiltonian critique boundary. Configuration selection is mathematically defined without requiring a complete atomic Hamiltonian. A Hamiltonian is required only when the claimed downstream result is an energy, force, time evolution, spectrum, or other operator-derived observable. It is not permitted to define retroactively which C is admissible or entropy-selected unless that dependence is explicitly included in S

$$C$$

and frozen before comparison.

18.1 Existing Hamiltonian correspondence and ratio

The Hamiltonian layer is a downstream representation of the already-selected complete state; it is not a new molecular force law. The canonical correspondence is

$$H_S(C^*) = \hat{H}_{\text{SEAM}}[C^*].$$

For comparison with a nonzero reference Hamiltonian that represents the same requested physical projection,

$$R_H(C^*; H_{\text{ref}}) = \frac{H_S(C^*)}{H_{\text{ref}}}, \quad H_{\text{ref}} \neq 0.$$

The correspondence $H(S^*)$ denotes the Hamiltonian consequence of the complete selected state C^* , with S^* retained as one evaluated property of that state. Entropy alone does not replace the configuration.

The same correspondence is used when C^* is atomic, molecular, composite, or macroscopic. Therefore molecular continuation imports the already-defined interaction/Hamiltonian operator unchanged:

$$\begin{aligned} C_{XY}^* &\rightarrow \{u_{XY}^*(N_r)\} \\ &\rightarrow S_{XY}^*(N_r) \rightarrow \hat{H}_{\text{SEAM}}[C_{XY}^*] \\ &\rightarrow \Delta H_{XY}^{(J)}(N_r) \rightarrow F_{XY}(N_r). \end{aligned}$$

The native Hamiltonian, pair-energy, and force quantities are consequently **defined operator quantities** once their complete retained state is supplied. A conventional numerical correspondence remains downstream and requires a declared same-projection reference/unit adapter; absence of that adapter is not absence of the Hamiltonian or force quantity.

18.2 Canonical v18.6 variational-field realization

For each occupied shell of each retained entity,

$$\Omega_{i,n}(C, N_r) = \{\xi : n - 1 \leq \|\xi - X_i\| < n\}.$$

The admissible interacting-field class is

$$U_{i,n}(C, N_r) = \left\{ u \geq 0 : \text{supp } u \subseteq \Omega_{i,n}(C, N_r), \int_{\Omega_{i,n}} |u|^2 d^3\xi = 1 \right\}.$$

The isolated-state and fixed-branch baseline is

$$u_{i,n}^{(0)}(\xi) = \frac{\mathbf{1}_{\Omega_{i,n}}(\xi)}{\sqrt{V_n}}.$$

For an interacting configuration at fixed native separation N_r , the field profile is selected variationally from the complete shell-constrained product family:

$$\{u_{i,n}^*(N_r)\} = \arg \max_{\{u_{i,n}\} \in \prod_{i,n} U_{i,n}(C, N_r)} S[C; N_r, \{u\}].$$

The shell support, shell occupancy, configuration topology, entropy composition, coupling law, comparator firewall, and declared run contract remain frozen while the normalized internal field profile is selected. Center weighting, boundary weighting, interaction-axis weighting, depletion, or concentration may emerge only through this entropy maximization. No profile may be inserted or tuned from comparator evidence.

The resolved complete field is

$$F_C(\xi; N_r) = \sum_{i,n} \sqrt{N_{i,n}} u_{i,n}^*(\xi; N_r).$$

Its normalized density is

$$q_C(\xi; N_r) = \frac{|F_C(\xi; N_r)|^2}{\int |F_C(\xi; N_r)|^2 d^3\xi}.$$

For each shell field,

$$p_{i,n}(\xi; N_r) = |u_{i,n}^*(\xi; N_r)|^2,$$

and the retained overlap is

$$O_{ij}^{nm}(N_r) = \int \sqrt{p_{i,n}(\xi; N_r) p_{j,m}(\xi; N_r)} d^3\xi.$$

The entropy components remain

$$S_{\text{field}}[C; N_r] = -k_B \int q_C \ln q_C d^3\xi,$$

$$S_{\text{coupling}}[C; N_r] = k_B \sum_{i < j} \sum_{n, m} \rho_{i, n} \rho_{j, m} O_{ij}^{nm}(N_r),$$

with the complete selected entropy

$$S^*(N_r) = \max_{\{u\} \in \prod U_{i, n}} [S_{\text{config}} + \lambda_{\text{field}} S_{\text{field}} + \lambda_{\text{coupling}} S_{\text{coupling}}].$$

18.2.1 Deterministic numerical realization of the variational selector

The mathematical selector remains the continuous shell-constrained variational definition above. Canonical numerical evaluation is not implementation-free. A numerical value reported as $S^*(N_r)$ is canonical only when produced by the active numerical contract or by an explicitly versioned successor contract.

The active evaluator is:

```
runtime_id      = VF-REGION-LBFGSB-R4
implementation  = Evidence/Runtime/variational_field_evaluator.py
Python         = 3.13.5
NumPy          = 2.3.5
SciPy          = 1.17.0
arithmetic     = binary64 / float64
```

At fixed N_r , each occupied shell support is partitioned into its exclusive region and every geometrically distinct intersection with an occupied shell of the opposing node. Since shells belonging to one node are mutually disjoint, this partition is complete for the two-node shell-support geometry. Each field is represented by one nonnegative constant amplitude per nonzero support-membership region, with shell normalization enforced through normalized region masses.

For shell **n** and shell **m**, the exact intersection volume is

$$V_{nm}(N_r) = V(B_n \cap B_m) - V(B_{n-1} \cap B_m) - V(B_n \cap B_{m-1}) + V(B_{n-1} \cap B_{m-1}),$$

using the exact two-sphere lens formula implemented in the bound evaluator. No voxel grid, cylindrical grid, sampled quadrature mesh, empirical bond distance, or comparator-conditioned field is admitted by this contract.

The optimization contract is fixed:

```
method         = L-BFGS-B
starts         = 5 deterministic starts
start scales   = [0.00, 0.35, 0.70, 1.05, 1.40]
ftol           = 1.0e-15
```

```

gtol          = 1.0e-9
maxiter       = 10000
maxls        = 100
accept gradient norm= 2.0e-5
accept top-two delta= 2.0e-6
winner        = highest complete S[C]; lower start index breaks exact ties

```

The five starts are deterministic perturbations of the uniform region-mass state using the fixed category patterns encoded in the bound implementation. Random initialization is prohibited.

A point terminates `RESOLVED` only when both numerical acceptance gates pass. Otherwise it terminates `INDETERMINATE_OPTIMIZER_NOT_CONVERGED`; the executor may not substitute a lower local maximum, change the grid/basis, add starts, loosen tolerances, or repair the run without a new versioned contract.

The canonical reference execution $Cu2$, $N_r = 6.20$, $\lambda_{\text{field}} = 1.0$, $\lambda_{\text{coupling}} = 0.1$ gives

$$S^*(6.20)/k_B = 12.502276251235417$$

with selected gradient norm `5.024295863460037e-7` and top-two objective difference `2.048518332742333e-8`. These are execution values bound to `VF-REGION-LBFGSB-R4`; they do not replace the complete selected field state.

No separate native electric/magnetic/mixed-channel decomposition is required to close **this v18.6 variational molecular branch**. Its radial response is the selected variational state itself. This statement is branch-scoped and does not supersede the separately declared long-range attraction Hamiltonian used by the macroscopic/long-range attraction branch, whose radial factor is `eta(N_R)/N_R`.

18.3 Radial response, Hamiltonian difference, and force

The native radial response is the resolved variational entropy curve itself:

$$R_{\text{SEAM}}(N_r) \equiv S^*(N_r).$$

A finite selected separation, when the active run contract requires one, is a relational coordinate of the complete selected state:

$$N_r^* = \arg \max_{N_r} S^*(N_r).$$

For an interior extremum,

$$\left. \frac{dS^*}{dN_r} \right|_{N_r^*} = 0, \quad \left. \frac{d^2 S^*}{dN_r^2} \right|_{N_r^*} < 0.$$

The complete configuration remains the physical state; N_r^* and S^* are retained properties of that state and do not replace it.

For every resolved pair state $C_{XY}^*(N_r)$, the pair Hamiltonian difference is

$$\Delta H_{XY}^{(J)}(N_r) = E_H[C_{XY}^*(N_r)] - E_H[C_X^* \oplus C_Y^*].$$

The pair-energy magnitude is

$$E_{XY}(N_r) = |\Delta H_{XY}^{(J)}(N_r)|.$$

The inherited Hamiltonian/Joule correspondence may be written

$$H^{(J)} = E_0 \tau_{\text{SEAM}} + \int T dS.$$

For molecular execution, τ_{SEAM} is supplied by the declared Cs-133 event interval when a time-dependent Hamiltonian projection is requested. Static pair structure and elapsed event time are distinct admitted inputs. The pair-state/event metrology path is closed by the current run contract.

and no numerical ΔH_{total} or total force is emitted. In particular, $\Delta \tau_{\text{SEAM}} = 0$ is never assumed from silence or from an entropy plateau.

The entropy/thermal contribution remains directly defined and executable:

$$\Delta H_{\text{th},XY}(N_r) = \int_{S_\infty}^{S^*(N_r)} T dS,$$

and at constant T ,

$$\Delta H_{\text{th},XY}(N_r) = T [S^*(N_r) - S_\infty].$$

Because physical separation is $R = L_{XY} N_r$, the force is generated directly by the resolved-state energy curve:

$$F_{XY}(N_r) = -\frac{1}{L_{XY}} \frac{d\Delta H_{XY}^{(J)}(N_r)}{dN_r}.$$

This is the current native interaction path:

$$\begin{aligned} C_{XY}^*(N_r) &\rightarrow \{u_{i,n}^*(N_r)\} \\ &\rightarrow F_C(\xi; N_r) \rightarrow q_C(\xi; N_r) \\ &\rightarrow S^*(N_r) \rightarrow E_H[C_{XY}^*(N_r)] \\ &\rightarrow \Delta H_{XY}^{(J)}(N_r) \rightarrow F_{XY}(N_r). \end{aligned}$$

Channel-energy decomposition, inserted phase/orientation terms, or a separately fitted pair-energy coefficient are not required native operators in this **v18.6 molecular path**. The separately declared attraction Hamiltonian is not imported into the v18.6 optimizer and the v18.6 `TAU_PAIR_MAPPING_UNBOUND` terminal is not exported into the attraction/long-range attraction branch. Conventional electric, magnetic, or force-channel descriptions may be used only as downstream projections where separately declared.

Chapter 19

Engine constraint requirements

Requirement	Operational requirement
ARCV construction	Construct the finite projection representation carrying structure, relations, bounds, and intent.
Atomic constructor	Use $Z(n)$, c_shell , c_n , and fill/skip conservation where atomic shell state is required.
Configuration constructor	Build $C=(\{S_i\},\mathcal{R})$, with $\mathcal{R}=\{r_{ij},chi_{ijkl}\}$, for configuration problems.
Entropy selection	Apply the same S

$$C$$

selection rule over the admissible candidate family. | | Direct interaction | Compare Q-ARC with candidate/reference ARCV structures through engine-defined interaction. | | Complete projection use | Evaluate every projection required by the active contract. | | Full-function evaluation | Invoke all applicable functions before a full-engine refusal. | | Determinism | Use canonical tie-breaking and stable serialization. | | Metrology | Resolve physical time through Cs-133 count metrology. | | Evidence boundary | Do not reveal held-out comparator data until native output is frozen. | | Transcript | Emit sufficient intermediate state to reproduce the result. |

The engine mathematics is fixed. Runners select input files, questions, held-out sets, and requested projections. Runners must not change engine equations or insert target-specific tuning.

Part III

Continuum, Manifold, and Reference Formation

Chapter 20

Continuum retention and Manifold normalization

The Continuum is persistent retained structural reference. The Manifold is a normalized comparison space derived from that retained state. A configuration selected by native SEAM mathematics can be retained directly or through an explicitly declared projection.

validated C*/projection -> Continuum K_t -> normalization -> Manifold M_t

The shell/configuration mathematics preserves this distinction and makes the retained atomic object explicit: the reference carries a shell state, relational state, evidence record, and permitted projections rather than an opaque label.

Property	Continuum	Manifold
role	persistent retained reference	normalized query/comparison space
update	validation-controlled	derived from retained state
canonical equation		derived from retained canonical state
query specificity	low / persistent	high / projection dependent

Chapter 21

Persistent representation and runtime projection

A persistent representation stores enough information to reproduce the supported projection without reconstructing the answer from prose. Runtime projection is a deterministic operation over retained structure.

K_t(object) -> Phi_f -> v_f

21.1 Schema/contract binding

schema(reference) compatible with schema(engine)

If not compatible, the runtime must transform, quarantine, or refuse the artifact under a logged migration rule.

Chapter 22

Reference genesis

Reference genesis handles the first validated instance of a structure or question class. The declared source is either a native SEAM derivation or an admissible empirical observation, and the role of every field remains explicit.

native structural result + evidence record -> validation -> retained reference

An external empirical measurement can seed an evidence/reference record, but its conventional explanatory formula is not thereby admitted as a native generator.

22.1 Atomic reference genesis

$Z(n) \rightarrow \mathcal{S}_i \rightarrow C \rightarrow C^* \rightarrow$ validated atomic reference

Once admitted, this reference supports reverse queries through the Manifold without requiring the runner to reconstruct shell mathematics from a label.

Chapter 23

Manifold completion

Manifold completion is measured over the defined representation and admitted evidence set, not by a claim that every possible physical phenomenon has been solved. The active artifact declares coverage, schema, hashes, canonical equation, and any unavailable evidence fields.

All operative mathematical definitions are contained in this specification. An unavailable evidence artifact is reported as EVIDENCE_UNAVAILABLE; it is never reclassified as a missing canonical equation.

Status	Meaning
CANONICAL_IN_DOCUMENT	definition or deterministic generating equation is stated in this specification
AUTHORIZED_EXTERNAL_INPUT	external numerical observation, evidence, or comparator field explicitly admitted by the active contract
ABSENT_FROM_CANONICAL	required mathematical object is neither stated nor generated in this specification; the contract terminal state fires

Evidence availability and mathematical-definition status are separate fields.

Part IV

Ingest, Coalescence, and Composition Accounting

Chapter 24

Primary ingest function

Ingest receives a source record and creates a normalized evidence/reference contribution with evidence record. The ingest process does not normalize away empirical content, and it does not convert conventional interpretation into native SEAM mathematics.

source -> parse -> evidence record -> normalize fields -> structural/evidence mapping -> validation -> disposition

Raw observables such as counts, positions, spectra, wavelengths, timestamps, uncertainties, and source conditions are retained as evidence fields. Any SEAM-derived shell, configuration, entropy, or projection value is produced by the engine or a documented native operator.

24.1 Time-bearing evidence

If a source time is used as a physical model input rather than external metadata, it is converted or resolved through the declared Cs-133 metrology path before entering native rate calculations.

Chapter 25

Contribution dispositions

Disposition	Meaning
RETAIN	admissible contribution participates in active reference construction
QUARANTINE	but excluded until a named issue is resolved
REJECT	fails source/evidence record/admissibility requirements
REPLACE_ACTIVE_RECORD	replaced by a source while model composition is retained
COMPARATOR_ONLY	used only after native result for evidence comparison

The disposition is attached to the contribution record; silent data loss and silent promotion are both prohibited.

Chapter 26

Composition ledger invariant

$$N_{in} = N_{retain} + N_{quarantine} + N_{reject} + N_{replace} + N_{comparator}$$

The composition ledger guarantees that every ingested contribution has an accounted disposition. If one raw record expands into several normalized records, the expansion relationship is recorded so the ledger remains auditable.

The invariant is documentary and computational: the same source count must be recoverable from the transaction log, index state, and retained/quarantined/rejected collections.

Chapter 27

Ingest transaction

A complete ingest transaction binds source identity, parse result, normalized record, evidence record, validation result, disposition, and indexes.

1. Hash or otherwise identify the source artifact.

2. Parse without changing empirical meaning.
3. Separate raw observation from source-provided interpretation.
4. Normalize only through documented deterministic transforms.
5. Map eligible structural fields to SEAM representations.
6. Run source and structural validation.
7. Assign an explicit disposition.
8. Commit the normalized record, ledger entry, and indexes atomically.

Chapter 28

Atomicity, idempotency, and replay

Ingest_v(source_hash) -> deterministic transaction result

The same source under the same ingest contract identity must not create uncontrolled duplicate active records. When the ingest contract changes, the contract identity changes and the earlier transaction remains traceable as an evidence transaction.

Replay is a reproducibility feature: a reviewer can re-run the transaction and determine whether the same source identity, ingest contract identity, and declared transforms reproduce the same normalized record and disposition.

Chapter 29

Ledger integrity and indexes

Indexes exist to locate retained structures, source records, transaction states, and projection/reference keys. They must not become a second untracked source of mathematical truth.

Index	Minimum key
source index	source hash / source ID
entity/reference index	stable structural/reference identifier
transaction index	ingest contract identity + transaction ID
projection index	reference ID + projection/operator ID
status index	disposition / validation state

Index repair is derived from retained authoritative records; index contents must not override them.

Chapter 30

Completion measurement over ingest

Four completion questions are kept separate:

Question	Measure
Were all source records processed?	ingest completion
Are required empirical fields present?	evidence coverage
Are native operators executable?	mathematical/runtime closure
Are required structural branches retained?	Continuum/Manifold coverage

A complete ingest does not imply a complete theory, and an incomplete external dataset does not imply an undefined native operator.

Part V

Production Execution and Artifact Contracts

Chapter 31

Manifold artifact gate

The active Manifold is a frozen functional artifact for a production run. Its identity, schema, and hash are recorded before execution. The runner must not decode the Manifold outside the engine and substitute external comparison logic for the engine call.

question -> engine(active manifold) -> engine output

A missing, corrupt, incompatible, or unverified Manifold triggers an artifact gate failure rather than a reconstructed substitute.

Chapter 32

Required validation interface

The validation interface must distinguish mathematical execution from external evidence agreement.

Gate	Question
structural	were primitives and configuration constructed under the declared rules?
admissibility	did the candidate satisfy count, capacity, relation, and query constraints?
reference	was the active reference frozen and valid?
metrology	were dimensional time-dependent quantities bound to the declared primitive?
evidence record	can every non-derived input be traced to source?
blindness	was comparator evidence withheld until native result freeze where required?
falsification	is a mismatch retained rather than tuned away?

Validation yields one declared terminal state: PASS, FAIL, INDETERMINATE, SYMBOLIC_ONLY, or another state explicitly named by the active contract. These states must not be collapsed into one generic 'did not work' label.

32.1 Run-contract completeness gate

Before execution, the validator checks the named run contract against this specification. Required fields are: contract ID; objective; this specification/hash; exact inputs; exact equations; permitted perturbation; prohibited inputs; coordinate/unit contract; coefficient values or extraction rules; domain/search bounds; numerical method; tolerances; terminal-state rules; required artifacts; comparator location; comparator access event; and disposition rule.

If any required field is absent, execution does not start. Terminal state: INDETERMINATE: CONTRACT_INCOMPLETE.

No software default satisfies a missing field. No executor interpretation satisfies a missing field. No source outside the declared contract is consulted during native execution.

32.2 Terminal-state finality gate

A terminal state is an execution boundary, not a request for repair. When any terminal condition named by the active contract is satisfied, mathematical and numerical execution under that contract ends immediately.

The only authorized post-terminal operations are artifact completion from values already produced, transcript finalization, hashing, sealing, and stop. Post-terminal artifact completion does not evaluate a new equation, call a new numerical method, change a parameter, or extend the search domain.

The executor has no repair authority. It does not change quadrature, node count, tolerance, optimizer, coordinate system, domain partition, smoothing rule, boundary treatment, initialization, coefficient, field equation, or any other declared method after a terminal condition occurs.

The executor does not ask the operator to choose an undeclared repair while the run is active or after it has terminated. A question such as “should I switch methods?” has no branch in the active contract and therefore has no execution status.

A proposed repair is recorded only as `FUTURE_CONTRACT_CANDIDATE` metadata after the run is sealed. It is not executed, tested, substituted, or treated as a continuation of the terminated run.

Controlling sequence:

```
TERMINAL_STATE -> COMPLETE_DECLARED_ARTIFACTS -> HASH -> SEAL ->  
RETURN_TO_WORKFLOW_SUPERVISOR
```

Any computation performed after `TERMINAL_STATE` other than the declared artifact-completion operations is `INVALID_RUN: POST_TERMINAL_EXECUTION`.

32.3 Run-terminal finality versus workflow successor dispatch

A terminal state closes the named run. It does not by itself close the root objective. After the run is sealed, the workflow supervisor evaluates root-objective closure and declared dependency state before deciding whether workflow execution ends.

```
RUN_A -> TERMINAL_STATE -> COMPLETE_ARTIFACTS -> HASH -> SEAL ->  
DECLARED_SUCCESSOR_EDGE -> RUN_B
```

A successor edge is executable only when all of the following fields are declared before `RUN_A` starts: predecessor contract ID; exact triggering terminal state; successor contract ID; successor contract identity/hash; carried-forward frozen objects; changed object; prohibited objects; comparator state; and workflow disposition after successor completion.

The workflow supervisor performs successor dispatch mechanically after the predecessor seal. The predecessor executor does not repair the run. `RUN_B` receives a new UID, new transcript, new hashes, and its own terminal state.

If no explicit contract-specific successor edge matches the sealed terminal state, the supervisor next evaluates the global root-objective dependency rule in §32.7. `WORKFLOW_STOP` is authorized only after the root objective is complete or declared dependency exhaustion has itself reached a root terminal.

```
NO_CONTRACT_SPECIFIC_EDGE -> ROOT_OBJECTIVE_CHECK ->  
DEPENDENCY_DISPATCH_OR_WORKFLOW_STOP
```

32.4 Sealed-result verification versus open-workflow interpretation

A sealed terminal artifact is immediately eligible for verification. Verification includes independent reproduction, arithmetic or numerical recomputation, contract-compliance audit, internal-consistency audit, artifact/hash verification, and scope-accurate reporting of the terminal result.

A sealed link inside a larger unfinished workflow retains its own scoped meaning. The open downstream portion of the workflow has no result and receives no scientific characterization.

During an unfinished workflow, the executor or reviewer reports only execution state: active contract ID, completed operation, sealed terminal code, artifact state, and automatic successor dispatch already declared by the frozen graph. It does not infer downstream outcomes, characterize unresolved layers, recommend model changes, or propose a continuation.

A continuation exists only through a frozen contract-specific successor edge or the globally predeclared dependency-exhaustion edge in §32.7. The dependency edge does not invent a new mechanism; it recursively resolves only the named required objects already emitted by the sealed child contract.

Controlling interpretation rule:

SEALED_RESULT -> VERIFY_AND_REPORT_WITHIN_SCOPE

OPEN_WORKFLOW -> REPORT_EXECUTION_STATE_ONLY

ROOT_OBJECTIVE_COMPLETE + NO_DECLARED_SUCCESSOR -> STOP_WITHOUT_PROPOSAL

32.5 Canonical-model composition resolution before external ingress

Before any object is classified as external ingress, the executor checks this specification. Every object consumed during execution is classified only by its stated role here, not by external naming, provenance, or reviewer interpretation.

Every object consumed during execution receives exactly one authority class: CANONICAL_IN_DOCUMENT; AUTHORIZED_EXTERNAL_INPUT; or EXTERNAL_NONAUTHORITATIVE.

CANONICAL_IN_DOCUMENT objects execute directly from this specification.

AUTHORIZED_EXTERNAL_INPUT objects can supply only the observation, comparator, or evidence fields named by the active run contract. EXTERNAL_NONAUTHORITATIVE objects have zero execution authority.

CANONICAL_AUTHORITY_CHECK -> EXTERNAL_INGRESS_CLASSIFICATION

CANONICAL_IN_DOCUMENT -> DIRECT_EXECUTION_AUTHORITY

EXTERNAL_PROVENANCE != EXECUTION_AUTHORITY

32.6 External-ingress firewall and guardrail precedence

External observation, comparator, and evidence material is readable and auditable but has zero operator-definition authority. It enters execution only when the active contract names the exact field, role, deterministic transform, scope, and reveal state.

The executable state is constructed only from the active canonical baseline, the explicit delta or override named by the immutable run contract, and predecessor fields explicitly canonical through a declared successor edge.

EXECUTABLE_STATE = CANONICAL_BASELINE $\oplus$ DECLARED_CONTRACT_DELTA $\oplus$
DECLARED_PREDECESSOR_FIELDS

The operator $\oplus$ is defined only for fields named by the active contract. It does not authorize semantic matching, symbol matching, external substitution, dimensional matching, or reviewer inference.

An external input enters execution only when the contract declares all of the following: input identity/hash; exact consumed field; target contract field; deterministic mapping or transform; role; override status; permitted scope; anti-circularity status; and validation condition. Missing any field rejects the ingress.

UNDECLARED_EXTERNAL_OBJECT -> EXTERNAL_INGRESS_REJECTED

CONSUMED_UNDECLARED_EXTERNAL_OBJECT -> INVALID_RUN:
UNDECLARED_EXTERNAL_INGRESS

Guardrail precedence is strict: the run contract and its declared canonical baseline control active execution; the declared successor graph controls continuation; explicitly canonical sealed fields are read-only inputs; named reference artifacts supply only their declared fields; external/evidence record material is read-only; comparator/evidence material is inaccessible until its reveal event. A lower-precedence layer never fills, overrides, or repairs a higher-precedence omission.

A guardrail is restrictive, not generative. A requirement such as “metrology required,” “Hamiltonian binding required,” “reference required,” or “complete state required” does not authorize an external or conventional substitute. If the required canonical object is absent, the declared terminal state fires.

32.7 Root-objective exhaustion and mandatory dependency dispatch

Every workflow that seeks closure of a named scientific or metrological objective has a ROOT_OBJECTIVE_ID, ROOT_SUCCESS_CRITERION, ROOT_REQUIRED_OBJECTS, and ROOT_TERMINAL_STATES. A terminal code emitted by a child contract is not automatically a root terminal.

A sealed child result that explicitly names unresolved required objects is a DEPENDENCY_TERMINAL. It records valid information about that child run, but it cannot terminate the workflow while the root objective still requires those objects.

CHILD_TERMINAL + UNRESOLVED_REQUIRED_OBJECTS != ROOT_TERMINAL

The following states are dependency-discovery states whenever they carry a nonempty unresolved-object ledger required by the root objective: SYMBOLIC_ONLY;

SYMBOLIC_CLOSED_NUMERIC_FACTORS_UNRESOLVED; UNRESOLVED_REQUIRED_OBJECT;
NUMERIC_BINDING_ABSENT; and any contract-specific state explicitly marked DEPENDENCY_TERMINAL.

After the child seal, the supervisor dispatches RUN-SEAM-DECLARED-DEPENDENCY-EXHAUST-01 or a narrower predeclared specialization. This global edge is part of the frozen canonical workflow and therefore does not require operator permission or a newly authored successor edge at runtime.

DEPENDENCY_TERMINAL -> COMPLETE_ARTIFACTS -> HASH -> SEAL -> DEPENDENCY_QUEUE ->
DECLARED_DEPENDENCY_EXHAUSTION

Dependency exhaustion is strictly non-generative. It follows only deterministic definitions, identities, constants, and mappings stated in this specification and admitted numerical run inputs. It cannot introduce a new physical mechanism, comparator, fitted value, software default, external substitute, or reviewer proposal.

Each unresolved object is expanded recursively through its canonical definition in this specification. Newly exposed dependencies are appended to the queue. The queue is processed until every required object is numerically resolved, identified as an authorized run input that is genuinely absent, classified as an undefined required native mapping/operator, rejected by a canonical-definition conflict, or shown to participate in a dependency cycle.

The executor does not stop merely because one child run emitted SYMBOLIC_ONLY or SYMBOLIC_CLOSED. It does not ask the operator whether to continue. Those states cause automatic dependency dispatch when the unresolved object belongs to ROOT_REQUIRED_OBJECTS.

A root workflow stops only on one of the following: ROOT_RESOLVED; ROOT_REJECTED under an explicit falsification predicate; ROOT_SYMBOLIC_ONLY after declared dependency exhaustion proves that one or more explicitly authorized numerical run inputs are absent while the symbolic map is complete; ROOT_INDETERMINATE after exhaustive detection of an undefined required native primitive/operator, canonical-definition conflict, or dependency cycle; or INVALID_ROOT_WORKFLOW.

SEALED + ROOT_OBJECTIVE_COMPLETE + NO_SUCCESSOR_EDGE -> WORKFLOW_STOP

SEALED + ROOT_OBJECTIVE_OPEN + UNRESOLVED_REQUIRED_OBJECTS ->
MANDATORY_DEPENDENCY_DISPATCH

Chapter 33

Production transcript

The transcript is part of the result. It shows what the engine did, not merely what the final answer was.

question -> query/reference -> primitive -> $\mathcal{S}_i$ -> C candidates -> S

C

-> C* -> projection -> interaction -> native result -> hash -> comparator -> residual

For questions that do not require atomic reconstruction, the transcript begins from the relevant retained structural primitive and records why that entry point is admissible.

Transcript field	Required content
run identity	UID, date/runtime metadata, implementation and orchestration identities
input	exact question and primitive/reference input
candidate generation	candidate count and constraints
selection	entropy/coherence/discrepancy values as applicable
terminal	native result before comparator
evidence reveal	time/order of comparator access
comparison	residual/disposition

33.1 Contract ledger

The transcript begins by reproducing the complete run contract exactly as executed. The transcript then records each consumed input and identifies the contract field that authorized it.

An input with no contract authorization is a run-invalidating error: `INVALID_RUN: UNDECLARED_INPUT_CONSUMED`. A declared input that was not consumed remains visible in the transcript.

33.2 Terminal transcript and sealing record

When a terminal condition occurs, the transcript records the exact condition, the exact values that triggered it, the last authorized equation or numerical operation, and the execution stop point.

The transcript then records artifact serialization, hashes, and seal identity. It records zero post-terminal model evaluations.

If the executor identifies a possible remedy, the transcript labels it `FUTURE_CONTRACT_CANDIDATE` and records it after the sealed native result. That label is descriptive metadata only. It supplies no permission to execute the remedy.

A cold-run package is incomplete until the native terminal record, required artifacts available at termination, transcript, and hashes are sealed. Explanation without sealing is not contract completion.

Every branch decision records the equation or explicit contract rule that selected the branch. “Reasonable,” “standard,” “conventional,” “default,” and equivalent unstated justifications are invalid branch sources.

33.3 Successor-dispatch transcript

When a sealed run triggers a predeclared successor, the workflow transcript records: predecessor UID; predecessor seal hash; triggering terminal code; successor-edge ID; successor contract ID/hash; dispatch timestamp/order; comparator-access state; and successor UID.

The successor transcript starts from its own immutable contract. It does not append new model evaluations to the terminated predecessor transcript.

A workflow-level conclusion is emitted only after the declared successor graph reaches a node with no matching successor edge or reaches an explicit workflow terminal disposition.

33.4 Dependency-exhaustion transcript

When §32.7 dispatches dependency exhaustion, the workflow transcript records the `ROOT_OBJECTIVE_ID`, child contract/seal, triggering dependency-terminal code, exact `UNRESOLVED_REQUIRED_OBJECTS` list, dependency-queue order, every definition used to expand an object, canonical-status authority for that definition, and the final disposition of each dependency.

The transcript distinguishes `DISCOVERED_DEPENDENCY` from `NEW_ASSUMPTION`. A dependency produced algebraically by an admitted definition is authorized. An object introduced because the executor believes it would be useful is unauthorized.

Every dependency factor receives an explicit terminal disposition in the ledger. The current terminal classes distinguish resolved values, authorized run-input boundaries, definition conflicts, dependency cycles, and other contract-declared outcomes. No dependency is silently dropped.

Chapter 34

Boundary trace

Every value entering a production calculation is assigned a source class:

Class	Definition
Primitive	declared first-principles or metrological starting quantity
Derived	computed from primitives and derived state
Reference	retrieved from validated Continuum/Manifold structure
Evidence	raw external observation
Comparator	external formula/value used after native output
Authorized external input	observation/evidence/comparator field admitted explicitly by the active contract
Implementation	software tolerance, iteration cap, serialization, runtime-only setting

The boundary trace is what makes anti-circularity auditable. If a target evidence value appears upstream of the native result, the blind test is invalid.

Part VI

Mathematical and Documentary Binding

Chapter 35

Formalization-to-engine binding

Every equation presented as executable SEAM mathematics must bind to an implementation or be marked as a formal specification awaiting executable binding. Conversely, every runtime operator that affects native output must identify the mathematical rule it implements.

canonical equation <-> operator implementation <-> test <-> artifact

Beginning with canonical, the entropy functional and shell-overlap operator are documentary-bound in §4.15. Remaining implementation obligations are runtime binding of the active structural field, frozen spatial partition, coefficient record, and test transcript. State-dependent Hamiltonian coefficients and relation kernels remain separate downstream obligations.

Chapter 36

Documentation-engine boundary

The document defines semantics, mathematics, contracts, canonical equation, and audit boundaries. The engine defines what was actually executed in a specific run. Neither is permitted to impersonate the other.

Statement	Proper status
equation in document, no implementation	formal/documentary
function in active source, tested	runtime-bound
required mathematical object absent from this specification	declared contract terminal state
worked comparator equation	comparator-only unless independently derived

A reviewer can state that a supplied artifact is insufficient to verify an empirical or implementation claim. That statement does not alter any mathematical definition stated in this specification.

Chapter 37

Valid critique boundary

A valid critique names the boundary it has actually tested.

Finding	Meaning
DOCUMENTARY_BINDING_ABSENT	supplied document does not bind a claim to source or transcript
ARTIFACT_NOT_SUPPLIED	required executable/data artifact is absent from the supplied package
RUNTIME_FAILURE	supplied executable ran correctly and failed at a defined boundary
ENGINE_CAPABILITY_ABSENT	inspection/execution establishes that capability is absent from that artifact
PRODUCTION_REFUSAL	full engine/Manifold execution completed and no admissible closure was found

A valid critique separates canonical-definition status, runtime status, and empirical-evidence status. An absent evidence artifact is not an absent canonical equation, and an absent canonical equation is not repaired from outside material.

A downstream disagreement must also be classified correctly. Because SEAM operates upstream of conventional observable equations, a reviewer must distinguish (a) a different SEAM-native state that conventional physics subsequently describes from (b) a genuinely incompatible observable prediction. Only the latter is a direct physical contradiction. This distinction does not excuse disagreement: the SEAM prediction must be frozen before projection and remains falsified if its downstream observable conflicts with evidence.

37.1 Contract-specific critique rule

Every execution critique names the exact run contract and evaluates only what that contract declares. A critique does not rescue a failed run by adding an unstated mechanism, and it does not invalidate a run by requiring an undeclared downstream conventional primitive.

A claim that “the model instead uses X” has no execution status until X is written as an explicit new contract. A claim that “the model must have assumed X” is false unless X appears in the canonical equations, run contract, or transcript.

37.2 Executor-compliance boundary

A correct diagnosis does not cure a contract violation. An executor can correctly identify why a run terminated and still violate the run by continuing with an undeclared repair.

Critique therefore separates `RESULT_VALIDITY` from `EXECUTOR_COMPLIANCE`. The terminal result is evaluated under the frozen contract. Every action after the terminal event is separately checked against the terminal-state finality gate.

A proposed superior numerical method, physical model, or mathematical representation is not evidence against the sealed terminal result and is not an authorized continuation. It is a candidate object for a different contract.

37.3 Numerical-evaluator failure boundary

`INDETERMINATE: QUADRATURE_NOT_CONVERGED` establishes failure of the declared numerical evaluator to satisfy its convergence gate. It does not establish rejection of the physical profile and does not establish project-level mathematical indeterminacy.

When the frozen workflow contains a successor evaluator for that terminal code, the current run is sealed at its declared terminal and the successor executes as a separately declared adjudication. The predecessor result is complete at its own contract scope.

`NUMERICAL_EVALUATOR_INDETERMINATE + DECLARED_SUCCESSOR -> SEAL_CURRENT_RUN -> EXECUTE_SUCCESSOR`

A successor numerical evaluator changes only the declared evaluation operator. It does not change the physical field profile, entropy law, coefficient values, support, search domain, comparator boundary, or terminal-test semantics unless the successor contract explicitly lists a separate changed object.

37.4 Cold-reviewer non-ingress rule

A reviewer tests sealed artifacts without becoming an input source to the run. Verification comments, alternative models, conventional correspondences, and proposed next steps remain outside executable state.

A sealed terminal result is reproduced and interpreted within its declared scope. Any declared downstream successor is a separate run contract and receives no inferred result from the reviewer.

During successor-workflow review, statements about what the process “is heading toward,” what parameter “should be added,” or what contract “should come next” have no execution status. The declared successor graph alone determines continuation.

External material used for an evidence record cannot cross the critique boundary into active operator definition by being described as an assumption, analogy, likely mapping, or obvious equivalent. Ingress requires the complete declaration defined in §32.6.

Chapter 38

Resolver boundary

The resolver receives the primitive output of the structural calculation. It formats, explains, or translates that result while preserving support and terminal state.

$$P_X \rightarrow \text{Resolver} \rightarrow A_X$$

The resolver must not decode an opaque artifact outside the engine, manufacture a candidate, substitute a conventional answer, or reinterpret an indeterminate result as a resolved one.

For one-question-per-run transcripts, the resolver output is linked to the same UID and engine result that produced the primitive.

Chapter 39

Complete closure operator

The canonical composed operator remains:

$$\text{Closure_SEAM}(X) = \text{VC}(\text{MC}(\text{CC}(\text{TC}(\text{SC}(\mathfrak{R}(X))))))$$

Operator	Canonical role
$\mathfrak{R}$	representation constructor; for atomic reconstruction begins from $Z(n)$, creates $\mathfrak{S}_i$ and admissible C candidates
SC	configuration/structural closure: applies $C^* = \arg \max S$

$$C$$

over admissible physical configurations where required | | TC | transfer/trajectory stage; is represented by when declared as $\hat{H}_{\text{SEAM}}$

$$C^*$$

when operator notation clarifies the selected-state interaction/evolution | | CC | internal structural/continuity coherence Coh_struct, computed before Manifold comparison | | MC | Manifold comparison; computes Coh_M for the retained projection and $\Delta_M = 1 - \text{Coh_M}$ | | VC | validation closure over residual, contradiction, support,

model composition, metrology, and admissibility gates |

$X \rightarrow O(X) \rightarrow \text{structural primitive} \rightarrow C \rightarrow C^* \rightarrow \Phi(C^*) \rightarrow \text{configuration closure} \rightarrow \text{transfer/operator stage} \rightarrow$
 $\text{Coh_struct} \rightarrow \text{Manifold interaction} \rightarrow \text{Coh_M} \rightarrow \Delta_M \rightarrow \text{validation} \rightarrow P_X \rightarrow A_X \text{ or refusal}$

The closure operator is retained exactly as defined here. Its structural content enters through $\mathfrak{R}$ and SC, and time/metrology binding is explicit wherever a transfer or observable is time-dependent.

Chapter 40

Canonical definition

SEAM is the operational layer of a structure-first mathematical framework. It represents an input as observable relations or a declared structural primitive; for atomic construction it begins from $Z(n)$, distributes the count into the fixed seven-shell architecture, constructs entity shell states and complete relational configurations, selects the admissible configuration under the entropy extremum, projects the retained structure into coherent facets, compares those projections against retained Continuum state through a normalized Manifold, calculates closure/coherence/residual under the active engine contract, and permits a readable answer only after support, continuity, evidence record, metrology, and falsification gates succeed.

$Z(n) \rightarrow \mathcal{S}_i \rightarrow C \rightarrow C^* \rightarrow$

optional $\hat{H}_S EAM \setminus [C \setminus ^*$

operator representation] $\rightarrow \Phi(C^*) \rightarrow \text{Coh_struct} \rightarrow \text{Coh_M}/\Delta_M \rightarrow \text{validation} \rightarrow P_X \rightarrow A_X$

No label, template, candidate injection, documentary omission, external interpretation, or held-out answer substitutes for that execution. Physical time is resolved through Cs-133 count metrology wherever the native calculation requires a time quantity.

$T(X) = n_Cs(X)/9,192,631,770$

Appendix A — Canonical Authority Boundary

This specification is self-contained and canonical as written. Only mathematics, constants, definitions, branch rules, run contracts, and terminal rules stated here have execution authority.

Outside material, project memory, reviewer knowledge, and software defaults cannot define or repair a mathematical object. External observations, comparators, and evidence enter only through explicitly declared contract interfaces.

A mathematical object absent from this specification is absent for execution. The executor reports the applicable terminal state rather than constructing a replacement from outside material.

A.1 Canonical-object authority

This specification is a complete canonical object. Only mathematics, definitions, constants, branch rules, run contracts, and terminal rules stated in this document have execution authority. A reader does not need development provenance to execute it. If an equation is stated here, it is canonical for this specification; if it is absent, it is absent for execution.

PRESENT_HERE -> CANONICAL_FOR_THIS_SPECIFICATION; NOT_PRESENT_HERE ->
NO_EXECUTION_AUTHORITY

Appendix B — Canonical Mathematical Registry

Object	Canonical form	Role
Atomic baseline	$Z(n)$	baseline number of electrons to distribute
Shell capacities	$c_{\text{shell}}=$	

$$2, 8, 18, 32, 18, 32, 18$$

$$|$$

$$| \text{ fixed seven-shell capacity vector}$$

| Shell fraction | $\rho_{i,n}=N_{i,n}/c_n$ | normalized occupancy | | Entity state | $\mathcal{S}_i=\{\rho_{i,1},\dots,\rho_{i,7}\}$ | seven-shell structural state | | Pair distance | $r_{ij}=|r_j-r_i|$ | rigid-motion invariant pair relation | | Parity | $\chi_{ijkl}=\text{sgn}$

$$(r_j - r_i) \cdot ((r_k - r_i) \times (r_l - r_i))$$

$$| \text{ reflection/chirality relation}$$

$$|$$

| Configuration | $C=(\{\mathcal{S}_i\},\mathcal{R}), \mathcal{R}=\{r_{ij},\chi_{ijkl}\}$ | complete configuration representation | | Selection | $C^*=\arg \max_{\{C \text{ in } \mathcal{A}\}} S$

$$C$$

$$|$$

$$| \text{ entropy-selected admissible configuration}$$

| Time | $T(X)=n_{\text{Cs}}(X)/9,192,631,770$ | Cs-133 count-based physical interval | | Manifold residual | $\Delta_M=1-\text{Coh}_M$ | normalized Manifold residual for the scalar-coherence instantiation | | Engine closure | Closure_SEAM=VC(MC(CC(TC(SC($\mathcal{R}(X)$)))))) | composed operational closure; CC yields Coh_struct and MC yields Coh_M/ Δ_M | | Configuration entropy | $S_{\text{config}}=k_B \sum_i \sum_n \ln C(c_n, N_i, n)$ | combinatorial shell-state entropy using integer occupancies | | Field entropy | $S_{\text{field}}=-k_B \sum_{\alpha} p_{\alpha} \ln p_{\alpha}$ | Shannon entropy of frozen normalized structural-field distribution | | Shell overlap | $O_{ij}^{\{nm\}}=\sum_{\alpha} \sqrt{p_{i,n,\alpha} p_{j,m,\alpha}}$ | symmetric normalized inter-shell spatial overlap in

0, 1

|

| Coupling entropy | $S_{\text{coupling}}=k_B \sum_{\{i<j,n,m\}} \rho_{i,n} \rho_{j,m} O_{ij}^{\{nm\}}$ | pair-counted structural coupling contribution | | Entropy functional | $S=S_{\text{config}}+\lambda_{\text{field}} S_{\text{field}}+\lambda_{\text{coupling}} S_{\text{coupling}}$ | canonical configuration-ordering functional |

B.1 Same rule across scale

$S_{\text{rule atomic}} = S_{\text{rule molecular}} = S_{\text{rule amalgamum}}$

The equality means one governing entropy functional rule is applied to different configurations. Numerical entropy values and forces are not asserted to be equal.

B.1.1 Single entropy-governed interaction law across scale

The interaction that binds atomic structure, atoms into molecules, and molecules into molecular amalgamums is one and the same entropy-governed interaction law. Scale changes the admissible configuration and the realized interaction geometry; it does not authorize a new fundamental force law.

$F_{\text{int}}^{\text{(atomic)}} \equiv F_{\text{int}}^{\text{(molecular)}} \equiv F_{\text{int}}^{\text{(amalgamum)}}$

The identity denotes common governing law/operator. It does not assert equal numerical force, equal entropy value, equal range, or equal geometry across configurations.

B.2 Full versus partial closure

Full entropy closure corresponds to a highly stable admissible entity/configuration. Partial closure retains structural differential participating in bonding, reconfiguration, reactivity, or volatility when the active structural contract contains the required interaction terms.

B.3 Canonical spatial-field realization

$\Omega_{i,n} = \{\xi: n-1 \leq ||\xi-X_i|| < n\}; V_n=(4\pi/3)$

$$n^3 - (n - 1)^3$$

$u_{i,n} = 1_{\{\Omega_{i,n}\}}/\sqrt{V_n}; F_C(\xi)=\sum_{i,n} \sqrt{V_n} u_{i,n}(\xi)$

$q_C=|F_C|^2/\int |F_C|^2; S_{field}=-k_B \int q_C \ln q_C d^3\xi$

$O_{ij}^{\{nm\}}=Vol(\Omega_{i,n} \cap \Omega_{j,m})/\sqrt{V_nV_m}$

Status: canonical mathematical definition. The field/overlap constructor is completely defined by the equations above. The uniform-shell assumption is frozen for each named run and is rejected when executed native evaluation under that run contract fails the declared closure requirement.

Appendix C — Operational Interface and Execution Contract

This appendix defines the operational interface implied by the foundations. It does not include engine or runner source code. The implementation layer orchestrates only the inputs and questions declared by the run contract; the mathematical order and operator meanings are fixed by the preceding foundations.

- One question per run.
- The orchestration layer passes the question and active Manifold/reference artifact to the operational layer without modifying the mathematical content.
- The orchestration layer does not preselect the winning operator from a keyword or expected answer.
- All applicable functions are evaluated before a universal production refusal.
- Implementation and reference identities are recorded sufficiently to reproduce the execution context.
- Physical time dependencies expose the Cs-133 metrology baseline/contract identity.
- Native result is frozen before held-out comparator reveal.
- Transcript includes intermediate structural and arbitration states.

C.1 Required artifact set

Artifact	Purpose
question record	exact input and UID
implementation identity	hash or equivalent identity of the implementation used; implementation cannot change canonical mathematics

Artifact	Purpose
orchestration record	input selection and execution-context record; orchestration cannot change canonical mathematics
manifold/reference artifact	frozen comparison object
metrology record	time primitive binding if required
candidate/intermediate log	reconstructible structural path
native-result artifact	machine-readable native result
resolver-output artifact	human-readable expression linked to the native result
manifest + hashes	artifact identity

C.2 Operational binding checklist

The canonical layout requires the documentary mathematics to bind to a concrete operational sequence. The checklist below preserves that technical progression while remaining implementation-neutral: it specifies what an implementation must accomplish, not how its source code is written.

1. Construct a structured query from representation and intent.
2. Construct or load a compliant ARCV record without inventing missing fields.
3. Construct the atomic shell state from $Z(n)$ when the question requires fresh atomic resolution.
4. Assemble complete configuration relations r_{ij} and chi_{ijkl} where geometry is part of the candidate state.
5. Evaluate the active entropy functional over the admissible configuration family and retain C^* .
6. Compare the question ARCV with every applicable candidate/reference ARCV.
7. Calculate the complete interaction record, discrepancy, closure, and admissibility values required by the runtime contract.
8. Resolve the strongest admissible structural lock deterministically and record tie-break behavior.
9. Preserve intermediate configurations and projections for trajectory requests.
10. Verify transference accountability between adjacent retained states.
11. Refresh a singular retained reference from an admissible inbound structure without append-only duplication.
12. Calculate observed Manifold-update magnitude and any declared completion estimator.
13. Apply Hamiltonian or conventional correspondence only after native structural resolution unless the active operator is independently bound as canonical.
14. Keep answer formation downstream of the primitive/resolver boundary while orchestration remains non-mathematical.
15. Emit implementation identity, reference, metrology, and output identifiers sufficient to reproduce the execution context.

C.3 Artifact execution boundary

ARCV and Manifold artifacts are functional objects under the operational contract. Documentary inspection can audit their structure, but it is not a substitute for applying the defined operators to them.

VALID: question -> defined operational layer(manifold/reference artifact) -> native output

INVALID AS A SEAM EXECUTION: decode artifact externally -> invent comparison -> report it as native SEAM output

C.4 Deterministic arbitration record

A scoped implementation orders candidate rows only by the admissibility, discrepancy, normalized discrepancy, canonical label, component tuple, and intent fields declared in its operational contract. Deterministic arbitration and explicit tie-breaking are mandatory; source-code structure remains implementation-specific.

C.5 Implementation identity boundary

The mathematical and execution semantics are complete in this specification. A concrete software implementation, when used, is identified by hash or equivalent implementation identity for reproducibility; software structure has no authority to change the equations or operational order stated here.

A cold mathematical execution does not require implementation source. Software is an optional realization of the canonical interface, not a source of mathematical definitions.

Appendix D — Entropy and Configuration Definitions

This appendix restates the canonical entropy and configuration equations used by the run contracts. Every operative object is defined directly here or in §4.15.

D.1 Canonical three-term entropy composition

The governing functional is the explicit three-term configuration entropy below. The configuration, field, and coupling terms are evaluated from the same canonical state C .

S

C

= S_config

C

- $\lambda_{\text{field}} S_{\text{field}}$

$$C$$

- $\lambda_{\text{coupling}} S_{\text{coupling}}$

$$C$$

S_{config}

$$C$$

$$= k_B \sum_i \sum_n \ln \text{binom}(c_n, N_{i,n})$$

$$q_C(\xi) = |F_C(\xi)|^2 / \int |F_C(\xi)|^2 d^3\xi$$

S_{field}

$$C$$

$$= -k_B \int q_C(\xi) \ln q_C(\xi) d^3\xi$$

$$O_{ij}^{\{nm\}}(C) = \text{Vol}(\Omega_{i,n} \cap \Omega_{j,m}) / \sqrt{V_n V_m}$$

S_{coupling}

$$C$$

$$= k_B \sum_{\{i < j\}} \sum_{\{n,m\}} \rho_{i,n} \rho_{j,m} O_{ij}^{\{nm\}}(C)$$

$$C^* = \arg \max_{\{C \in A\}} S$$

$$C$$

λ_{field} and $\lambda_{\text{coupling}}$ are run-bound operator normalizations. A named run uses numerical values only when its contract states and freezes them; the H2 reference contracts state $\lambda_{\text{field}}=1.0$ and $\lambda_{\text{coupling}}=0.1$.

D.2 configuration type

$$C = (\{\mathcal{S}_i\}, \mathcal{R}), \mathcal{R} = \{r_{ij}, \chi_{ijkl}\}$$

The canonical representation is shell/relational. No alternative coordinate system is required at the configuration layer unless a named run contract explicitly adds one as a noncanonical perturbation.

D.3 Canonical field and overlap equations

The field basis $u_{i,n}$, normalized density q_C , and overlap $O_{ij}^{\{nm\}}$ are defined explicitly in §4.15.3-§4.15.7 and are executable without any external mathematical source.

D.4 Canonical field-constructor boundary

The canonical field realization is the uniform normalized basis on each occupied native shell support. Shell overlap is the normalized geometric intersection of those supports. These definitions are mathematical facts of this specification for the named baseline contracts.

An alternative field constructor has no execution authority unless a named run contract states its complete equation, freezes the unchanged baseline objects, and identifies the changed object explicitly.

canonical equation -> frozen operator -> native execution -> seal -> comparator

Coefficient status is explicit: λ_{field} and $\lambda_{\text{coupling}}$ are run-bound operator normalizations. Every numerical run states and freezes their values; no unstated coefficient value is permitted.

D.5 Field-constructor execution status

canonical field concept + canonical field constructor -> executable named run

The field constructor is explicitly defined in §4.15.4. Execution determines whether that constructor satisfies or fails a stated run contract; outside reconstruction is neither required nor permitted.

Appendix E — Metrology Binding

SEAM's physical-time resolver is fixed to the Cs-133 hyperfine transition count:

$$T(X) = n_{\text{Cs}}(X) / 9,192,631,770$$

Consequences:

- Frequency is cycles per Cs-resolved interval, not an independent primitive.
- Rates inherit the time resolver.
- Trajectory derivatives inherit the time resolver when they represent physical evolution.
- Force/energy/Hamiltonian expressions that contain time or rate must expose the same dependency chain.
- Wall-clock execution time is reproducibility metadata and does not enter physical calculations.
- Missing required metrology produces a fail-closed metrology status rather than an invented projection.

E.1 Non-reciprocal resolver rule

Once a quantity is resolved from the Cs-133 primitive, a comparison to a conventional second does not modify the primitive count relation. The comparison is downstream evidence, not a feedback calibration loop.

E.2 Spatial metrology status and dimensionalization gate

The canonical atomic spatial coordinate ξ and every separation formed as a norm in that coordinate system are native dimensionless spatial quantities. A native distance count is already a SEAM distance at the native layer. Dimensional baseline-length projection is the separate operation $L=N_L L_A$, where the baseline is generated upstream by $L_A=N_A \lambda_A$.

The empirical normalized-observable adapter is validation-only. It cannot provide L_A , calibrate L_A , or supply a first-principles spatial scale.

The controlling spatial-metrology workflow is defined directly in Appendix M. Appendix N records the executed hash-locked metrology-closure test and its PASS result. The sealed H2 extrema remain native distance-count results and consume L_A only after the atomic baseline is frozen.

E.2.1 Four-layer separation

Layer 1 — native distance closure: for native spatial coordinates X_i , define $N_D(X_i, X_j) = ||X_j - X_i||$. For H2, $N_D = N_r = ||X_B - X_A||$. No SI length, empirical observable, or external comparator is required.

$$N_D(X_i, X_j) = ||X_j - X_i||; N_{D,H2} = N_r$$

Layer 2 — SEAM baseline-length projection: the atomic metrology layer defines $L_A = N_A \lambda_A$, $\tau_A = M_A / v_A$, and $V_A = L_A / \tau_A$ before molecular construction. The universal map is $L = N_L L_A$ and $L_{H2} = N_r L_A$. No molecular configuration-to-baseline generator is required.

$$L_A = N_A \lambda_A; \tau_A = M_A / v_A; V_A = L_A / \tau_A; L = N_L L_A; L_{H2} = N_r L_A$$

Layer 3 — empirical validation adapter: normalized observational coordinates I_{norm} , W_{norm} , P_{norm} , and Π_{norm} can be transformed into Θ_{obs} for validation studies. This adapter is non-generative and is prohibited from defining L_A or any native state.

$$\Theta_{\text{obs}} =$$

$$\max(I_{\text{norm}} - \Pi_{\text{norm}}, 10^{-10})$$

$$^{0.5} \exp$$

$$-(P_{\text{norm}} + W_{\text{norm}})$$

$$\max(|I_{\text{norm}} - W_{\text{norm}}|, 10^{-10})$$

Layer 4 — conventional-unit reporting: meters, ångströms, Bohr radii, or another external unit are downstream evidence/reporting quantities. They remain sealed until an explicit post-native comparison contract authorizes access.

Appendix F — Test Evidence and Artifact Registry

This appendix records sealed run outcomes generated by contracts in this specification. The mathematical meaning of every result is defined here; evidence artifacts provide reproducibility records but no operator definitions.

F.1 Evidence-direction rule

SEAM first principles -> native SEAM result -> projection onto observable -> external evidence comparison

The reverse direction is not accepted as a native derivation:

SEAM contract -> native state -> downstream conventional observable is allowed; conventional target -> retune SEAM contract is not.

SEAM is evaluated one representational layer upstream of the conventional observable description. A changed native contract produces its own downstream state and observation through the declared projection. The conventional measurement relation itself is not modified by that upstream change. The native contract, constructor, coefficients, and projection map are frozen before comparator access.

F.2 Validation states

State	Meaning
RESOLVED	native calculation produces a supported terminal result
SYMBOLIC_ONLY	all required objects exist symbolically but unresolved functions/constants remain
INDETERMINATE	an upstream primitive/mapping is undefined so no unique result can be constructed
REJECTED	a complete calculation violates a declared mathematical/admissibility condition
ARTIFACT_NOT_SUPPLIED	required source/runtime object is missing from the reviewed package

F.3 canonical numerical-resolution evidence

RUN-H2-RADIAL-PROFILE-SET-01 produced sealed INDETERMINATE: QUADRATURE_NOT_CONVERGED terminal records under the H.5 tensor-product evaluator. Those records establish evaluator insufficiency only; they are not profile rejection.

RUN-H2-RADIAL-SUPPORT-SET-01 evaluates the same frozen linear, quadratic, and edge profiles with the support-aligned evaluator declared in H.13. The dimensional H2 comparator remains inaccessible throughout native execution.

Sealed support-aligned artifact hashes recorded in H.18: linear
b9548c8899ad446266b888c9d64f3bc010c8bca9b107df36445c6fc5b5859496; quadratic
32379a4a9b4doeid3335600369414797da462a7dfdc53627c8c762b7fd4cf4ae; edge
638b7do87c7f06f8b242ecb8e941a42d10bb29d8c5a79abe97729b8418bce803.

F.4 Embedded offline evidence cards

These cards embed the external datum needed to check each empirical worked example without network access. They are evidence records only. They have zero authority to define native SEAM mathematics, select a candidate, tune a coefficient, or populate a missing native input. A blind native run keeps the relevant card sealed until its declared reveal event.

Evidence ID	Worked example / check	Offline datum embedded here	Source citation	Execution role
EVID-CS-TIME-001	Cs-133 time resolver	The unperturbed ground-state hyperfine transition frequency of 133Cs is exactly 9 192 631 770 Hz; equivalently, one SI second contains 9 192 631 770 periods.	BIPM, SI base unit: second (s), SI Brochure definition of ΔvCs.	Post-definition metrology corroboration; exact external reference.
EVID-H-ATOM-001	Hydrogen baseline	Neutral hydrogen ground-state configuration: 1s (one electron in the first shell).	NIST Physics, Cross Section Atom Information, Holdings for Hydrogen.	External corroboration of the worked Z(n)=1 shell example.
EVID-HE-ATOM-001	Helium baseline	Neutral helium ground-state configuration: 1s ² .	NIST Physics, Cross Section Atom Information, Holdings for Helium.	External corroboration of first-shell closure.
EVID-LI-ATOM-001	Lithium baseline	Neutral lithium ground-state configuration: 1s ² 2s.	NIST Physics, Cross Section Atom Information, Holdings for Lithium.	External corroboration of closed first shell plus one outer electron.
EVID-CO2-GEOM-001	CO2 geometry discrimination	Experimental CO2 geometry: r_CO=1.162 Å, O-C-O angle=180°, Cartesian O-O separation=2.3242 Å.	NIST Computational Chemistry Comparison and Benchmark Database (CCCBDB), Experimental data for CO2 (CAS 124-38-9), geometric data; reference 1966 Herzberg.	External geometry check only; geometry values do not define the SEAM relational constructor.

Evidence ID	Worked example / check	Offline datum embedded here	Source citation	Execution role
EVID-H2-BOND-001	H2 bond comparator	Experimental H2 H-H distance recorded by NIST CCCBDB: 0.7414 Å (experimental geometry record).	NIST CCCBDB, Experimental data / experimental geometry for H2 (CAS 1333-74-0).	Sealed dimensional comparator; inaccessible during native H2 optimization.
EVID-BOHR-001	H2 conventional-unit cross-check	2022 CODATA Bohr radius $a_0 = 5.291\,772\,105\,44(82) \times 10^{-11}$ m. Therefore $0.7414\text{ Å} / a_0 = 1.40104295$ Bohr.	NIST, 2022 CODATA recommended value: Bohr radius.	Unit-conversion check after comparator reveal; never a native spatial generator.
EVID-NE-ATOM-001	Ne-like count 10	Neutral neon ground-state configuration: $1s^2 2s^2 2p^6$; ten electrons complete the first and second occupied shells.	NIST Physical Measurement Laboratory, Elemental Data Index: 10 Neon; ground-state configuration $1s^2 2s^2 2p^6$.	External corroboration of the $Z(n)=10$ two-shell-closure illustration; never consumed by the shell constructor.

Offline arithmetic checks

Cs check: $9,192,631,770$ transition periods / $9,192,631,770 = 1$ resolved conventional second.

H2 unit check: $0.7414\text{ Å} = 7.414 \times 10^{-11}\text{ m}$; $(7.414 \times 10^{-11}\text{ m}) / (5.29177210544 \times 10^{-11}\text{ m}) = 1.40104295\text{ a}_0$.

The arithmetic above verifies only the external reporting conversion. It does not determine N_r^* , calibrate the physical magnitude of L_A , select a radial profile, or define any native SEAM quantity.

Appendix G — Worked Atomic Configuration and H2 Full-Chain Evidence

Hydrogen baseline

$Z(n)=1 \rightarrow$ shell occupancy

1, 0, 0, 0, 0, 0, 0

$\rightarrow \rho =$

$1/2, 0, \dots$

.

Offline evidence check: EVID-H-ATOM-001 records the NIST neutral-hydrogen ground-state configuration $1s$. The NIST datum is corroboration only; the SEAM shell state is generated from $Z(n)=1$ and the canonical capacity vector.

Helium baseline

$Z(n)=2 \rightarrow$

2, 0, 0, 0, 0, 0, 0

$\rightarrow$ first-shell closure.

Offline evidence check: EVID-HE-ATOM-001 records the NIST neutral-helium ground-state configuration $1s^2$. The evidence is not consumed by the shell constructor.

Lithium baseline

$Z(n)=3 \rightarrow$

2, 1, 0, 0, 0, 0, 0

$\rightarrow$ closed first shell plus partial outer shell.

Offline evidence check: EVID-LI-ATOM-001 records the NIST neutral-lithium ground-state configuration $1s^2 2s$. The evidence is not consumed by the shell constructor.

Two-entity configuration

$\{\mathcal{S}_A, \mathcal{S}_B\}$ plus complete r_{AB} defines C. Section G.5 carries this construction through the explicit canonical entropy functional for H_2 and records the native evidence contract.

CO₂ geometry discrimination

Complete pairwise distances include O1-O2 and therefore distinguish linear from bent geometry even when C-O distances are fixed.

Offline evidence check: EVID-CO2-GEOM-001 records NIST CCCBDB experimental CO₂ geometry with O-C-O=180 degrees and O-O=2.3242 Angstrom. The card checks the relational example after construction; it does not supply r_{ij} to a blind native derivation.

Chirality extension

Distance matrices fix geometry only up to reflection; χ_{ijkl} supplies the required parity relation.

G.5 H₂ full-chain worked evidence

Purpose. This worked example carries H₂ from the atomic count baseline through the explicit canonical shell-to-field constructor, analytical overlap/field entropy, complete entropy objective, and falsification result without using the known bond distance as an input. It is evidence of executable mathematical closure and of the rule that a closed operator is rejected rather than tuned when it fails its contract.

Offline evidence boundary: EVID-H2-BOND-001 embeds the experimental H-H distance and EVID-BOHR-001 embeds the CODATA Bohr radius. Both cards remain downstream evidence and are prohibited from native optimization until the declared reveal event.

G.5.1 Input and atomic construction

$Z_A(n)=1, Z_B(n)=1$

$c_{\text{shell}}=$

2, 8, 18, 32, 18, 32, 18

$N_A=$

1, 0, 0, 0, 0, 0, 0

, $N_B=$

1, 0, 0, 0, 0, 0, 0

$\rho_A=\rho_B=$

1/2, 0, 0, 0, 0, 0, 0

$\mathcal{S}_A=\mathcal{S}_B=\{1/2, 0, 0, 0, 0, 0, 0\}$

No element label is required to construct these shell states after the electron count is supplied. Count conservation and shell capacity are satisfied identically.

G.5.2 Relational configuration and admissible family

$\mathcal{R}(r)=\{r_{AB}=r\}, r>0$

$C_{H_2}(r)=(\{\mathcal{S}_A, \mathcal{S}_B\}, \mathcal{R}(r))$

A two-entity configuration has no four-point chirality coordinate. The active admissible family contains only the electron-configuration variations and N_r values explicitly declared by the run contract. Every candidate preserves electron count, shell bounds, declared spatial/field constraints, and frozen test domain. The bond comparator is not part of $\mathcal{A}$.

G.5.3 Configuration-entropy term

$$S_{\text{config},\text{H}_2} = k_B$$

$$\ln \text{binom}(2, 1) + \ln \text{binom}(2, 1)$$

$$= 2 k_B \ln 2$$

All unoccupied-shell factors contribute $\ln \text{binom}(c_n, 0) = 0$. This term is therefore fully numerical before any intermolecular geometry is chosen.

G.5.4 Native H₂ spatial-field realization

$$N_r = ||X_B - X_A||; V_1 = 4\pi/3; u_A(\xi) = 1_{\{||\xi - X_A|| < 1\}}/\sqrt{V_1}; u_B(\xi) = 1_{\{||\xi - X_B|| < 1\}}/\sqrt{V_1}$$

$$F_{\text{H}_2}(\xi; N_r) = u_A(\xi) + u_B(\xi)$$

For hydrogen only shell $n=1$ is occupied, so the constructor reduces to two normalized unit-ball shell amplitudes in native coordinate ξ . No external bond length or conventional orbital is used. The dimensional bond-length comparator is downstream of the native normalized result.

G.5.5 Closed first-shell overlap and field entropy

$$O_{11}(N_r) = ((4 + N_r)(2 - N_r)^2)/16 \text{ for } 0 \leq N_r \leq 2; O_{11}(N_r) = 0 \text{ for } N_r > 2$$

$$S_{\text{field},\text{H}_2}(N_r)/k_B = \ln$$

$$V_1(1 + O_{11})$$

•

$$(1 - 3O_{11})/(1 + O_{11})$$

$$\ln 2$$

The overlap formula follows from the exact intersection volume of two equal unit spheres divided by V_1 . The field-entropy expression follows by splitting native space into the two exclusive regions and the overlap region, where $|F_{\text{H}_2}|^2$ is respectively $1/V_1$ and $4/V_1$ before normalization. Both are analytic functions of N_r ; no unspecified F_C or spatial partition remains in the H₂ chain.

$$S_{\text{coupling},\text{H}_2}(N_r) = (k_B/4) O_{11}(N_r)$$

All other shell-pair terms vanish because their shell fractions are zero.

G.5.6 Complete native H₂ objective and analytical test

$$S_{\text{H}_2}(N_r)/k_B = 2 \ln 2 + \lambda_{\text{field}} \{ \ln$$

$$V_1(1 + O_11)$$

•

$$(1 - 3O_11)/(1 + O_11)$$

$$\ln 2\} + (\text{lambda_coupling}/4)O_{11}$$

$$N_{r^*} = \arg \max_{\{N_r \text{ in } A_r\}} S_{H2}(N_r)$$

$$d(S_{H2}/k_B)/dO_{11} = \text{lambda_field}$$

$$1/(1 + O_11) - 4\ln 2/(1 + O_11)^2$$

- $\text{lambda_coupling}/4$

A finite bond requires a strict interior maximum in N_r ; a plateau or boundary maximum is not bond closure.

The H_2 geometry question is therefore completely explicit in the minimal field realization. Because O_{11} decreases monotonically from 1 to 0 as N_r increases from 0 to 2, the sign and curvature of the closed expression determine whether a finite bond can exist.

G.5.7 Analytical result for the declared run coefficients

$$\text{lambda_field} = 1.0, \text{lambda_coupling} = 0.1$$

declaredruncoefficients

For these frozen values, $d(S_{H2}/k_B)/dO_{11}$ is negative throughout $0 \leq O_{11} \leq 1$. Therefore S_{H2} increases as overlap decreases, reaches its maximum when $O_{11}=0$ at $N_r=2$, and remains constant for larger separations in the translation-invariant native domain.

$$S_{H2}(N_r=0)/k_B = 2.8437063194$$

$$S_{H2}(N_r \geq 2)/k_B = 3.5118535000$$

$\arg \max S_{H2} = \{N_r: N_r \geq 2\}$; no unique finite interior H_2 bond maximum

Disposition: REJECTED_H2_UNIFORM_SHELL_CONTRACT. Under RUN-H2-BASELINE-UNIFORM-SHELL-01, the canonical uniform-shell field constructor with $\text{lambda_field}=1.0$ and $\text{lambda_coupling}=0.1$ does not produce a unique finite interior H_2 entropy maximum. The comparator did not enter the native derivation.

Result boundary: this run rejects the tested uniform-shell H_2 contract. It does not identify a replacement field, radial law, phase term, orientation term, coefficient change, or other mechanism. None of those objects is implied by the failure.

G.5.8 SEAM-to-observable projection boundary for H₂

The native result N_r^* is a resolved SEAM distance count because $N_r = ||X_B - X_A||$ is explicitly the norm of the native spatial separation. Under the canonical universal distance-count metrology, its SEAM baseline-length projection is $L_{H2} = N_r^* L_A$. This projection does not require a separate H₂-specific identity merely because the external reach-count symbol also used N_r . Conventional-unit reporting remains downstream and requires the applicable canonical baseline/reporting bindings; it must not be used to define or retroactively validate the native distance count.

SEAM field/entropy contract $\rightarrow N_r^* \rightarrow$ freeze $\rightarrow$ dimensional projection $\rightarrow$ H₂ bond-length evidence

G.5.8.1 canonical spatial-scale gate

G.5.9 Contract binding

- The executable baseline contract is Appendix H.2, RUN-H₂-BASELINE-UNIFORM-SHELL-01. The executor receives the contract, canonical equations, and native inputs. The executor does not receive the comparator artifact.
- Native execution follows H.2 exactly. No additional field term, radial profile, occupancy-support rule, dimensional scale, conventional orbital, or target-derived quantity is permitted.
- The transcript records every value consumed, every equation evaluated, every branch taken, S

C

, gradient or analytic derivative, configuration, N_r , terminal state, and artifact hash.

- The native result, transcript, contract, implementation identity, and hashes are sealed before comparator access.
- Comparator execution is a separate post-seal step. The blind execution bundle does not contain the dimensional H₂ target. The evidence package loads the comparator only after native sealing and applies the independently frozen projection contract.
- Disposition is determined solely by the terminal rules in H.2. No post-run reinterpretation changes the terminal state.

G.5.10 Evidence-status separation

External comparator record: EVID-H₂-BOND-001 embeds the NIST CCCBDB H-H distance 0.7414 Angstrom. EVID-BOHR-001 embeds the 2022 CODATA Bohr radius, giving 1.40104295 Bohr by direct unit conversion. These values are evidence only. They are not evidence that RUN-H₂-BASELINE-UNIFORM-SHELL-01 passed; that run produces a separation plateau after overlap vanishes and is rejected for finite H₂ bond closure. The comparator cards remain separate from native execution.

formal field chain closed + analytical minimal test executed $\neq$ H₂ bond law validated

native N_r^* / terminal state -> freeze -> dimensional projection -> reveal 1.40 Bohr comparator -> residual -> validation disposition

G.5.11 Executed occupancy-volume support perturbation

Contract ID: RUN-H2-RHO-VOLUME-01. Status: explicitly noncanonical perturbation of the frozen canonical H₂ baseline. This run changes one object only: the fixed uniform first-shell support radius.

Declared transform: $V_{\text{occ}} = \rho_{\text{H},1} V_1$; $R_{\text{rho}} = \rho_{\text{H},1}^{(1/3)}$. For hydrogen $\rho_{\text{H},1} = 1/2$, so $R_{\text{rho}} = 2^{(-1/3)} = 0.7937005260$ and zero-overlap begins at $N_r = 2R_{\text{rho}} = 1.587401052$.

All other equations and coefficients remain identical to RUN-H2-BASELINE-UNIFORM-SHELL-01. The dimensional H₂ comparator, Bohr radius, conventional orbitals, fitted exponents, phase/orientation terms, and coefficient changes are prohibited.

For two identical normalized uniform spheres, changing fixed radius changes only the map $N_r \rightarrow O_{11}$ and adds the constant $\ln(V_R/V_1)$ to S_{field} . It does not change dS/dO_{11} . With $\lambda_{\text{field}} = 1.0$ and $\lambda_{\text{coupling}} = 0.1$, dS/dO_{11} remains negative for $0 \leq O_{11} \leq 1$.

Native result: $\arg \max S_{\text{H2}} = \{N_r : N_r \geq 1.587401052\}$. There is no unique finite interior maximum.

Disposition: REJECTED_H2_FIXED_SUPPORT_RESCALING. The run rejects occupancy-volume support rescaling as a finite-bond solution under the frozen uniform-field entropy contract.

Class result: any fixed uniform-support radius $R = R(\rho)$ with the same normalized basis, entropy functional, and frozen coefficient pair leaves the S-versus-overlap derivative unchanged. Fixed support rescaling changes the separation scale but cannot create an interior entropy maximum.

No replacement mechanism is inferred. A subsequent run must declare an explicit equation that changes more than fixed support scale; otherwise no subsequent run exists.

Appendix H — Explicit Run Contracts, Blind Tests, and Falsification

H.1 Universal execution contract

Every native run is identified by a unique contract ID. The contract is immutable after execution begins.

Required contract fields: OBJECTIVE; CANONICAL_SPECIFICATION; INPUTS; EQUATIONS; PERMITTED_PERTURBATION; PROHIBITED_INPUTS; COORDINATE_AND_UNIT_CONTRACT; COEFFICIENTS; DOMAIN; NUMERICAL_OR_ANALYTIC_METHOD; TOLERANCES; TERMINAL_STATES; REQUIRED_ARTIFACTS; COMPARATOR_LOCATION; COMPARATOR_ACCESS_EVENT; DISPOSITION_RULE.

No implicit assumptions rule: any value, function, transform, default, branch, unit, scale, boundary condition, initialization, tolerance, or external source not listed in the contract is prohibited. Omission never authorizes a default.

Missing required contract field -> INDETERMINATE: CONTRACT_INCOMPLETE. Missing required native mathematical object -> INDETERMINATE: UNDECLARED_INPUT. Consumption of an undeclared input -> INVALID_RUN: UNDECLARED_INPUT_CONSUMED.

Comparator isolation rule: a blind executor does not receive the comparator artifact. The comparator is loaded by a separate post-seal process only after native outputs and hashes exist. A document that contains evidence record comparator values is not itself a blind execution bundle.

Perturbation rule: a noncanonical exploratory run is permitted only when its changed equation is written explicitly in PERMITTED_PERTURBATION and every unchanged baseline equation is frozen by reference. The perturbation does not become canonical because it was executed or because it passed.

One-change rule: unless the contract explicitly declares a factorial/multi-change experiment, one perturbation contract changes one mathematical object. This prevents hidden compensating assumptions.

H.1.1 Terminal-state supremacy and no-repair rule

Every TERMINAL_STATES field is operative. When one terminal predicate becomes true, the run reaches terminal state at that operation. No mathematical or numerical branch exists unless the same contract explicitly declares that branch before execution.

Required terminal sequence:

DETECT_TERMINAL -> RECORD_TRIGGER -> COMPLETE_AVAILABLE_REQUIRED_ARTIFACTS -> HASH -> SEAL -> STOP

AVAILABLE_REQUIRED_ARTIFACTS means every required artifact constructible solely from values already generated before termination. A missing downstream artifact that requires additional model execution is marked NOT_GENERATED_DUE_TO_TERMINAL_STATE; it is not fabricated and does not authorize more computation.

PROHIBITED_AFTER_TERMINAL: alternate quadrature; increased node count; relaxed or tightened tolerance; alternate optimizer; domain splitting; domain extension; smoothing; extrapolation; interpolation not already declared; changed coordinate system; modified boundary treatment; substituted coefficient; modified profile; added state variable; comparator access; external lookup; operator consultation for a repair decision.

A remedy proposed after terminal state receives exactly one status: FUTURE_CONTRACT_CANDIDATE. The run remains closed. The proposed remedy exists as executable SEAM work only after a separate immutable contract declares it before execution.

An executor response that ends with a request such as “should I change the method?” is noncompliant with a cold-run contract. The compliant response seals the terminal run and stops.

Violation after a valid terminal event does not retroactively alter the terminal predicate that was reached. It creates a separate compliance status: INVALID_RUN: POST_TERMINAL_EXECUTION for the continued execution branch.

H.2 RUN-H2-BASELINE-UNIFORM-SHELL-01

OBJECTIVE: Determine whether the canonical uniform-shell field constructor with the frozen external coefficient pair produces a unique finite interior H2 entropy maximum.

CANONICAL_SPECIFICATION: canonical §4.10-§4.15 and Appendix G.5 equations.

INPUTS: $Z_A=1$; $Z_B=1$; $c_1=2$; $N_{A,1}=N_{B,1}=1$; $\rho_{A,1}=\rho_{B,1}=1/2$; all other shell counts and fractions zero; two entity centers separated by native dimensionless N_r .

EQUATIONS: $\Omega_{i,1}=\{\xi: ||\xi-X_i||<1\}$; $V_1=4\pi/3$; $u_{i,1}=1_{\{\Omega_{i,1}\}}/\sqrt{V_1}$; $F_{H2}=u_A+u_B$; $O_{11}=((4+N_r)(2-N_r)^2)/16$ for $0\leq N_r\leq 2$ and $O_{11}=0$ for $N_r>2$; $S_{\text{config},H2}=2k_B \ln 2$; $S_{\text{field},H2}/k_B=\ln$

$$V_1(1 + O_1)$$

•

$$(1 - 3O_1)/(1 + O_1)$$

$\ln 2$; $S_{\text{coupling},H2}=(k_B/4)O_{11}$; $S_{\text{total}}=S_{\text{config}}+\lambda_{\text{field}} S_{\text{field}}+\lambda_{\text{coupling}} S_{\text{coupling}}$.

PERMITTED_PERTURBATION: NONE.

PROHIBITED_INPUTS: 0.74 Å; 1.40 Bohr; Bohr radius; Coulomb potential; conventional hydrogen orbital; target-derived radial profile; rho-dependent support scaling; phase; orientation; fitted coefficients; fitted distance scale; post-reveal equation changes.

COORDINATE_AND_UNIT_CONTRACT: native dimensionless ξ and N_r only. L_A and all dimensionalization are absent from native optimization.

COEFFICIENTS: $\lambda_{\text{field}}=1.0$; $\lambda_{\text{coupling}}=0.1$.

DOMAIN: $N_r\geq 0$. The analytic expression controls. A numerical implementation must include $N_r=0$, $N_r=2$, and values beyond 2 sufficient to establish the plateau.

METHOD: evaluate $S_{H2}(N_r)$ or analytically evaluate dS/dO_{11} and monotonic $O_{11}(N_r)$. No alternate optimizer or smoothing function changes the mathematical objective.

TERMINAL_STATES: RESOLVED_INTERIOR_MAXIMUM when one strict finite interior maximum exists; REJECTED_H2_UNIFORM_SHELL_CONTRACT when no strict finite interior maximum exists; INDETERMINATE only for a declared missing artifact or incomplete execution.

REQUIRED_ARTIFACTS: contract copy; implementation identity; exact equations; evaluated derivative or grid; S-versus- N_r record; terminal state; native result; hashes.

COMPARATOR_LOCATION: separate sealed evidence artifact. COMPARATOR_ACCESS_EVENT: after native result and transcript hashes are sealed.

DISPOSITION_RULE: no strict finite interior maximum -> REJECTED_H2_UNIFORM_SHELL_CONTRACT. No reinterpretation converts a plateau or boundary maximum into a bond.

EXECUTED RESULT: $d(S_{H2}/k_B)/dO_{11} < 0$ for $0 \leq O_{11} \leq 1$; $\arg \max S_{H2} = \{N_r: N_r \geq 2\}$; terminal state REJECTED_H2_UNIFORM_SHELL_CONTRACT.

H.3 RUN-H2-RHO-VOLUME-01

OBJECTIVE: Test whether using the already-defined shell fraction ρ as occupied-volume fraction changes the H_2 extremum under the otherwise frozen uniform-field contract.

CANONICAL_SPECIFICATION: canonical baseline equations frozen; this run is NONCANONICAL_PERTURBATION.

INPUTS: identical to H.2.

PERMITTED_PERTURBATION: replace first-shell unit radius by $R_{\rho} = \rho_{H,1}^{1/3}$, derived from $V_{occ} = \rho_{H,1} V_1$. For $\rho_{H,1} = 1/2$, $R_{\rho} = 0.7937005260$. Use normalized uniform basis on that support. No other equation changes.

PROHIBITED_INPUTS: all H.2 prohibited inputs plus any fitted exponent in $R(\rho)$, any second structural variable, and any coefficient change.

COORDINATE_AND_UNIT_CONTRACT: native dimensionless separation only.

COEFFICIENTS: $\lambda_{field} = 1.0$; $\lambda_{coupling} = 0.1$.

METHOD: substitute R_{ρ} into the exact equal-sphere overlap. Evaluate the unchanged S-versus-overlap derivative.

TERMINAL_STATES: RESOLVED_INTERIOR_MAXIMUM if one strict finite interior maximum appears; REJECTED_H2_FIXED_SUPPORT_RESCALING otherwise.

EXECUTED RESULT: zero overlap begins at $N_r = 2R_{\rho} = 1.587401052$. The S-versus-overlap derivative is unchanged and remains negative across the full overlap interval. $\arg \max S_{H2} = \{N_r: N_r \geq 1.587401052\}$. Terminal state REJECTED_H2_FIXED_SUPPORT_RESCALING.

CLASS DISPOSITION: fixed uniform-support rescaling $R = R(\rho)$ changes only the separation-to-overlap map and an additive field-entropy constant. Under the H.2 entropy law and coefficient pair it cannot create a strict interior maximum. This entire fixed-support-rescaling class is rejected for H_2 finite-bond closure under the frozen contract.

H.4 Admission rule for the next run

No next mechanism is assumed by H.2 or H.3. A next run exists only after a complete explicit equation is written for the object being changed.

The next contract must identify exactly one changed mathematical object relative to H.2, state its complete formula and domain, prove that no prohibited comparator quantity entered its construction, and freeze all unchanged objects.

Statements such as “add radial structure,” “include phase,” “use orientation,” “change the coefficient,” “let the field respond,” or “use the Manifold” are not executable instructions. They have no model status until converted to explicit equations and contract fields.

An executor presented with such prose terminates INDETERMINATE: NONEXECUTABLE_PROPOSAL.

H.5 Common contract — H₂ fixed-support radial-profile discriminator set

CONTRACT_SET_ID: RUN-H₂-RADIAL-PROFILE-SET-01. STATUS:

NONCANONICAL_PERTURBATION_SET. EXECUTION_STATUS: SEALED. The H.6-H.8 cold runs terminated under the declared 192/256 tensor-product convergence gate and H.9 emitted

INDETERMINATE_RADIAL_PROFILE_SET_01. These terminal records remain final for H.5-H.9.

OBJECTIVE: Determine whether changing only the normalized radial intensity profile inside the fixed first-shell support changes the topology of S_{H₂}(N_r) and produces one strict finite interior entropy maximum under the otherwise frozen H.2 entropy contract.

BASELINE: RUN-H₂-BASELINE-UNIFORM-SHELL-01 in H.2. Every H.2 equation, coefficient, shell count, shell fraction, support radius, entropy term, comparator restriction, and terminal-state rule remains frozen except the single basis-function equation u_{i,1} explicitly replaced by H.6, H.7, or H.8.

INPUTS: Z_A=1; Z_B=1; c₁=2; N_{A,1}=N_{B,1}=1; rho_{A,1}=rho_{B,1}=1/2; all other shell counts and fractions zero; entity centers X_A=(0,0,-N_r/2) and X_B=(0,0,+N_r/2); native dimensionless separation N_r.

FROZEN SUPPORT: Ω_{i,1}={ξ: ||ξ-X_i|| ≤ 1}. The support radius is exactly 1. Occupancy-dependent support scaling is prohibited in this contract set.

CHANGED MATHEMATICAL OBJECT: u_{i,1} only. For profile k, define r_i=||ξ-X_i|| and

u_{i,1}^(k)(ξ) = A_k f_k(r_i) 1_{0 ≤ r_i ≤ 1}, A_k =

$$4\pi \int_0^1 r^2 f_k(r)^2 dr$$

^(-1/2)

The normalization condition is integral |u_{i,1}^(k)(ξ)|² d³ξ=1. H.6-H.8 supply f_k and the exact closed-form A_k. No fitted parameter exists in any profile.

FIELD EQUATIONS: F_{H₂}^(k)(ξ;N_r)=u_{A,1}^(k)(ξ)+u_{B,1}^(k)(ξ); q_k(ξ;N_r)=|F_{H₂}^(k)|² / integral |F_{H₂}^(k)|² d³ξ; use 0 ln 0 = 0 in the field-entropy integral.

S_{field}^(k)(N_r) = -k_B integral q_k(ξ;N_r) ln q_k(ξ;N_r) d³ξ

OVERLAP EQUATION: retain the defining integral from §4.15.6 and discard only the uniform-volume closed-form equality, which is inapplicable after u changes:

O₁₁^(k)(N_r) = integral sqrt(|u_{A,1}^(k)(ξ)|² |u_{B,1}^(k)(ξ)|²) d³ξ

Because all three declared profiles are real and nonnegative on their support, O₁₁^(k)=integral u_{A,1}^(k) u_{B,1}^(k) d³ξ.

FROZEN ENTROPY EQUATIONS: S_{config,H₂}=2 k_B ln 2; S_{coupling,H₂}^(k)=(k_B/4) O₁₁^(k); lambda_{field}=1.0; lambda_{coupling}=0.1;

S_{H₂}^(k)(N_r) = 2 k_B ln 2 + S_{field}^(k)(N_r) + 0.1 S_{coupling,H₂}^(k)(N_r)

PROHIBITED_INPUTS: 0.74 Å; 1.40 Bohr; Bohr radius; Coulomb potential; conventional hydrogen orbital; target-derived radial profile; target-derived exponent; fitted radius; fitted coefficient; fitted distance scale; rho-dependent support scaling; phase; orientation; polarity; state-response term; Manifold-derived correction; Hamiltonian correction; post-reveal equation change; any profile other than H.6-H.8.

COORDINATE_AND_UNIT_CONTRACT: ξ is a native dimensionless spatial coordinate and $N_r = ||X_B - X_A||$ is its native spatial-distance count. Baseline-length and conventional-unit projection are excluded from native optimization. This exclusion isolates the optimization layer; it does not mean N_r lacks distance status. The comparator artifact is not available to the executor.

SEPARATION DOMAIN: $0 \leq N_r \leq 3$. The compact supports are disjoint for $N_r > 2$; points beyond 2 are retained to establish any post-overlap plateau explicitly.

INTEGRATION COORDINATES: use cylindrical coordinates (s, ϕ, z) around the internuclear axis with $d^3\xi = ds d\phi dz$. Axisymmetry removes ϕ analytically, giving factor 2π . For each N_r integrate over $0 \leq s \leq 1$ and $-(1+N_r/2) \leq z \leq (1+N_r/2)$. Values outside both unit supports are exactly zero.

NUMERICAL QUADRATURE: tensor-product Gauss-Legendre quadrature. Primary calculation uses 192 nodes in s and 192 nodes in z . Verification calculation uses 256 nodes in s and 256 nodes in z . A point is numerically admissible only when $|S_{256}/k_B - S_{192}/k_B| \leq 1.0e-8$ and $|O_{256} - O_{192}| \leq 1.0e-8$. Failure of either condition terminates INDETERMINATE: QUADRATURE_NOT_CONVERGED.

SEPARATION SEARCH: evaluate the verification calculation on $N_r = j/200$ for $j = 0, \dots, 600$. A strict interior candidate requires $S(N_{r,j}) > S(N_{r,\{j-1\}})$ and $S(N_{r,j}) > S(N_{r,\{j+1\}})$. Refine every such bracket

$$N_{r,j} - 1, N_{r,j} + 1$$

by deterministic bounded Brent maximization with absolute N_r tolerance $1.0e-10$ and maximum 500 iterations.

STRICT-MAXIMUM TEST: for each refined N_r^* , evaluate S at N_r^* , $N_r^* - h$, and $N_r^* + h$ with $h = 1.0e-4$. Define $D_2 =$

$$S(N_r^* + h) - 2S(N_r^*) + S(N_r^* - h)$$

$/h^2$. RESOLVED_INTERIOR_MAXIMUM requires $0 < N_r^* < 2$, $D_2 < 0$, and $S(N_r^*) - \max(S(N_r^* - h), S(N_r^* + h)) > 1.0e-10 k_B$. Otherwise the candidate is not a strict interior maximum.

TERMINAL_STATES: RESOLVED_INTERIOR_MAXIMUM when exactly one strict finite interior maximum satisfies the preceding test; RESOLVED_MULTIPLE_INTERIOR_MAXIMA when more than one strict finite interior maximum satisfies it; REJECTED_NO_INTERIOR_MAXIMUM when none exists and the numerical calculation converges; INDETERMINATE: QUADRATURE_NOT_CONVERGED when convergence fails; INDETERMINATE: OPTIMIZER_FAILURE when a declared bracket does not converge; INVALID_RUN: CONTRACT_VIOLATION when an undeclared input or equation enters execution.

TERMINAL FINALITY FOR H.5: if the 192/256 convergence gate fails at any required point, that profile run terminates INDETERMINATE: QUADRATURE_NOT_CONVERGED. The executor records the residuals, completes artifacts available from the declared calculations, seals the run, and stops. Domain splitting, higher node counts, changed tolerances, alternate quadrature, smoothing, and any other numerical repair are prohibited in H.5-H.8.

A proposed discontinuity-aligned or domain-split quadrature has status FUTURE_CONTRACT_CANDIDATE only. It is not H.5 execution and is not run until a separate contract freezes that numerical operator while identifying every unchanged H.5 object.

REQUIRED_ARTIFACTS PER RUN: immutable contract copy; implementation identity and hash; exact profile formula; exact normalization derivation; N_r grid; $O_{11}(N_r)$; $S_{\text{field}}(N_r)/k_B$; $S_{\text{coupling}}(N_r)/k_B$; $S_{\text{total}}(N_r)/k_B$; convergence residuals; optimizer brackets and refined coordinates; strict-maximum test values; terminal state; native result; transcript hash.

COMPARATOR_LOCATION: separate sealed evidence artifact. COMPARATOR_ACCESS_EVENT: after native result, transcript, and hashes are sealed. Comparator access before sealing terminates INVALID_RUN: COMPARATOR_LEAKAGE.

INTERPRETATION RULE: execution determines only whether the specified profile changes the native H_2 entropy topology under this frozen contract. A resolved interior maximum does not make the profile canonical and does not validate its dimensional agreement with H_2 . Canonical promotion requires a separate formal and independent justification.

H.6 RUN-H2-RADIAL-LINEAR-01

STATUS: NONCANONICAL_PERTURBATION. BASE CONTRACT: H.5. This run changes no object except $u_{i,1}$.

PROFILE: $f_{\text{lin}}(r)=1-r$ for $0 \leq r \leq 1$; $f_{\text{lin}}(r)=0$ outside the support.

$$\text{integral}_0^1 r^2(1-r)^2 dr = 1/30$$

$$A_{\text{lin}} = \sqrt{15/(2\pi)}$$

$$u_{i,1}^{\text{lin}}(\xi) = \sqrt{15/(2\pi)} (1-r_i) \mathbf{1}_{\{0 \leq r_i \leq 1\}}$$

PROHIBITED MODIFICATIONS: every modification not listed in this H.6 profile equation. The H.5 frozen equations and instructions control all remaining execution.

TERMINAL STATE: INDETERMINATE: QUADRATURE_NOT_CONVERGED. EXECUTED RESULT: 599/601 required separation points failed the H.5 convergence gate; $\max |\Delta S/k_B| = 4.40e-06$; $\max |\Delta O_{11}| = 5.30e-06$; comparator_accessed=false. Run sealed.

H.7 RUN-H2-RADIAL-QUADRATIC-01

STATUS: NONCANONICAL_PERTURBATION. BASE CONTRACT: H.5. This run changes no object except $u_{i,1}$.

PROFILE: $f_{\text{quad}}(r)=1-r^2$ for $0 \leq r \leq 1$; $f_{\text{quad}}(r)=0$ outside the support.

$$\text{integral}_0^1 r^2(1-r^2)^2 dr = 8/105$$

$$A_{\text{quad}} = \sqrt{105/(32\pi)}$$

$$u_{i,1}^{\text{quad}}(\xi) = \sqrt{105/(32\pi)} (1-r_i^2) \mathbf{1}_{\{0 \leq r_i \leq 1\}}$$

PROHIBITED MODIFICATIONS: every modification not listed in this H.7 profile equation. The H.5 frozen equations and instructions control all remaining execution.

TERMINAL STATE: INDETERMINATE: QUADRATURE_NOT_CONVERGED. EXECUTED RESULT: 595/601
required separation points failed the H.5 convergence gate; max $|\Delta S/k_B| = 5.30e-06$; max $|\Delta O_{11}| = 6.07e-06$;
comparator_accessed=false. Run sealed.

H.8 RUN-H2-RADIAL-EDGE-01

STATUS: NONCANONICAL_PERTURBATION. BASE CONTRACT: H.5. This run changes no object except $u_{i,1}$.

PROFILE: $f_{\text{edge}}(r) = r$ for $0 \leq r \leq 1$; $f_{\text{edge}}(r) = 0$ outside the support.

$$\int_0^1 r^4 dr = 1/5$$

$$A_{\text{edge}} = \sqrt{5/(4\pi)}$$

$$u_{i,1}^{\text{edge}}(\xi) = \sqrt{5/(4\pi)} r_{i,1} \{0 \leq r_{i,1} \leq 1\}$$

PROHIBITED MODIFICATIONS: every modification not listed in this H.8 profile equation. The H.5 frozen equations and instructions control all remaining execution.

TERMINAL STATE: INDETERMINATE: QUADRATURE_NOT_CONVERGED. EXECUTED RESULT: 601/601
required separation points failed the H.5 convergence gate; max $|\Delta S/k_B| = 2.02e-03$; max $|\Delta O_{11}| = 2.67e-03$;
comparator_accessed=false. Run sealed.

H.9 Radial-profile discriminator set disposition

This section is evaluated only after H.6, H.7, and H.8 each possess sealed native artifacts.

IF all three runs terminate REJECTED_NO_INTERIOR_MAXIMUM:

SET_DISPOSITION=REJECTED_RADIAL_PROFILE_SET_01. This disposition rejects exactly the three declared fixed-support static profiles under the H.5 frozen entropy contract. It does not reject every mathematically possible static radial function.

IF one or more runs terminates RESOLVED_INTERIOR_MAXIMUM or
RESOLVED_MULTIPLE_INTERIOR_MAXIMA:

SET_DISPOSITION=RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED. The result establishes that changing radial intensity shape changes the H2 entropy topology under the frozen contract. It does not establish canonical status, dimensional agreement, or physical correctness of the resolving profile.

IF any run terminates INDETERMINATE or INVALID_RUN:

SET_DISPOSITION=INDETERMINATE_RADIAL_PROFILE_SET_01. No set-level physical or mathematical conclusion is drawn until all three declared runs terminate validly.

COLD-RUN COMPLETION RULE: H.9 is evaluated from the sealed terminal records of H.6-H.8. An INDETERMINATE profile run remains a completed run record for set accounting. It is not reopened or repaired inside RUN-H2-RADIAL-PROFILE-SET-01.

COMPARATOR RULE: no set disposition uses the dimensional H2 comparator. Comparator projection occurs only after each resolving native run is independently sealed, and comparison results remain downstream evidence.

H.9.1 Executed set result

EXECUTED SET_DISPOSITION: INDETERMINATE_RADIAL_PROFILE_SET_01.

Reason: all three H.6-H.8 profile runs sealed INDETERMINATE: QUADRATURE_NOT_CONVERGED under the H.5 evaluator. This disposition establishes numerical-evaluator insufficiency only. It does not reject the three radial profiles.

COMPARATOR_ACCESS: false for H.6-H.9.

H.10 Admission gate for configuration-responsive radial fields

A configuration-responsive field is not part of H.5-H.9. It exists only after a separate run contract supplies an explicit equation of the form $u_{i,1}(\xi|C)=G(\xi,X_i,\{\mathcal{S}_j\},\mathcal{R})$ with every argument, coefficient, domain, boundary condition, normalization rule, and update rule written before execution.

The phrase “let the field respond to proximity” is NONEXECUTABLE_PROPOSAL. The phrase does not authorize dependence on N_r , overlap, phase, orientation, Manifold state, conventional orbital structure, or comparator data.

Admission requires: exactly one declared new mathematical dependency relative to H.5; an explicit closed formula; no target-derived quantity; frozen H.5 entropy and coefficient equations unless a different contract explicitly identifies a separate changed object; deterministic numerical instructions; complete terminal states; comparator isolation.

Until those fields exist in a written contract, execution terminates INDETERMINATE: NONEXECUTABLE_PROPOSAL. No implicit configuration-response law is permitted.

H.11 Deterministic successor edge for H.5 numerical indeterminacy

SUCCESSOR_EDGE_ID: EDGE-H2-RADIAL-QUADRATURE-01.

TRIGGER: a sealed H.6, H.7, or H.8 terminal state equal to INDETERMINATE: QUADRATURE_NOT_CONVERGED.

SUCCESSOR: launch the matching support-aligned run H.14, H.15, or H.16 after the predecessor run is sealed. No operator consultation occurs. The predecessor run remains unchanged and retains its own terminal state.

CARRIED FORWARD FROZEN: Z values; shell capacities; shell counts; rho values; support radius 1; exact H.6-H.8 radial profile formula; field superposition; entropy equations; $\lambda_{\text{field}}=1.0$; $\lambda_{\text{coupling}}=0.1$; separation domain $0 \leq N_r \leq 3$; grid spacing 0.005; optimizer rule; strict-maximum test; prohibited comparator list; comparator seal.

CHANGED OBJECT: numerical integration operator only.

PROHIBITED: changing a radial profile; changing support; changing coefficients; changing entropy terms; smoothing a profile; importing a conventional orbital; importing a dimensional distance; accessing the H2 comparator; modifying a sealed predecessor artifact.

H.12 Support-aligned integration geometry

Let $d=N_r$. Use spherical coordinates centered on entity A. Let $r=||\xi-X_A||$, $\mu=\cos\theta$ relative to the A-to-B axis, and $q(r,\mu;d)=\sqrt{r^2+d^2-2rd\mu}=||\xi-X_B||$.

$$q(r,\mu;d) = \sqrt{r^2 + d^2 - 2 r d \mu}$$

For $d=0$, the complete A support is also inside B. For $0< d < 1$, the radial interval $0 \leq r \leq 1-d$ is full overlap and $1-d < r \leq 1$ is partial overlap. For $1 \leq d < 2$, $d-1 \leq r \leq 1$ is partial overlap. For $d \geq 2$, overlap is empty.

On a partial-overlap radial interval define

$$\mu_O(r,d) = (r^2 + d^2 - 1)/(2 r d)$$

The support boundary $q=1$ is therefore an integration boundary rather than a discontinuity crossed by a tensor grid.

Define the support-aligned overlap operator I_d

$$g$$

by the following exact piecewise domains:

$$0 < d < 1: I_d$$

$$g$$

$$= 2\pi$$

$$\int_0^{1-d} (1-d)r^2 dr \int_{-1}^1 g d\mu + \int_{1-d}^1 (1-d)^1 r^2 dr \int_{\mu_O}^1 g d\mu$$

$$1 \leq d < 2: I_d$$

$$g$$

$$= 2\pi \int_{d-1}^1 r^2 dr \int_{\mu_O}^1 g d\mu$$

$$d=0: I_O$$

$$g$$

$$= 2\pi \int_0^1 r^2 dr \int_{-1}^1 g(r,r) d\mu; d \geq 2: I_d$$

$$g$$

$$=0$$

H.13 Common successor contract — support-aligned radial-profile evaluator

CONTRACT_SET_ID: RUN-H2-RADIAL-SUPPORT-SET-01. STATUS: NONCANONICAL_NUMERICAL_SUCESSOR_SET.

OBJECTIVE: resolve the H.5 numerical indeterminacy without changing any H.5-H.8 physical equation or radial profile.

For each profile $u(r)$, define

$$O(d) = I_d$$

$$u(r)u(q)$$

$$J = 4\pi \int_0^1 r^2 u(r)^2 \ln(u(r)^2) dr$$

$$K(d)=I_d$$

$$(u(r) + u(q))^2 \ln((u(r) + u(q))^2) - 2u(r)^2 \ln(u(r)^2)$$

Use $x \ln x = 0$ at $x=0$. Because each profile is normalized, the field-intensity normalization is exactly

$$I_2(d) = \int |u_A + u_B|^2 d^3\xi = 2 + 2O(d)$$

The H.5 field entropy is evaluated by the algebraically identical expression

$$S_{\text{field}}(d)/k_B = \ln(I_2(d)) -$$

$$2J + K(d)$$

$$/I_2(d)$$

$$S_{\text{coupling}}(d)/k_B = O(d)/4$$

$$S_{H2}(d)/k_B = 2 \ln 2 + S_{\text{field}}(d)/k_B + 0.025 O(d)$$

These equations are a support-aligned evaluation of the same H.5 whole-space integrals. They introduce no field smoothing, no new physical term, and no target-derived quantity.

NUMERICAL STAGE A: Gauss-Legendre quadrature is applied separately on every analytic r and μ interval in H.12. Primary order is 128×128 and verification order is 192×192 . J uses 128 and 192 radial nodes respectively. At every required N_r point and every strict-maximum test point require $|\Delta S_{\text{total}}/k_B| \leq 1.0e-8$, $|\Delta S_{\text{field}}/k_B| \leq 1.0e-8$, and $|\Delta O| \leq 1.0e-8$.

DECLARED FALLBACK B: if Stage A fails any convergence gate, execute the same support-aligned equations at primary 192×192 and verification 256×256 . This is an internal declared branch, not a terminal repair.

DECLARED FALLBACK C: if Stage B fails, execute primary 256×256 and verification 384×384 . If Stage C fails, terminate INDETERMINATE: SUPPORT_ALIGNED_QUADRATURE_NOT_CONVERGED, seal, and stop.

SEPARATION SEARCH AND OPTIMIZER: retain H.5 exactly: $N_r = j/200$ for $j=0, \dots, 600$; strict grid candidate; deterministic bounded Brent refinement with absolute tolerance $1.0e-10$ and maximum 500 iterations; $h=1.0e-4$ strict-maximum test; $D2 < 0$; strict margin $> 1.0e-10$ k_B ; finite interior condition $0 < N_{r^*} < 2$.

TERMINAL STATES: RESOLVED_INTERIOR_MAXIMUM; RESOLVED_MULTIPLE_INTERIOR_MAXIMA; REJECTED_NO_INTERIOR_MAXIMUM; INDETERMINATE: SUPPORT_ALIGNED_QUADRATURE_NOT_CONVERGED; INDETERMINATE: OPTIMIZER_FAILURE; INVALID_RUN: CONTRACT_VIOLATION; INVALID_RUN: COMPARATOR_LEAKAGE.

REQUIRED ARTIFACTS: successor contract; predecessor seal hash; implementation identity/hash; exact profile; stage used; every convergence residual; N_r grid; O; S_{field}/k_B ; S_{coupling}/k_B ; S_{total}/k_B ; optimizer brackets; refined points; $D2$; strict margin; terminal state; transcript; hashes; comparator_accessed flag.

H.14 RUN-H2-RADIAL-LINEAR-SUPPORT-01

PREDECESSOR: RUN-H2-RADIAL-LINEAR-01. PROFILE FROZEN: $u(r) = \sqrt{15/(2\pi)}(1-r)$ for $0 \leq r \leq 1$ and zero outside. CHANGED OBJECT: H.13 numerical evaluator only.

EXECUTED TERMINAL STATE: RESOLVED_INTERIOR_MAXIMUM.

Stage A converged over the complete 601-point grid: $\max |\Delta S_{\text{total}}/k_B| = 5.4041260355575105e-09$; $\max |\Delta S_{\text{field}}/k_B| = 5.438363981369321e-09$; $\max |\Delta O| = 1.3714737123748932e-09$.

$N_{r^*} = 1.6008992282173213$

$S(N_{r^*})/k_B = 2.779426544263546$; $D2 = -0.32176754594104295$; strict margin $= 1.6087633447625649e-09$ k_B ; comparator_accessed=false.

SEALED ARTIFACT SHA-256: b9548c8899ad446266b888c9d64f3bc010c8bca9b107df36445c6fc5b5859496.

H.15 RUN-H2-RADIAL-QUADRATIC-SUPPORT-01

PREDECESSOR: RUN-H2-RADIAL-QUADRATIC-01. PROFILE FROZEN: $u(r) = \sqrt{105/(32\pi)}(1-r^2)$ for $0 \leq r \leq 1$ and zero outside. CHANGED OBJECT: H.13 numerical evaluator only.

EXECUTED TERMINAL STATE: RESOLVED_INTERIOR_MAXIMUM.

Stage A converged over the complete 601-point grid: $\max |\Delta S_{\text{total}}/k_B| = 7.041034422172743e-12$; $\max |\Delta S_{\text{field}}/k_B| = 7.04092339987028e-12$; $\max |\Delta O| = 5.218048215738236e-15$.

$N_{r^*} = 1.7017708805671177$

$S(N_{r^*})/k_B = 2.969779994124098$; $D2 = -0.2599694681748588$; strict margin $= 1.2994023634860241e-09$ k_B ; comparator_accessed=false.

SEALED ARTIFACT SHA-256: 32379a4a9b4doe1d3335600369414797da462a7dfdc53627c8c762b7fd4cf4ae.

H.16 RUN-H2-RADIAL-EDGE-SUPPORT-01

PREDECESSOR: RUN-H2-RADIAL-EDGE-01. PROFILE FROZEN: $u(r)=\sqrt{5/(4\pi)}r$ for $0\leq r\leq 1$ and zero outside. CHANGED OBJECT: H.13 numerical evaluator only.

EXECUTED TERMINAL STATE: REJECTED_NO_INTERIOR_MAXIMUM.

Stage A converged over the complete 601-point grid: $\max |\Delta S_{\text{total}}/k_B|=2.8954882935749993e-10$; $\max |\Delta S_{\text{field}}/k_B|=2.83725709593341e-10$; $\max |\Delta O|=4.532337283258414e-10$. No strict interior candidate exists. The entropy reaches its maximum at the zero-overlap boundary/plateau. `comparator_accessed=false`.

SEALED ARTIFACT SHA-256: 638b7d087c7f06f8b242ecb8e941a42d10bb29d8c5a79abe97729b8418bce803.

H.17 Support-aligned set disposition

IF at least one successor profile resolves a strict interior maximum:

SET_DISPOSITION=RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED.

IF all successor profiles converge and reject: SET_DISPOSITION=REJECTED_RADIAL_PROFILE_SET_01.

IF any required successor profile remains INDETERMINATE or INVALID_RUN after the declared H.13 ladder:

SET_DISPOSITION=INDETERMINATE_RADIAL_PROFILE_SET_02.

EXECUTED SET_DISPOSITION: RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED.

Meaning: under the frozen H.5 entropy law, fixed support, coefficients, and comparator isolation, changing only the static radial intensity profile changes the topology of $S_{H2}(N_r)$. The linear and quadratic profiles produce strict interior native maxima; the edge-weighted profile does not.

This disposition does not promote either resolving profile to canonical SEAM mathematics and does not establish dimensional agreement with the empirical H₂ bond. Comparator projection remains a separate post-seal operation.

H.18 Numerical-resolution workflow conclusion

The H.5 tensor evaluator remains sealed INDETERMINATE. The H.13 support-aligned successor resolves the same physical-profile question without modifying the failed runs. The workflow-level result is therefore determined by H.17 rather than by the numerical failure of H.5.

H.5 NUMERICAL INDETERMINACY -> SEAL -> H.13 SUCCESSOR -> H.17
RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED

Cold-run rule established by this sequence: evaluator terminal state is final for that evaluator; a predeclared successor evaluator continues the workflow under a new contract identity. No implicit repair and no operator intervention occurs.

Appendix I — Input Authority and Critique Protocol

Canonical mathematical authority

The definition, equation, constant, mapping, or branch rule appears in this specification and is executable as stated.

Authorized external input

An observation, evidence value, comparator, or other numerical field is external to the mathematics and enters only through an explicit run-contract field.

Evidence-artifact gap

A required evidence or reproducibility artifact is unavailable. This affects verification of that artifact only and does not alter canonical mathematics.

Implementation gap

A concrete implementation does not execute a capability required by this specification. The implementation fails its interface obligation; the mathematical definition remains unchanged.

Canonical-definition absence

A required mathematical object is absent only when it is neither stated nor generated anywhere in this specification. The active contract then emits its declared missing-object terminal state; no outside material fills the gap.

Appendix J — Full Definitions, Nomenclature, and Symbol Glossary

Run-contract binding: H.5-H.18 contain the complete radial-profile contracts, deterministic numerical successor graph, and sealed outcomes. Chapters 32-33, Appendix H, and J.0 control every named run. Noncanonical radial profiles remain noncanonical unless a formal contract explicitly promotes a fully defined constructor.

This glossary is normative for terminology and notation in this specification. Symbols used by external evidence have no authority to redefine the canonical symbols below.

Term / symbol	Canonical name	Definition and use
ESIH	Electron-Structured Interaction Hypothesis	Principles layer. The name retains “Hypothesis,” but the explicitly declared ESIH premises are the canonical axiomatic constraints of the model under test.
ESAM	Empiric Structural Atomic Model	Mathematical-formulation layer. Expresses ESIH constraints as explicit shell/configuration mathematics and selects admissible configurations by S

C

. | | SEAM | Systemic Empirical Atomic Model | Executable mathematical engine. Executes ESAM under ESIH constraints; handles projection, retained reference, comparison, closure, validation, and answer support. | | Z(n) | Atomic baseline electron count | Number of electrons to be distributed before downstream charge/isotope/state modifications. | | c_shell | Seven-shell capacity vector | $c_shell=(c_1,...,c_7)=$

$2, 8, 18, 32, 18, 32, 18$

. | | c_n | Scalar shell capacity | Capacity of shell n; never the whole capacity vector. | | N_i,n | Shell occupancy count | Number of electrons assigned to shell n for entity i. | | rho_i,n | Normalized shell occupancy | $\rho_{i,n}=N_{i,n}/c_n$. | | $\mathcal{S}_i$ | Entity shell state | Seven-component structural state $\{\rho_{i,1},...,\rho_{i,7}\}$. Reserved for the canonical shell state. | | r_ij | Pairwise relation | Distance/separation between represented entities i and j. | | chi_ijkl | Reflection/chirality parity | Derived sign of the oriented volume for four points; distinguishes mirror configurations. | | $\mathcal{R}$ | Relational state | Complete relation set $\{r_{ij},\chi_{ijkl}\}$. Reserved for relational configuration data. | | C | Configuration | Complete canonical physical configuration $C=(\{\mathcal{S}_i\},\mathcal{R})$. | | $\mathcal{A}$ | Admissible configuration set | Candidate configurations satisfying declared count, capacity, conservation, topology, and other first-principles constraints. | | S

C

| governing ESAM entropy functional

Primary entropy-first ordering functional over canonical configuration C. Explicitly bound in §4.15 as $S_{\text{config}} + \lambda_{\text{field}} S_{\text{field}} + \lambda_{\text{coupling}} S_{\text{coupling}}$.

| S_config

C

| Configuration entropy

$k_B \sum_i \sum_n \ln \text{binom}(c_n,N_{i,n})$; combinatorial entropy of admissible integer shell occupancies.

|

| $p_{\alpha}(C)$ | Normalized field probability | Fraction of total structural-field intensity in frozen spatial cell
 $V_{\alpha}; \sum p_{\alpha}=1.$ | | S_{field}

C

| Field entropy | $-k_B$
 $\sum_{\alpha} p_{\alpha} \ln p_{\alpha}$ on the frozen metrology-bound spatial partition.
|

| $O_{ij}^{\{nm\}}(C)$ | Normalized shell-overlap operator | $\sum_{\alpha} \sqrt{p_{i,n,\alpha} p_{j,m,\alpha}}$; symmetric
dimensionless overlap in

0,1

. | | S_{coupling}

C

| Coupling entropy | $k_B \sum_{i \setminus \langle j,n,m \rangle} \rho_{i,n} \rho_{j,m} O_{ij}^{\{nm\}}$; excludes self-interaction and pair double-counting.
|

| $\lambda_{\text{field}}, \lambda_{\text{coupling}}$ | Entropy operator normalizations | Run-bound coefficients frozen with
operator identity. The H₂ reference contracts explicitly use 1.0 and 0.1; no unstated values are permitted. | | C^* |
Selected configuration | $C^* = \arg \max_{\{C \in \mathcal{A}\}} S$

C

. | | configuration closure | Physical structural closure | Level-specific condition associated with the selected
admissible entropy extremum; not the downstream query-closure scalar. | | $\hat{H}_{\text{SEAM}}$

C^*

| SEAM Hamiltonian operator representation |
Downstream Hamiltonian operator representation of an already entropy-selected state C^* .
Used to describe interaction/evolution/observables without replacing S

C

as the selection principle. | | λ_k

Psi

| State-dependent effective coefficient architecture |
structural-field notation for effective coefficients determined from natural state where an
explicit non-circular extraction rule is supplied.
|

| $\mathcal{R}(X)$ | Representation constructor | Operator that constructs the represented structural object from the
permitted primitive/input. Distinct from $\mathcal{R}$. | | SC | Structural/configuration-closure operator | Applies structural
admissibility and entropy-selected configuration closure. | | TC | Transfer/trajectory operator stage | Applies
declared transfer/evolution accounting. Operator descriptions use $\hat{H}_{SEAM}$

C^*

only under an explicit contract. | | CC | Structural coherence operator | Computes Coh_struct before Manifold
comparison. | | MC | Manifold comparison operator | Compares retained projection to the active Manifold and
produces Coh_M and Δ_L . | | VC | Validation-closure operator | Applies contradiction, support, evidence record,
metrology, residual, and admissibility gates. | | Coh_struct | Structural/continuity coherence | Internal coherence
of the retained state to Manifold comparison. | | Coh_M | Manifold coherence | Coherence/alignment between a
retained projection and active Manifold reference. | | Δ_M | Manifold residual | $\Delta_M = 1 - \text{Coh}_M$ for the
normalized scalar-coherence instantiation. | | Q-ARC | Query ARCV | ARCV representation of the submitted query
and its admissibility bounds. | | A-ARC | Answer/candidate ARCV | ARCV representation of a candidate or
retained reference. | | ARCV | Finite engine projection representation | Structured projection object carrying the
components required by the active query/reference contract. | | Δ_{ARCV} | Structured ARCV discrepancy |
Tuple/record of differences between Q-ARC and candidate/reference ARCV. | | $D_Q(Q, Y)$ | Query discrepancy
scalar | $D_Q = \mathcal{N}_Q(\Delta_{ARCV})$; scalar aggregation defined by the active query contract, not by answer fitting. | |
 $\mathcal{N}_Q$ | Query normalization/aggregation operator | Deterministic contract-bound mapping from structured ARCV
discrepancy to D_Q . | | ϵ_Q | Query admissibility tolerance | Positive deterministic tolerance extracted by $\mathcal{E}_Q$
from active Q-ARC admissibility bounds; not a universal physical constant. | | $\mathcal{E}_Q$ | Tolerance extraction operator
| Maps the active Q-ARC admissibility specification to ϵ_Q . | | $\mathcal{C}_Q(Y|Q)$ | Operational/query closure coefficient |
Downstream comparison scalar in

0, 1

for candidate Y relative to query Q; distinct from physical configuration closure. | | Y_j | Generic query/reference
candidate | Reserved generic candidate symbol in ARCV/PLL contexts; avoids collision with canonical shell state
 $\mathcal{S}_i$. | | K_t | Continuum state | Persistent retained reference structure at retention cycle t. | | M_t | Manifold state
| Normalized comparison space derived from retained Continuum structure. | | Φ_f | Projection operator | Maps
retained configuration/reference structure to a question-relevant facet. | | T(X) | Physical time resolver |
 $T(X) = n_{Cs}(X)/9,192,631,770$. | | $n_{Cs}(X)$ | Cs-133 transition count | Number of canonical Cs-133 hyperfine

transition cycles associated with physical interval X. | | $\mathcal{T}$ | Transfer/transition operator symbol | Generic transfer operator where Hamiltonian representation is not specifically invoked. Reserved so T remains physical time. | | Primitive | Source class | Declared first-principles or metrological starting quantity. | | Derived | Source class | Quantity calculated only from primitives and derived state. | | Reference | Source class | Validated retained Continuum/Manifold structural object. | | Evidence | Source class | Raw external observation or measurement. | | Comparator | Source class | External value/formula used only after native result for comparison/falsification. | | Authorized external input | Input authority class | Observation, evidence, or comparator field admitted by exact run-contract role; no operator-definition authority. | | Implementation | Source class | Software tolerance/serialization/runtime setting that does not define physical law. | | CANONICAL_IN_DOCUMENT | Input status | Definition or deterministic generating equation is stated in this specification. | | AUTHORIZED_EXTERNAL_INPUT | Input status | External numerical field admitted explicitly by the active contract for observation, evidence, or comparator use. | | ABSENT_FROM_CANONICAL | Input status | Required mathematical object is neither stated nor generated in this specification; the declared terminal state fires. | | SYMBOLIC_ONLY | Terminal status | All required symbolic objects exist, but unresolved functions/constants prevent numerical closure. | | INDETERMINATE | Terminal status | An upstream primitive or mapping is undefined, so no unique symbolic/native result can be constructed. | | REJECTED | Terminal status | A complete calculation violates a declared mathematical/admissibility condition. | | I_norm | Empirical normalized ionization coordinate | Validation-adapter input only. It is observational/run-supplied and has no authority to construct C*, L_A, or another native first-principles quantity. | | W_norm | Empirical normalized width coordinate | Validation-adapter input only. Used only inside the empirical observation adapter; prohibited from native spatial-scale generation. | | P_norm | Empirical normalized polarizability/perturbability coordinate | Validation-adapter input only. Used only inside the empirical observation adapter; prohibited from native spatial-scale generation. | | Pi_norm | Empirical normalized polarizability/field-displacement coordinate | Validation-adapter input only. Used only inside the empirical observation adapter; prohibited from native spatial-scale generation. | | Q_raw^(obs) | Raw empirical shell-coherence coordinate | Q_raw^(obs)=I_norm-W_norm. Validation only. | | Q_shell^(obs) | Empirical shell-coherence coordinate | Q_shell^(obs)=max(|I_norm-W_norm|,10^-10). Validation only; not a native shell-state primitive. | | chi_e^(obs) | Empirical perturbability/divergence coordinate | chi_e^(obs)=P_norm+W_norm. Validation only. | | Xi_raw^(obs) | Raw empirical field-support coordinate | Xi_raw^(obs)=I_norm-Pi_norm. Validation only. | | Xi_F^(obs) | Empirical field-support coordinate | Xi_F^(obs)=max(I_norm-Pi_norm,10^-10). Validation only. | | a, b, q | Empirical-adapter exponents | a=0.5, b=1.0, q=1.0 for the validation-only observation adapter. | | Theta_obs | Empirical validation coordinate | Theta_obs=

$$Xi_F^{(obs)}$$

^0.5 exp(-chi_e^(obs)) Q_shell^(obs). It is downstream validation data and cannot define L_A. | | lambda_A | Atomic baseline length increment | Frozen atomic-metrology input used by L_A=N_A lambda_A. It is established upstream of molecular structure and is not set by C*, the empirical validation adapter, or a dimensional comparator. | | N_A | Atomic baseline length count | Frozen atomic-metrology count multiplying lambda_A under L_A=N_A lambda_A. It is an upstream metrology input and is not inferred from a downstream molecular target. | | tau_A | Atomic baseline interval | Atomic metrology interval satisfying tau_A=M_A/v_A. In a physical run, time remains count-resolved by the canonical Cs-133 resolver T(X)=n_Cs(X)/9,192,631,770; the closure test uses an exact synthetic fixture to verify the algebraic metrology chain without calibrating physical magnitude. | | L_A | Canonical atomic baseline length | L_A=N_A lambda_A. This baseline is generated at the atomic metrology layer before downstream structural or molecular projection and is consumed by L=N_L L_A and L_H2=N_r L_A. | | N_D(X_i,X_j) | Native spatial-distance count | N_D(X_i,X_j)=||X_j-X_i|| in the canonical native spatial coordinate. This native distance is resolved independently of dimensional projection. | | N_L | Generic native length count | Dimensionless native distance count multiplying L_A only when dimensional baseline projection is requested. | | N_r | H2 / radial

native separation count | For H2, $N_r = ||X_B - X_A||$. This is a resolved native distance count; dimensional projection, if available, is $L_{H2} = N_r L_A$. | | L | Baseline-length projection | Dimensional SEAM projection $L = N_L L_A$. Numerical evaluation requires a canonical numerical L_A . | | L_{H2} | H2 baseline-length projection | $L_{H2} = N_r L_A = N_r N_A \lambda_A$. The native H2 count N_r is consumed downstream of the atomic baseline. The executed closure test verifies this projection by exact round-trip arithmetic while the dimensional H2 evidence comparator remains sealed. | | f | Derived frequency | $f = N_T / T$ for N_T counted cycles in resolved interval T; for one cycle $f = 1/\tau$. | | v_A | Atomic reference cycle rate | Frozen atomic-metrology input used with M_A to form $\tau_A = M_A / v_A$. For a physical run it must be represented consistently with the canonical count-resolved time contract; the exact closure fixture tests only dependency and algebra. | | M_A | Atomic reference cycle count | Frozen count used with v_A to form $\tau_A = M_A / v_A$. It is established before downstream structural or molecular projection. | | V_A | Atomic baseline transfer constant | $V_A = L_A / \tau_A = (N_A \lambda_A) / (M_A / v_A)$. It is derived from the atomic metrology baseline and satisfies the exact reconstruction identity $V_A \tau_A = L_A$. |

J.o Execution-contract supremacy

For a named run, the canonical equations plus that run contract are the complete executable specification. No implicit assumption has model status. Silence is prohibition.

J.o.1 Cold-run terminal-state supremacy

Terminal state ends executor authority under the named contract. No implicit “try another method” branch exists.

After terminal state, the executor performs only declared artifact completion, transcript finalization, hashing, sealing, and stop. It does not repair, optimize, reinterpret, consult, or continue the model.

A remedy is a `FUTURE_CONTRACT_CANDIDATE` until a different immutable run contract explicitly defines it. Describing a remedy does not authorize it.

Cold-run compliance invariant:

`TERMINAL_STATE + undeclared continued computation = INVALID_RUN: POST_TERMINAL_EXECUTION`

J.o.2 Run-finality/workflow-finality separation

`RUN_TERMINAL` is final for the named run. `WORKFLOW_TERMINAL` occurs only when the sealed run has no matching predeclared successor edge or when an explicit workflow disposition is reached.

`RUN_TERMINAL != WORKFLOW_TERMINAL`

A successor edge does not reopen, repair, or mutate its predecessor. It starts a new immutable contract with a new UID and preserves the predecessor terminal record.

When a successor edge exists, the supervisor dispatches it automatically. Asking the operator whether to use the declared successor is noncompliant.

J.o.3 Numerical-resolution supremacy

A numerical evaluator that fails its convergence gate establishes only evaluator indeterminacy. Physical-profile rejection or acceptance requires a converged evaluator under the declared workflow.

If the document predeclares a numerical successor for the failed evaluator terminal code, the successor executes after sealing. No physical conclusion is emitted from the failed evaluator alone.

J.o.4 Sealed-result/open-workflow interpretation supremacy

A sealed terminal artifact is testable immediately and indefinitely. Verification does not reopen the run and does not extend its scope.

A sealed link in an unfinished workflow is reported only for what it actually establishes. Unresolved downstream stages remain uncharacterized until their own contracts reach terminal state. Absence of a contract-specific successor triggers the root-objective dependency rule; workflow stop occurs only after the root objective is complete.

The executor and cold reviewer do not infer a next contract. Continuation comes only from a predeclared successor edge. Absence of an edge means workflow stop.

J.o.5 Canonical-model composition precedence supremacy

Canonical authority is checked before every external-ingress decision. An object stated mathematically in this document is `CANONICAL_IN_DOCUMENT` and cannot be demoted by reviewer preference, symbol similarity, or absence of outside material.

`CANONICAL_IN_DOCUMENT > EXTERNAL_INGRESS_FIREWALL`

`FIREWALL_DEMOTION_OF_CANONICAL_IN_DOCUMENT -> INVALID_AUDIT:
CANONICAL_AUTHORITY_MISCLASSIFICATION`

J.o.6 External-ingress firewall supremacy

Executable admission requires explicit current-contract binding. The binding names the source identity/hash, object, consumed fields, target field, deterministic mapping, role, override status, scope, and validation condition.

Readability does not confer executability. Evidence Provenance does not confer executability. Dimensional compatibility does not confer executability. Symbol similarity does not confer executability. successful use does not confer executability.

If a required object is absent, the run terminates under its declared missing-object state. A guardrail never licenses external ingress to avoid termination.

Cold-run ingress invariant:

NOT_EXPLICITLY_ADMITTED = NOT_EXECUTABLE

UNDECLARED_EXTERNAL_INGRESS_CONSUMED = INVALID_RUN

J.1 Entropy-first operator hierarchy

$Z(n) \rightarrow \mathcal{S}_i \rightarrow C \rightarrow C^* = \arg \max_{\{C \in \mathcal{A}\}} S$

C

Operator representation is invoked only after C^* is selected for the selected-state interaction/evolution:

$C^* \rightarrow \hat{H}_{SEAM}$

C^*

$\rightarrow$ operator description / derived observable

Projection and reference comparison then remain downstream:

$C^* \rightarrow \Phi_f(C^*) \rightarrow \text{Coh_struct} \rightarrow \text{MC} \rightarrow \text{Coh_M} \rightarrow \Delta_M \rightarrow \text{VC}$

This hierarchy controls interpretation whenever Hamiltonian notation is used. Hamiltonian representation is downstream of entropy-selected configuration and cannot replace the entropy selection rule.

S

C

$= S_{\text{config}}$

C

- $\lambda_{\text{field}} S_{\text{field}}$

C

- $\lambda_{\text{coupling}} S_{\text{coupling}}$

C

The explicit entropy expansion in §4.15 is part of the controlling hierarchy. Active executions bind to those definitions unless a named noncanonical perturbation contract declares a complete alternative before execution.

J.2 Reserved-symbol rule

The following symbols are reserved in active prose: $\mathcal{C}$ for physical configuration; $\mathcal{R}$ for relational state; $\mathcal{S}_i$ for canonical shell state; $T(X)$ for physical time; $\hat{H}_{\text{SEAM}}$ for Hamiltonian operator representation; $\mathfrak{R}$ for representation construction; Δ_M for Manifold residual. The canonical meanings in this glossary control execution.

J.3 Canonical spatial-field constructor

$$\Omega_{i,n} = \{\xi: n-1 \leq ||\xi - X_i|| < n\}; u_{i,n} = 1_{\Omega_{i,n}} / \sqrt{V_n}; F_C(\xi) = \sum_{i,n} \sqrt{N_{i,n}} u_{i,n}(\xi)$$
$$q_C = |F_C|^2 / \int |F_C|^2; O_{ij}^{\{nm\}} = \text{Vol}(\Omega_{i,n} \cap \Omega_{j,m}) / \sqrt{V_n V_m}$$

These equations define the canonical uniform-shell spatial field. A different constructor requires a complete named perturbation contract declared before execution. Target evidence never retunes an executed constructor.

J.4 SEAM-to-conventional-physics boundary

SEAM structural contract $\rightarrow$ native state $\rightarrow$ projection $\rightarrow$ conventional observable

SEAM operates one representational layer upstream of the conventional observable description. A native contract change changes the state supplied to downstream projection. The conventional measurement relation remains downstream. The native prediction and projection are frozen before evidence comparison, and disagreement at the observable boundary is a failed prediction under the named run contract.

Forbidden: conventional target $\rightarrow$ retune native SEAM contract $\rightarrow$ claim prediction

Appendix K — Expanded Executed Transcript: RUN-H2-RADIAL-SUPPORT-SET-01

This appendix is an execution record of the canonical support-aligned successor run. It is not a hypothetical worked example. Every numerical result below is bound to the retained runner and sealed JSON artifacts. The purpose is to expose exactly how the declared assumptions were consumed to reach the workflow-level result without importing undeclared information.

Transcript status vocabulary: DECLARED = supplied by the controlling contract; DERIVED = calculated only from declared/derived state; EXECUTED = numerical operation authorized by the contract; PROHIBITED = value or operation unavailable to the run; OBSERVED = numerical output of an authorized operation; BRANCH = deterministic contract-directed transition; TERMINAL = named run terminal state; SEALED = serialized artifact and hash; WORKFLOW = set-level disposition after all required runs.

K.1 Evidence identity and immutability

Object	Identity / hash	Role
Executed runner	sealed artifact SHA-256 992c7cfb3e28c5a6563b3743b321059cc63b1d24cc848362208456da8a6b7677	Implements H.12-H.17 support-aligned successor contracts.
Linear artifact	b9548c8899ad446266b888c9d64f3bc010c8bca9b107df36445c6fc5b5859496	Sealed H.14 native run artifact.
Quadratic artifact	32379a4a9b4doe1d3335600369414797da462a7dfdc53627c8c762b7fd4cf4ae	Sealed H.15 native run artifact.
Edge artifact	638b7d087c7f06f8b242ecb8e941a42d10bb29d8c5a79abe97729b8418bce803	Sealed H.16 native run artifact.
Set summary	sealed artifact SHA-256 3b5bd12d2bbef9f47895eb33ff0b731f405a1c293ebae5478012bcdd79dd18b4	Sealed H.17 set disposition record.

The transcript below is subordinate to these artifacts. If a prose value conflicts with a sealed artifact, the sealed artifact controls.

K.2 Predecessor state and legal successor dispatch

K-000

DECLARED

The predecessor set H.6-H.8 was executed under the H.5 tensor-product evaluator and each run sealed INDETERMINATE: QUADRATURE_NOT_CONVERGED. H.9 therefore sealed INDETERMINATE_RADIAL_PROFILE_SET_01.

K-001

DECLARED

H.11 contains successor edge EDGE-H2-RADIAL-QUADRATURE-01. Its exact trigger is a sealed predecessor terminal state INDETERMINATE: QUADRATURE_NOT_CONVERGED.

K-002

BRANCH

Because the trigger matched, the workflow supervisor launched the matching H.14, H.15, and H.16 support-aligned successors. No predecessor artifact was reopened and no failed tensor run was repaired.

K-003

DECLARED

The successor changed one object only: the numerical integration operator. All physical/profile assumptions were carried forward frozen.

K-004

PROHIBITED

The successor was forbidden to change radial profiles, support radius, entropy terms, λ_{field} , $\lambda_{\text{coupling}}$, separation domain, search grid, optimizer rule, strict-maximum test, or comparator state.

K-005

PROHIBITED

No H₂ bond length, Bohr radius, Coulomb orbital, conventional radial distribution, fitted radial exponent, phase term, orientation term, or dimensional distance was available to the native run.

K-006

DECLARED

comparator_accessed=false was initialized as part of every successor artifact and remained false through sealing.

K.3 Frozen assumption-consumption ledger

Declared item	Value used by execution	Where consumed
Entity count	two identical H entities	H ₂ field superposition and pair overlap.
Atomic baseline	$Z_A=Z_B=1$	Shell-state model composition; not used as a fitted spatial parameter.
First-shell occupancy	$N_1=1, \rho_1=1/2$	Canonical SEAM state. No support rescaling was permitted in this set.
Support	$0 \leq r \leq 1; u(r)=0$ outside	Defines exact support-aligned geometric integration limits.
Linear profile	$u(r)=\sqrt{15/(2\pi)}(1-r)$	H.14 only.
Quadratic profile	$u(r)=\sqrt{105/(32\pi)}(1-r^2)$	H.15 only.
Edge profile	$u(r)=\sqrt{5/(4\pi)}r$	H.16 only.
Entropy coefficients	$\lambda_{\text{field}}=1.0; \lambda_{\text{coupling}}=0.1$	S_H2(d) calculation.
Separation domain	$0 \leq N_r \leq 3$	Native search domain.
Search grid	$N_r=j/200, j=0, \dots, 600$	601-point candidate scan.
Convergence gate	$ \Delta S_{\text{total}} , \Delta S_{\text{field}} , \Delta O \leq 1e-8$	Determines whether Stage A is admissible.

Declared item	Value used by execution	Where consumed
Optimizer	bounded Brent; xatol=1e-10; maxiter=500	Refines a strict grid candidate only.
Strict maximum test	h=1e-4; D ² <0; margin>1e-10; 0<N_r*<2	Determines RESOLVED_INTERIOR_MAXIMUM.
Comparator	sealed / inaccessible	No target-derived selection or tuning.

K.4 Mathematical evaluator actually executed

K-010

DECLARED

The A-centered coordinate system uses $r=||\xi-X_A||$ and $\mu=\cos\theta$ along the A-to-B axis. Separation is $d=N_r$.

$$q(r,\mu;d) = \sqrt{r^2 + d^2 - 2\,r\,d\,\mu}$$

K-011

DERIVED

The B-support condition $q\leq 1$ gives the exact angular boundary $\mu_0(r,d)$.

$$\mu_0(r,d) = (r^2 + d^2 - 1)/(2\,r\,d)$$

K-012

DECLARED

For $0<d<1$, $0\leq r\leq 1-d$ is full overlap and $1-d<r\leq 1$ is partial overlap. For $1\leq d<2$, $d-1\leq r\leq 1$ is partial overlap. For $d\geq 2$, overlap is empty.

K-013

EXECUTED

The support boundary was used as an integration boundary. The quadrature therefore never integrated one tensor interval across the compact-support discontinuity that caused the predecessor convergence failure.

$0<d<1$: I_d

g

$= 2\pi$

$$\int_0^1 (1-d)r^2 dr \int_0^1 g d\mu + \int_0^1 (1-d)^1 r^2 dr \int_0^1 (\mu_0)^1 g d\mu$$

$1 \leq d < 2$: I_d

$$g$$

$$= 2\pi \int_0^1 (d-1)^1 r^2 dr \int_0^1 (\mu_0)^1 g d\mu$$

$d=0$: I_0

$$g$$

$$= 2\pi \int_0^1 r^2 dr \int_0^1 g(r,r) d\mu; d \geq 2$$
: I_d

$$g$$

$=0$

K-014

DECLARED

The profile overlap $O(d)$, isolated-profile entropy integral J , and overlap correction $K(d)$ were evaluated from the same frozen profile $u(r)$.

$O(d) = I_d$

$$u(r)u(q)$$

$$J = 4\pi \int_0^1 r^2 u(r)^2 \ln(u(r)^2) dr$$

$K(d) = I_d$

$$(u(r) + u(q))^2 \ln((u(r) + u(q))^2) - 2u(r)^2 \ln(u(r)^2)$$

K-015

DECLARED

The endpoint convention $x \ln x = 0$ at $x=0$ was used exactly. No smoothing epsilon was inserted.

K-016

DERIVED

Because every candidate profile was normalized to unit L2 norm, direct evaluation of the retained two-entity field-intensity normalization gives exactly $2+2O(d)$; the complete field state remains operative.

$$I_2(d) = 2 + 2O(d)$$

K-017

EXECUTED

The field entropy, coupling entropy, and total H2 objective were then evaluated without any additional term.

$$S_{\text{field}}(d)/k_B = \ln(I_2(d)) -$$

$$2J + K(d)$$

$$/I_2(d)$$

$$S_{\text{coupling}}(d)/k_B = O(d)/4$$

$$S_{\text{H2}}(d)/k_B = 2 \ln 2 + S_{\text{field}}(d)/k_B + 0.025 O(d)$$

K.5 Numerical-stage transcript common to all three profiles

K-020

DECLARED

Stage A primary quadrature order =128×128 on each analytic r/μ subdomain. Verification order =192×192. J uses 128 and 192 radial nodes respectively.

K-021

EXECUTED

For every one of the 601 declared N_r grid values, the runner independently evaluated $O(d)$, $S_{\text{field}}(d)/k_B$, $S_{\text{coupling}}(d)/k_B$, and $S_{\text{total}}(d)/k_B$ at primary and verification orders.

K-022

EXECUTED

At every grid point the runner calculated $\text{residual}_S = |S_{\text{total}}^{192} - S_{\text{total}}^{128}|$, $\text{residual}_{S_{\text{field}}} = |S_{\text{field}}^{192} - S_{\text{field}}^{128}|$, and $\text{residual}_O = |O^{192} - O^{128}|$.

K-023

BRANCH

Stage A is accepted only if all required residuals stay $\leq 1e-8$. If Stage A fails, H.13 predeclares Stage B; if Stage B fails, Stage C; no operator choice is involved.

K-024

EXECUTED

After an admissible convergence stage, the 601-point verification curve was searched for strict grid candidates satisfying $S_j > S_{(j-1)}$ and $S_j > S_{(j+1)}$.

K-025

EXECUTED

Each grid candidate bracket was refined with bounded Brent optimization using the frozen tolerance and iteration cap.

K-026

EXECUTED

At each refined point x , $S(x-h)$, $S(x)$, and $S(x+h)$ were evaluated with $h=1e-4$. $D^2=$

$$S(x+h) - 2S(x) + S(x-h)$$

$/h^2$ and strict margin= $S(x)-\max(S(x-h),S(x+h))$ were calculated.

K-027

TERMINAL

A unique candidate becomes RESOLVED_INTERIOR_MAXIMUM only when optimizer_success=true, $0 < x < 2$, $D^2 < 0$, and strict margin $> 1e-10$ k_B. No relaxed or visual maximum test is permitted.

K.6 Expanded run transcript — H.14 linear center-weighted profile

L-000

DECLARED

Profile loaded exactly: $u(r)=\sqrt{15/(2\pi)}(1-r)$, $0 \leq r \leq 1$; $u=0$ outside. No profile parameter is free.

L-001

EXECUTED

The isolated-profile J integral converged: J_128=-0.6964935319693841; J_192=-0.6964935319734040; |ΔJ|=4.0199e-12.

L-002

EXECUTED

The complete 601-point Stage A scan converged. Maximum residuals over the entire scan were |ΔS_total/k_B|=5.4041260355575105e-09; |ΔS_field/k_B|=5.438363981369321e-09; |ΔO|=1.3714737123748932e-09. All are below 1e-8.

N_r	O_192	S_field/k_B	S_coupling/k_B	S_total/k_B	ΔS_total
1.595	0.0073572235463	1.39294253054	0.00183930588658	2.77942082225	3.964e-12
1.600	0.00701333333333	1.39295671929	0.00175333333333	2.77942641374	3.967e-12
1.605	0.00668130522839	1.39296248601	0.00167032630710	2.77942387976	3.969e-12

L-003

OBSERVED

The grid value at N_r=1.600 exceeded its immediate declared neighbors 1.595 and 1.605. The contract therefore admitted one and only one candidate bracket

1.595, 1.605

.

L-004

EXECUTED

Bounded Brent refinement on

1.595, 1.605

returned N_r*=1.6008992282173213 with optimizer_success=true.

L-005

OBSERVED

At N_{r^*} : $O=0.006952752080494968$; $S_{\text{field}}/k_B=1.3929583643416432$;
 $S_{\text{coupling}}/k_B=0.001738188020123742$; $S_{\text{total}}/k_B=2.779426544263546$.

L-006

EXECUTED

Primary-vs-verification S_{total} at N_{r^*} differed by $3.9675e-12$, remaining inside the $1e-8$ gate.

L-007

EXECUTED

Strict-maximum samples: $S(x-h)/k_B=2.779426542654783$; $S(x)/k_B=2.779426544263546$;
 $S(x+h)/k_B=2.779426542654634$.

L-008

DERIVED

$D^2=-0.32176754594104295 < 0$. Strict margin= $1.6087633447625649e-09$ $k_B > 1e-10$ k_B . The point is finite and interior because $0 < 1.6008992282173213 < 2$.

L-009

TERMINAL

All declared resolution conditions were true. Terminal state = RESOLVED_INTERIOR_MAXIMUM.

L-010

SEALED

sealed artifact SHA-256 `b9548c8899ad446266b888c9d64f3bc010c8bca9b107df36445c6fc5b5859496`.
`comparator_accessed=false`.

K.7 Expanded run transcript — H.15 quadratic center-weighted profile

Q-000

DECLARED

Profile loaded exactly: $u(r)=\sqrt{105/(32\pi)}(1-r^2)$, $0 \leq r \leq 1$; $u=0$ outside. No profile parameter is free.

Q-001

EXECUTED

The isolated-profile J integral converged: $J_{128}=-0.8886786210059963$; $J_{192}=-0.8886786210130339$; $|\Delta J|=7.0376e-12$.

Q-002

EXECUTED

The complete 601-point Stage A scan converged. Maximum residuals were $|\Delta S_{\text{total}}/k_B|=7.041034422172743e-12$; $|\Delta S_{\text{field}}/k_B|=7.040923399870280e-12$; $|\Delta O|=5.218048215738236e-15$.

N_r	O_192	S_field/k_B	S_coupling/k_B	S_total/k_B	\Delta S_total
1.695	0.00378668087102	1.58338481592	0.000946670217754	2.96977384406	6.996e-12
1.700	0.00355797246094	1.58339627266	0.000889493115234	2.96977958309	6.998e-12
1.705	0.00333934519409	1.58340081449	0.00083483629852 2	2.96977865924	7.001e-12

Q-003

OBSERVED

$N_r=1.700$ exceeded immediate neighbors 1.695 and 1.705. One candidate bracket

1.695, 1.705

was admitted.

Q-004

EXECUTED

Bounded Brent refinement returned $N_r^*=1.7017708805671177$ with `optimizer_success=true`.

Q-005

OBSERVED

At N_r^* : $O=0.003479400662609171$; $S_{\text{field}}/k_B=1.583398647987642$;
 $S_{\text{coupling}}/k_B=0.0008698501656522928$; $S_{\text{total}}/k_B=2.969779994124098$.

Q-006

EXECUTED

Primary-vs-verification S_{total} at N_r^* differed by $6.9997e-12$, inside the $1e-8$ convergence gate.

Q-007

EXECUTED

Strict-maximum samples: $S(x-h)/k_B=2.969779992823806$; $S(x)/k_B=2.969779994124098$;
 $S(x+h)/k_B=2.9697799928246957$.

Q-008

DERIVED

$D^2=-0.2599694681748588 < 0$. Strict margin= $1.2994023634860241e-09$ $k_B > 1e-10$ k_B . The point is finite and interior.

Q-009

TERMINAL

Terminal state = RESOLVED_INTERIOR_MAXIMUM.

Q-010

SEALED

sealed artifact SHA-256 32379a4a9b4doe1d3335600369414797da462a7dfdc53627c8c762b7fd4cf4ae.
comparator_accessed=false.

K.8 Expanded run transcript — H.16 edge-weighted profile

E-000

DECLARED

Profile loaded exactly: $u(r)=\sqrt{5/(4\pi)}r$, $0 \leq r \leq 1$; $u=0$ outside. No profile parameter is free.

E-001

EXECUTED

The isolated-profile J integral converged: J_128=-1.3215863345351968; J_192=-1.3215863345352010; |
ΔJ|=4.2188e-15.

E-002

EXECUTED

The complete 601-point Stage A scan converged. Maximum residuals were |
ΔS_total/k_B|=2.8954882935749993e-10; |ΔS_field/k_B|=2.837257095933410e-10; |
ΔO|=4.532337283258414e-10.

N_r	O_192	S_field/k_B	S_total/k_B	Execution reading
0.000	1.0000000000000	1.32158633454	2.73288069566	full overlap
0.500	0.512586805612	1.55212221732	2.95123124858	increasing
1.000	0.263888888946	1.70458505889	3.09747664223	increasing
1.500	0.100839120370	1.85030500785	3.23912034698	increasing
1.995	0.000015559993	2.01469976215	3.40099451227	approaching zero overlap
2.000	0	2.01473351510	3.40102787622	zero-overlap boundary
2.500	0	2.01473351510	3.40102787622	same plateau

E-003

OBSERVED

No verification-grid point inside $0 < N_r < 2$ satisfied the strict local candidate test. The objective increased to the zero-overlap boundary and remained constant for $d \geq 2$.

E-004

TERMINAL

Because no strict interior candidate existed, terminal state = REJECTED_NO_INTERIOR_MAXIMUM. No optimizer was invoked on a nonexistent bracket.

E-005

SEALED

sealed artifact SHA-256 638b7d087c7f06f8b242ecb8e941a42d10bb29d8c5a79abe97729b8418bce803.
comparator_accessed=false.

K.9 Set arbitration transcript

S-000

DECLARED

H.17 set rule: if at least one required successor resolves a strict interior maximum,
SET_DISPOSITION=RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED.

S-001

OBSERVED

Linear terminal = RESOLVED_INTERIOR_MAXIMUM.

S-002

OBSERVED

Quadratic terminal = RESOLVED_INTERIOR_MAXIMUM.

S-003

OBSERVED

Edge terminal = REJECTED_NO_INTERIOR_MAXIMUM.

S-004

BRANCH

The first H.17 condition was true because at least one resolving profile existed. No alternative set rule was evaluated as the selected disposition.

S-005

WORKFLOW

SET_DISPOSITION = RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED.

S-006

SEALED

sealed artifact SHA-256 3b5bd12d2bbef9f47895eb33ffob731f405a1c293ebae5478012bcdd79dd18b4.
comparator_accessed=false.

H.5 NUMERICAL INDETERMINACY -> SEAL -> H.13 SUPPORT-ALIGNED SUCCESSOR -> H.14/H.15
RESOLVED + H.16 REJECTED -> H.17 RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED

K.10 Assumption-use audit: what produced the successful workflow result

The workflow result did not arise from adding assumptions until a maximum appeared. It arose from consuming a fixed declared set in a controlled discriminator. The following causal audit is normative for reading this transcript.

Item	Used?	Effect on result
Linear/quadratic/edge radial shape	YES	The single physical discriminator. Two shapes generated interior maxima; one did not.
Support radius R=1	YES, FROZEN	Fixed spatial support for all three profiles; did not vary between runs.
$\lambda_{\text{field}}=1.0$	YES, FROZEN	Declared and frozen by the named run; weights field entropy identically across all three profiles.
$\lambda_{\text{coupling}}=0.1$	YES, FROZEN	Declared and frozen by the named run; produces 0.025 O contribution identically across all three profiles.
Support-aligned quadrature	YES, NUMERICAL ONLY	Resolved the predecessor evaluator indeterminacy without altering the physical integrals.
Comparator bond distance	NO	Unavailable before and during native execution.
Bohr radius / orbital profile	NO	Never consumed.
Coulomb law	NO	Never consumed.
ρ -dependent support scaling	NO	Explicitly frozen out of this discriminator set.
Phase / orientation	NO	Never consumed.
Fitted coefficient or radial exponent	NO	No fit parameter exists in any of the three profiles.

Therefore the successful set disposition is attributable to radial intensity shape sensitivity under the declared frozen entropy law and not to target leakage, support rescaling, coefficient tuning, numerical tolerance relaxation, or conventional orbital assumptions.

K.11 What “successful result” means in this transcript

SUCCESS does not mean that a dimensional H₂ bond length has been predicted or that the linear/quadratic profile has been promoted to canonical SEAM mathematics. The successful result is the narrower executed proposition encoded by H.17: under one frozen entropy/coupling contract and one fixed support, changing only

the static radial intensity profile changes the topology of the native H2 entropy objective, and at least one declared profile produces a strict finite interior maximum.

SUCCESSFUL WORKFLOW RESULT = RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED

The dimensional comparator remains a post-seal operation. Canonical promotion requires a separate formal determination and is not performed by this transcript.

K.12 Reproduction checklist for a cold executor

9. Load the immutable H.11-H.17 contract chain before loading any comparator.
10. Verify predecessor seals and trigger EDGE-H2-RADIAL-QUADRATURE-01 only from the exact predecessor terminal code.
11. Carry forward every H.11 frozen item unchanged and verify the changed-object ledger contains numerical evaluator only.
12. Set comparator_accessed=false and reject the run immediately if comparator data enters memory as a model input.
13. Load exactly one H.14-H.16 profile formula; derive nothing from a dimensional target.
14. Evaluate the H.12 piecewise support domains; do not integrate across the compact-support edge with an undeclared whole-domain tensor grid.
15. Evaluate O, J, K, I2, S_field, S_coupling, and S_H2 exactly as H.13 defines them.
16. Execute Stage A at 128/192 and apply all three 1e-8 convergence gates at every required point.
17. Use declared Stage B/C only if a declared stage fails. Do not invent another numerical method.
18. Scan the fixed 601-point grid and admit only strict grid candidates.
19. Refine only admitted brackets using the declared bounded Brent optimizer.
20. Apply the exact h=1e-4 second-difference and strict-margin tests.
21. Assign the named terminal state mechanically from the test outputs.
22. Serialize every required artifact, record comparator_accessed, hash, and seal before set arbitration.
23. After all three required runs are sealed, apply H.17 exactly and seal the set disposition.

K.13 Transcript closure

This appendix closes the documentary gap between “the contract exists” and “the run was performed correctly.” It records the complete authorized reasoning path from predecessor numerical indeterminacy to the sealed workflow result. Future cold-run transcripts must preserve the same distinction among declared assumption, derived quantity, authorized operation, observed result, branch rule, terminal state, and seal.

Appendix L — External Ingress Firewall and Cold-Run Guardrail Supremacy

L.1 Controlling cold-run rule

Before the cold-run decision matrix evaluates external ingress, it checks this specification.

CANONICAL_IN_DOCUMENT objects enter directly; external objects remain nonauthoritative unless a run contract admits exact numerical fields.

PRECHECK_o = CANONICAL_AUTHORITY_CHECK

PRECHECK_o failure or ambiguity -> EXTERNAL_NONAUTHORITATIVE; do not silently classify the object as external.

The executor performs only the active immutable contract and declared successor graph. The reviewer verifies only sealed artifacts. Neither role completes omitted mathematics from context.

DECLARE -> FREEZE -> EXECUTE -> TERMINAL -> COMPLETE_ARTIFACTS -> HASH -> SEAL

SEALED + MATCHING_SUCCESSOR_EDGE -> DISPATCH_NEW_CONTRACT

SEALED + UNRESOLVED_REQUIRED_OBJECTS -> DISPATCH_DECLARED_DEPENDENCY_EXHAUSTION

SEALED + ROOT_OBJECTIVE_COMPLETE + NO_SUCCESSOR_EDGE -> WORKFLOW_STOP

L.2 Verification permission and interpretation limit

After a terminal artifact is sealed, independent reproduction is authorized. The reproducer can recompute equations, rerun the declared numerical method, verify hashes, inspect assumption consumption, and test terminal predicates.

Verification does not add a new model branch. A verifier does not repair, complete, reinterpret, or continue the sealed run.

A sealed result inside an open workflow remains a valid result at its own layer. It does not create a result for any downstream layer that has not itself executed.

During open-workflow execution, the only permitted commentary is execution-state reporting. Scientific characterization of unresolved downstream objects is prohibited. A dependency-terminal state is reported and sealed, then the supervisor continues through the predeclared dependency-exhaustion edge without requesting operator permission.

L.3 External-ingress definition

EXTERNAL_INGRESS is consumption of any object not mathematically defined in this specification. Such an object is non-executable unless the active run contract explicitly admits named fields as observation, comparator, or evidence input.

- excluded or struck-through equations;
- external aliases and similarly named symbols;
- formulas from outside this self-contained specification;
- development notes, conversation commentary, reviewer proposals, and inferred mappings;
- conventional-physics equations or constants used to fill a native SEAM field;
- old metrology proposals or external-equivalence maps;
- software/library defaults not frozen by contract;
- comparator or held-out evidence before the declared reveal event.

L.4 Required ingress-admission record

- INGRESS_SOURCE_ID and source identity/hash;
- OBJECT_ID or exact symbol;
- SOURCE_STATUS_CLASS;
- CONSUMED_FIELDS;
- TARGET_CONTRACT_FIELD;
- DETERMINISTIC_MAPPING_OR_TRANSFORM;
- ROLE_CLASS: primitive, derived, reference, implementation, external, comparator, or explicitly declared perturbation;
- OVERRIDE_STATUS and exact object overridden, if any;
- PERMITTED_SCOPE;
- ANTI_CIRCULARITY_STATUS;
- VALIDATION_CONDITION.

An incomplete ingress-admission record has no partial authority.

MISSING_INGRESS_FIELD -> EXTERNAL_INGRESS_REJECTED

EXTERNAL_INGRESS_REJECTED + CONSUMPTION -> INVALID_RUN:
UNDECLARED_EXTERNAL_INGRESS

L.5 Guardrail precedence

The following precedence controls cold execution. Lower layers never repair or populate higher layers:

1. Active immutable run contract, including its explicitly declared perturbation or override fields.
2. Active canonical baseline for every object not explicitly changed by the run contract.
3. Frozen successor graph for control-flow continuation only.
4. Sealed predecessor fields explicitly canonical by the successor contract; all other predecessor content remains read-only evidence.
5. Named Continuum/Manifold/reference artifacts, limited to fields declared by the active contract.
6. Declared-reference and evidence record material: read-only, zero executable authority without ingress admission.
7. Comparator and external evidence: inaccessible until the contract reveal event and never retroactive inputs to native execution.

A conflict is resolved by the higher-precedence active specification. An omission is not resolved by falling to a lower-precedence layer; the omission triggers the contract terminal rule.

L.6 Symbol and role non-equivalence rule

Names do not establish identity. Similar symbols, dimensions, domains, or external purposes do not create a mapping.

`SYMBOL_SIMILARITY != CANONICAL_BINDING`

`DIMENSIONAL_COMPATIBILITY != CANONICAL_BINDING`

`EXTERNAL_USE != CANONICAL_BINDING`

Similar symbol names, dimensions, or domains do not establish identity. A canonical universal quantity-class rule is a deterministic mapping and therefore binds every object of the declared type without requiring an additional symbol-level identity.

L.6.1 Universal quantity-class rule

A canonical universal rule applies to every active object that satisfies its declared type unless the active contract explicitly overrides or excludes that object. The executor does not demand a redundant object-specific identity after type membership and the universal map are explicit.

`TYPE_MEMBERSHIP + UNIVERSAL_TYPE_MAP -> CANONICAL_BINDING`

`SYMBOL_SIMILARITY alone -> NO_BINDING`

For spatial metrology, the canonical rule $L=N_L\ L_A$ applies to every operational native distance count. If an active contract defines $N_D=||\Delta X||$ as a native spatial-distance count, then its baseline projection is $L=N_D\ L_A$ by the universal rule. Requiring an additional sentence equating that N_D to an externally named reach-count

symbol is a redundant-binding error.

REDUNDANT_OBJECT_SPECIFIC_BINDING_REQUIRED_AFTER_UNIVERSAL_MAP ->
INVALID_CONTRACT_TEST

This rule does not authorize conventional-unit projection, comparator access, or import of an external object. It only prevents a canonical universal map from being disabled by a redundant symbol-level admission test.

L.7 Guardrails are restrictive, never generative

A guardrail states what execution must possess or must not consume. It does not generate the missing object. The phrases “metrology required,” “complete state required,” “Hamiltonian binding required,” “reference required,” and “comparator required after seal” do not import any external or conventional implementation of those objects.

REQUIRED_OBJECT_ABSENT -> DECLARED_TERMINAL_STATE

REQUIRED_OBJECT_ABSENT + EXTERNAL_SUBSTITUTION -> INVALID_RUN

L.8 Cold-reviewer behavior

- Verify sealed result: permitted.
- Reproduce sealed result under the same contract: permitted.
- Report exactly what the sealed result establishes: permitted.
- Report active execution state for an unfinished workflow: permitted.
- Infer an unresolved downstream result: prohibited.
- Recommend a model change during an executing run contract: prohibited as part of the run record.
- Infer an undeclared scientific successor after root-objective closure: prohibited. Mechanical dependency exhaustion under §32.7 is already declared and is not reviewer inference.
- Use reviewer commentary as an execution input: prohibited.

L.9 Cold-run decision matrix

SEALED_RESULT + verification request -> verify and report within sealed scope.

SEALED_RESULT + declared downstream successor -> report sealed scope only; successor is adjudicated under its own contract.

ACTIVE_RUN -> execute contract; report execution state only.

RUN_TERMINAL + declared successor -> seal predecessor; dispatch successor automatically.

RUN_TERMINAL + no contract-specific successor -> seal; evaluate §32.7 root-objective dependency rule. Stop only if the root objective is complete.

MISSING_CURRENT_OBJECT -> fire declared missing-object terminal state.

EXTERNAL_OBJECT_FOUND -> keep read-only unless complete ingress-admission record exists.

UNDECLARED_EXTERNAL_OBJECT_CONSUMED -> INVALID_RUN:

UNDECLARED_EXTERNAL_INGRESS.

L.10 Normative supremacy statement

For every cold run, this specification and the active contract graph control execution and continuation. Sealed artifacts control what can be verified and reported. External material controls nothing unless the active contract explicitly admits it as observation, comparator, or evidence. Reviewer commentary controls neither execution nor continuation.

CONTRACT_GRAPH -> EXECUTION

SEALED_ARTIFACT -> VERIFICATION_AND_SCOPED_REPORTING

EXTERNAL_MATERIAL -> READ_ONLY_UNLESS_CONTRACT_ADMITS

COMPARATOR -> POST_SEAL_COMPARISON_ONLY

REVIEWER_COMMENTARY -> ZERO_EXECUTABLE_AUTHORITY

L.11 Cold-executor reproduction checklist

24. Load this specification and the immutable named run contract before reading comparator or evidence material.
25. Construct EXECUTABLE_STATE only from the canonical baseline, declared contract delta, and predecessor fields explicitly carried by the declared successor edge.
26. Reject every attempted external ingress without a complete ingress-admission record.
27. Execute the contract until its first declared terminal predicate fires.
28. Complete only declared artifacts, hash, and seal.
29. Verify and report the sealed result only within its scope.
30. If a matching frozen successor edge exists, dispatch it mechanically under a new UID.
31. If no contract-specific successor exists, evaluate the globally declared dependency-exhaustion edge. Stop without proposing an undeclared scientific continuation only after the root objective is complete.
32. Do not characterize unresolved downstream layers.
33. Do not use external, conventional, comparator, or reviewer content to fill a missing field.

L.12 Root-terminal supremacy over child-terminal stop

The cold-run STOP rule is evaluated at root-objective scope. A valid child terminal remains final for that child contract, but it does not terminate the root workflow when its own artifact names required unresolved dependencies.

SYMBOLIC_ONLY is therefore not synonymous with WORKFLOW_STOP. If the root objective requires numerical closure and the sealed child artifact names the unresolved factors, the supervisor must dispatch dependency exhaustion.

The executor cannot use “no successor edge” as a loophole to stop before a declared dependency chain has been exhausted. Dependency exhaustion remains mandatory when a run artifact actually names unresolved required objects; the metrology test in Appendix N does not create an unresolved baseline dependency because the atomic baseline equations are explicit.

The executor cannot use dependency exhaustion to broaden the task. It follows only equations and dependencies stated in this specification for the named unresolved objects. An absent authorized run input yields ROOT_SYMBOLIC_ONLY only when the contract explicitly declares that input. An undefined required native primitive or mapping yields ROOT_INDETERMINATE. Neither state authorizes improvised calibration or comparator-derived substitution.

Appendix M — Canonical Atomic Metrology and Universal Native-Distance Binding

M.1 Native spatial-distance law

The native coordinate geometry defines distance before dimensionalization. For any represented entities i and j:

$$N_D(X_i, X_j) = ||X_j - X_i||$$

For H₂, $N_D = N_r = ||X_B - X_A||$. This count is a complete native SEAM distance result.

M.2 Canonical atomic metrology law

The atomic baseline is constructed before C* is selected. Its primitive tuple is { λ_A , N_A , v_A , M_A }.

$$L_A = N_A \lambda_A$$

$$\tau_A = M_A / v_A$$

$$V_A = L_A / \tau_A = (N_A \lambda_A) / (M_A / v_A)$$

The dependency direction is atomic metrology $\rightarrow \{L_A, \tau_A, V_A\} \rightarrow$ native spatial projection. The molecular configuration consumes the baseline and does not regenerate it.

M.3 Universal dimensional projection

Every native distance count projects through the already-frozen atomic baseline:

$$L = N_D L_A$$

$$L_{H2} = N_r L_A = N_r N_A \lambda_A$$

The molecular configuration supplies N_r ; it does not supply or retune L_A .

M.4 Empirical adapter firewall

The empirical validation adapter in §4.16.3 is downstream-only. Its output Θ_{obs} can be compared with observations or used in explicitly declared validation studies, but it has zero authority to construct λ_A , N_A , v_A , M_A , L_A , τ_A , V_A , N_r , C^* , or the entropy extremum.

EMPIRICAL_ADAPTER -> VALIDATION_ONLY; EMPIRICAL_ADAPTER -/-> ATOMIC_BASELINE

M.5 Spatial-metrology execution contract

CONTRACT_ID = RUN-SEAM-NATIVE-ATOMIC-METROLOGY-CLOSURE-01

ROOT_OBJECTIVE_ID = ATOMIC_METROLOGY_EXECUTABLE_CLOSURE

The contract freezes the atomic primitive fixture before execution, computes the baseline quantities, verifies exact reconstruction and inversion, then permits the previously sealed H₂ native N_r^* to consume L_A . The dimensional H₂ comparator remains sealed.

M.6 Mandatory execution order

1. Freeze and hash $\{\lambda_A, N_A, v_A, M_A\}$.
2. Compute $L_A = N_A \lambda_A$.
3. Compute $\tau_A = M_A / v_A$.
4. Compute $V_A = L_A / \tau_A$.
5. Verify $V_A \tau_A = L_A$, $L_A / N_A = \lambda_A$, and $M_A / \tau_A = v_A$.
6. Only after the baseline is frozen, consume the sealed native H₂ N_r^* .
7. Compute $L_{H2} = N_r^* L_A$ and verify the round-trip $N_r^* = L_{H2} / L_A$.
8. Evaluate the Cs count checkpoint $T = n_{\text{Cs}} / 9,192,631,770$.
9. Verify empirical-adapter access=false, dimensional-H₂-comparator access=false, and downstream molecular selected-state data used to construct or tune the atomic baseline=false.
10. Seal the artifact and hashes.

M.7 Pass/fail predicate

PASS requires every exact arithmetic identity and every ingress prohibition in M.6 to hold. Any failed equality, hash mismatch, empirical backflow, comparator ingress, or use of downstream molecular selected-state data to construct or tune the atomic baseline yields FAIL: CLOSURE_INVARIANT_VIOLATION.

M.8 Executed disposition

Appendix N records the completed execution. Terminal state: PASS:
NATIVE_ATOMIC_METROLOGY_CHAIN_NUMERICALLY_CLOSED.

Appendix N — Executed Native Atomic Metrology Closure Test — PASS

N.1 Test objective

Demonstrate by executable exact arithmetic that the canonical atomic primitive tuple generates L_A , τ_A , and V_A upstream of C^* , and that a sealed native H2 separation count projects through $L_{H2}=N_r L_A$ without consuming an empirical validation adapter or dimensional H2 comparator.

N.2 Frozen canonical equations

$$L_A = N_A \lambda_A$$

$$\tau_A = M_A / v_A$$

$$V_A = L_A / \tau_A$$

$$L = N_L L_A$$

$$L_{H2} = N_r L_A$$

$$T(X) = n_{Cs}(X) / 9,192,631,770$$

The test fixture is deliberately exact-arithmetic and synthetic at the atomic length/time scale. Its purpose is to test executable closure and dependency direction, not to calibrate a physical atomic length.

N.3 Contract freeze

CONTRACT_ID = RUN-SEAM-NATIVE-ATOMIC-METROLOGY-CLOSURE-01

CONTRACT_SHA256 = bf246cce3d09a23b85d9bcb54b3b69a3e2f8c2744da7c1c1feb0bfba98c4b86b

FROZEN_INPUT_SHA256 = bec4coe1769b7517f75219dbeb8a5e96dc7dcbe5b0c86616e951c0ca0c46e939

Frozen fixture: $\lambda_A=3/7$ fixture_length; $N_A=14$; $v_A=11$ cycles/fixture_time; $M_A=22$ cycles. The previously sealed linear-profile H2 discriminator result $N_r^*=1.6008992282$ is downstream input and is not used until after L_A is computed.

Prohibited inputs: dimensional H2 bond length, Bohr radius, I_{norm} , W_{norm} , P_{norm} , Π_{norm} , Θ_{obs} , and any downstream molecular target or selected-state value used to construct or tune the atomic baseline.

N.4 Executed transcript

01 FREEZE input fixture — PASS

02 $L_A=N_A \lambda_A = 14 \times 3/7 = 6$ fixture_length — PASS

03 $\tau_A=M_A/v_A = 22/11 = 2$ fixture_time — PASS

04 $V_A=L_A/\tau_A = 6/2 = 3$ fixture_length/fixture_time — PASS

05 $V_A \tau_A \rightarrow L_A = 3 \times 2 = 6$ — PASS

06 recover $\lambda_A=L_A/N_A = 6/14 = 3/7$ — PASS

07 recover $v_A=M_A/\tau_A = 22/2 = 11$ — PASS

08 consume sealed $N_r^*=1.6008992282$ only after baseline construction — PASS

09 $L_{H2}=N_r^*L_A = 1.6008992282 \times 6 = 9.6053953692$ fixture_length — PASS

10 round-trip $N_r=L_{H2}/L_A = 9.6053953692/6 = 1.6008992282$ — PASS

11 Cs checkpoint $T=9,192,631,770/9,192,631,770 = 1$ — PASS

12 empirical validation adapter consumed=false — PASS

13 dimensional H2 comparator accessed=false — PASS

14 molecular selected-state used to construct or tune atomic baseline=false — PASS

TERMINAL = PASS: NATIVE_ATOMIC_METROLOGY_CHAIN_NUMERICALLY_CLOSED

N.5 Result ledger

Check	Expected	Observed	Status
L_A construction	6	6	PASS
τ_A construction	2	2	PASS
V_A construction	3	3	PASS
$V_A \tau_A$ reconstruction	L_A	$6 = L_A$	PASS
λ_A inversion	$3/7$	$3/7$	PASS
v_A inversion	11	11	PASS

Check	Expected	Observed	Status
H2 projection round-trip	1.6008992282	1.6008992282	PASS
Cs time checkpoint	1	1	PASS
Empirical adapter consumed	false	false	PASS
Dimensional H2 comparator accessed	false	false	PASS
Molecular selected-state used to construct/tune atomic baseline	false	false	PASS

N.6 Narrative interpretation

The passing test establishes the dependency direction of the canonical spatial metrology. L_A is not generated by the selected molecular configuration. It is generated upstream from the frozen atomic metrology primitives through $L_A = N_A \lambda_A$. The selected molecular state contributes its native spatial count N_r , and the universal projection multiplies that count by the already-existing baseline.

Because L_A was constructed and verified before N_r^* was consumed, downstream molecular selected-state data did not participate in baseline construction or tuning. The exact reconstruction $V_A \tau_A = L_A$ and the inverse recoveries $L_A/N_A = \lambda_A$ and $M_A/\tau_A = v_A$ demonstrate that the baseline tuple is algebraically closed and reversible under the declared equations.

The H2 portion then exercised the interface rather than redefining it: the sealed native count $N_r^* = 1.6008992282$ projected to 9.6053953692 fixture-length units and returned exactly to the same N_r^* when divided by L_A . This verifies the quantity-class rule $L_{H2} = N_r L_A$ as an executable downstream map.

The pass therefore establishes the required dependency direction directly: the atomic baseline is a metrology construction that exists before molecular configuration selection. A molecular calculation consumes the baseline; it does not construct, recreate, or tune it.

The result does not claim that the synthetic fixture value $L_A = 6$ is the physical atomic baseline length, and it does not compare 9.6053953692 fixture-length units with the measured H2 bond length. Those are separate empirical-calibration/comparison questions and remain outside this test by contract.

N.7 Offline evidence checkpoint

EVID-CS-001 — Bureau International des Poids et Mesures (BIPM), “SI base unit: second (s).” The BIPM fixes the unperturbed ground-state hyperfine transition frequency of caesium-133 at exactly 9,192,631,770 Hz. This source is used only to check the Cs count checkpoint $T = n_{Cs}/9,192,631,770$; it does not define or calibrate λ_A , N_A , M_A , L_A , V_A , N_r , or the H2 projection.

Independent source check: <https://www.bipm.org/en/si-base-units/second>

N.8 Artifact record

The reproducibility package contains contract.json, input_fixture.json, run_native_atomic_metrology_closure.py, result.json, transcript.txt, console.txt, OFFLINE_EVIDENCE.md, README.md, and manifest_sha256.txt. The package is hash-sealed separately from this document.

Result JSON SHA256 = 4eca6977baeadf04325869a05caf93e98deccc725a86601f832fd9729575a836

Transcript SHA256 = ba2fcf3a52d2d4c9da10da2141fcb07849a99d705ef9276064082f24c128cc8b

Package SHA256 = 0334419221bcae01e8b4c6c8cbb6365d3ca309e1aca63346ee9c50bed973b4of

N.9 Canonical disposition

PASS: NATIVE_ATOMIC_METROLOGY_CHAIN_NUMERICALLY_CLOSED

The passing disposition is evidence for executable closure of the canonical atomic metrology chain and the downstream native-distance projection. It is not evidence of empirical H₂ dimensional agreement until a separate contract freezes a physical atomic baseline and authorizes comparator reveal.

Appendix O — Current Canonical Completion Reconciliation

This appendix binds the current canonical results into the same mathematical and execution authority. The equations stated here are controlling.

O.1 Signed shell-field operator

For an atomic shell occupancy with total electron count E , shell occupancies N_n , and shell support volumes V_n , the canonical field entropy admits the exact decomposition:

For the disjoint-support single-atom case, direct evaluation of the retained field entropy admits the exact analytical evaluation identity:

$$S_{\text{field}}[C]/k_B = \ln E - (1/E) \sum_n N_n \ln(N_n/V_n).$$

Define the causal/backward shell-response operator explicitly as

$O(Z) = S_{\text{field}}(Z) - 2S_{\text{field}}(Z - 1) + S_{\text{field}}(Z - 2)$. Over the complete architectural domain $Z = 3..128$, 126 states are evaluable and execution gives zero sign violations. The current empirical-comparison subset $Z = 3..118$ contains 116 evaluable states, also with zero violations. The durable invariant is the operator-level second-difference sign structure; a fitted fixed scalar or subshell-offset scalar is not the canonical invariant.

O.2 Set-valued neutron and mass state

Electron-structural resolution does not require scalar collapse of a nuclear stability coordinate. Where neutron support is an admissible interval, the native result remains set-valued:

$N_Z =$

N_{low}, N_{high}

.

Empirical mass adjudication preserves that coordinate: $M = M(Q_{res}, N)$. The comparator is a set join over admitted nuclide records, not a forced scalar mapping from Q_{res} to one mass.

The active nuclear formation authority is the frozen $\Gamma_{Z,N}$ construction. Its certified execution supplies the current formation envelope and associated outer selectors.

O.3 Entropy-resolution state evolution

The physical evolution loop remains distinct from downstream operational validation:

$C_n \rightarrow Xi_n \rightarrow \Delta C_n \rightarrow C_{\{n+1\}} \rightarrow Xi_{\{n+1\}} \rightarrow$ entropy-directed evolution.

Xi is the current entropy-resolution state of the realized structure. The governing selector remains $C^* = \arg \max_{\{C \in A\}} S$

C

. The complete operational closure operator audits representation, structural closure, transfer, coherence, Manifold comparison, and validation without replacing the native evolution that produced the state.

O.4 Macroscopic interaction, finite-body aggregation, and Cavendish projection

The macroscopic attraction branch is a body-scale continuation of the same Continuum/SEAM structural law, but its executable source operator is **not inferred from the v18.6 two-body molecular Hamiltonian merely because both branches use interaction language or share symbols**. The active macroscopic chain is retained in its own native form:

```
resolved constituent / body composition
-> retained body structural source
-> S_A, S_B (or S_M for one source body)
-> macroscopic attraction law
-> finite-body / continuum geometry
```

-> apparatus projection
 -> comparator only after native result freeze

A molecular selected-state result such as C_{XY*} , $S * (N_r)$, ΔH_{XY} , or a τ_{SEAM} pair mapping enters this branch only if an explicit canonical binding states that it is the constructor for the macroscopic source term. Shared notation alone is not such a binding.

O.4.1 Recovered macroscopic structural-source constructor

The macroscopic source is constructed from the retained body state before any conventional-unit or apparatus projection. The recovered body-level aggregation operator is

$$\mathfrak{A}(B) = \sum_{a \in B} \mu_a - \mathcal{C}_{\text{internal}}(B), \quad S_M(B) \equiv \mathfrak{A}(B).$$

For two bodies, $S_A = A(A)$ and $S_B = A(B)$. μ_a is the retained primitive field contribution of constituent a ; $\mathcal{C}_{\text{internal}}(B)$ is retained internal electric/magnetic cancellation. Neither is defined by conventional gravitational mass.

The native spatial reconstruction uses the native coordinate $\mathbf{x}_i$ and a source density satisfying

$$S_M(B) = \int_{\Omega_B^{(s)}} \sigma_B^{(s)}(\boldsymbol{\xi}) d^3\xi.$$

The continuum/reporting representation

$$\sigma_B^{(obs)}(\mathbf{x}) = \rho_{m,B}(\mathbf{x}) \Xi_B(\mathbf{x})$$

is retained only as a downstream projection for comparison with conventional material descriptions. It does not define the native source in a SEAM calculation. Likewise, the representative Ξ expansion records candidate structural coordinates; its coefficients have no authority inside the native interaction derivation unless independently bound upstream.

The recovered aggregation record fixes the same-state accounting rule

$$\mathfrak{A}(cB) \approx c\mathfrak{A}(B)$$

and the native body-interaction interface

$$\mathfrak{F}_{12}(N_R) = \mathcal{I}_{N_R}[\mathfrak{A}(B_1), \mathfrak{A}(B_2)].$$

Universal raw constituent count is not itself the source constructor; the retained primitive contribution and internal cancellation remain part of the source state.

O.4.2 Finite-body aggregation

For two finite bodies, the native discrete interaction is

$$\mathfrak{F}_{A \leftarrow B} = \sum_{i \in A} \sum_{j \in B} \mathfrak{f}_{ij}.$$

This is a mathematical definition, not a requirement for brute-force enumeration. When a continuum representation is useful, the exact same native aggregate is written

$$\mathfrak{F}_{A \leftarrow B} = \iint \sigma_A^{(S)}(\boldsymbol{\xi}) \sigma_B^{(S)}(\boldsymbol{\eta}) \mathbf{k}_S(\boldsymbol{\xi}, \boldsymbol{\eta}) d^3\eta d^3\xi.$$

For a spherical exterior source, using the resolved native separation count N_R ,

$$\Lambda(N_R) \propto -\frac{S_M}{N_R}$$

and therefore

$$-\frac{d\Lambda}{dN_R} \propto -\frac{S_M}{N_R^2}.$$

For two retained body sources, the native long-range response is the aggregate of their already-resolved constituent/pair responses:

$$\mathfrak{F}_{A \leftarrow B}(N_R) = \sum_{i \in A} \sum_{j \in B} \mathfrak{f}_{ij}(N_{R,ij}).$$

In the homogeneous exterior limit, that aggregate retains the established source-population product and N_R^{-2} radial dependence. Its magnitude is inherited from the resolved constituent/pair response. No independent body-scale amplitude is introduced.

O.4.3 Torsion-apparatus projection

Torsion geometry is a **post-native projection**. First compute and freeze the native distributed interaction response. Then a separate apparatus adapter may map the frozen response and retained geometry to a conventional torque/displacement observable.

No torsion-balance target, conventional lever-arm unit, measured force, target acceleration, or conventional gravitational constant may enter the native source/field calculation.

O.4.4 Blind mathematical Cavendish execution contract

A blind SEAM Cavendish run follows this order:

1. Freeze the external apparatus/body descriptions as ingress evidence.
2. Translate them to retained constituent states and native geometry/count coordinates; seal the ingress translation.
3. Construct S_A , S_B , and any required native source-density representation.
4. Evaluate $\mathfrak{F}(N_R)$ and finite-body aggregation entirely in native coordinates.
5. Freeze the complete native SEAM result.
6. Only after freeze, apply an apparatus/conventional-unit adapter and reveal the experimental comparator.

The native run fails the contract if a target force, `G`, meter-valued distance, newton-valued force, conventional acceleration, or measured torque is used to construct an upstream native result.

The contract also prohibits silently substituting the v18.6 molecular branch for the macroscopic body-source constructor. `TAU_PAIR_MAPPING_UNBOUND` remains a molecular-branch boundary only.

O.4.5 Operator-boundary and continuity rule

The following remain distinct mathematical representations unless an explicit canonical relation binds them:

- a complete atomic selected state;
- a molecular selected-state/Hamiltonian representation;
- the retained per-constituent native field/interaction response;
- the macroscopic body aggregate;
- downstream conventional reporting.

The rule prevents **substitution**, not **continuation**. Where SEAM has already established the native constituent field response and its atomic→molecular equivalence, that resolved response is the aggregation unit carried to larger populations. The macroscopic calculation does not reopen the constituent amplitude and does not introduce a new macroscopic fitted coefficient merely because the population is larger.

Accordingly, the valid scale chain is

```
resolved atomic constituent state
-> resolved native constituent field / pair response
-> atomic-to-molecular equivalence
-> repeated equivalent-state aggregation
-> finite-body geometry in native coordinates
-> native time-space evolution
-> freeze
-> optional conventional projection/comparison
```

A molecular quantity that is not part of that declared equivalence may not be substituted into the body source merely because it shares notation. Conversely, an already-resolved constituent response may not be relabelled as "unbound" at macroscopic scale when the body construction is exactly its repeated aggregation.

O.4.5.1 Retained Fe–Fe molecular result

The v18.6 Fe–Fe molecular result remains a valid molecular-branch result. Under `VF-REGION-LBFGSB-R4`, Fe uses $N = [2, 8, 16]$, $c = [2, 8, 18]$, with $S_{\text{inf}}/k_B = 15.455879927694806$; the reproduced Δs values are negative inside overlap and reach zero at $N_r = 6.00$. This result is not used as a replacement for the native macroscopic field constructor.

O.4.6 Native constituent equivalence and arbitrary-population lift

Let `x` denote a resolved constituent type. A macroscopic body is a retained population of constituent states, not a new separate gravitational primitive. The recovered body constructor remains

$$\mathfrak{A}(B) = \sum_{a \in B} \mu_a - \mathcal{C}_{\text{internal}}(B).$$

For homogeneous repeated state, the body construction is extensive under the closed native constituent interaction constructor. For mixtures, the complete retained composition remains explicit. The constituent contribution and internal-cancellation accounting are populated directly from the retained atomic/pair state during execution; they are not separate laws awaiting external binding.

The radial pair law is declared by the attraction-Hamiltonian branch in O.4.8. Native magnitude is inherited from the resolved pair consequence defined by the two-stage construction in O.4.8.0A; no third attraction-factor binding is inserted. μ_a and equivalent per-constituent quantities are retained state products used by aggregation, not observation-specific free parameters.

No kilogram, Newtonian `G`, fitted $\mathcal{G}_S$, SI force, comparator value, or regression is permitted to define the native constructor.

O.4.7 Native time-space ratio continuation

SEAM distance and evolution remain count-resolved. Native radial separation is N_R ; physical ordering is carried by the canonical transition-count time resolver. Ratios therefore propagate without regression or refitting:

$$\frac{R_2}{R_1} = \frac{N_{R,2}}{N_{R,1}}, \quad \frac{\Delta t_2}{\Delta t_1} = \frac{n_{Cs,2}}{n_{Cs,1}}.$$

A conventional specimen description may be used only at ingress to identify a retained constituent population. It is not a causal input to the native interaction derivation.

O.4.8 Native long-range attraction Hamiltonian and observational gravity projection

O.4.8.o Operator authority and radial-law scope

For the long-range attraction branch, the radial factor $\text{eta}(N_R)/N_R$ is part of the **declared native attraction Hamiltonian itself**. It is not imported from the v18.6 molecular variational branch, and the proof obligation for this branch is not to rediscover its radial factor by executing a different operator.

The canonical scope split is:

```
v18.6 variational operator -> molecular selected-state / finite-support branch
H_attr                    -> native long-range attraction branch
finite-body/orbital       -> continuation of H_attr
derived structural attraction -> finite-body exterior response -> orbital continuation
gravity / Newton / GR labels -> downstream observational and mathematical representations
```

Consequently, the compact-support terminal of v18.6 does not modify, null, supersede, or adjudicate the exterior behavior of H_{attr} . Conversely, H_{attr} is not imported into the v18.6 optimizer. The operators coexist under distinct declared scopes. An executor must use the operator declared for the requested branch and may not replace it with the other merely because both describe interactions.

The R24 no-reduction runtime performs an explicit authority audit against this section before executing the long-range attraction law.

The long-range attraction branch is governed by the declared native attraction Hamiltonian. The macroscopic source is constructed from the retained attraction state and is not defined by a unit-amplitude surrogate.

For resolved constituents **a** and **b** at native separation $N_{R,ab}$, the attraction Hamiltonian is

$$H_{\text{attr}} = - \sum_{a,b} E_{ab} g_{\Theta,ab} \frac{\eta_{ab}(N_{R,ab})}{N_{R,ab}} \mathcal{S}_{ab}.$$

The factors are native state quantities: E_{ab} is the state-derived pair-energy magnitude, $g_{\Theta,ab}$ is the declared structural/orientation transfer factor, $\mathcal{S}_{ab}$ is the retained pair structural factor, and eta_{ab} is the short-range regularizer of the attraction branch. None is Newton's G , an SI calibration, or a body-scale fitted macroscopic coefficient.

O.4.8.oA Two-stage resolved-state interaction construction

The native interaction is evaluated in two stages. First, each constituent is resolved independently into its complete retained state:

$$C_a \rightarrow \Sigma_a, \quad C_b \rightarrow \Sigma_b.$$

Second, the already-resolved states are evaluated together at native separation N_R :

$$\boxed{(\Sigma_a, \Sigma_b, N_R) \rightarrow C_{ab}^*(N_R).}$$

For the active shell-constrained variational realization, the pair configuration produces

$$\{u_{i,n}^*(N_R)\} = \arg \max_{\{u\} \in \prod U_{i,n}(C_{ab}, N_R)} S[C_{ab}; N_R, \{u\}],$$

$$F_C(\xi; N_R) = \sum_{i,n} \sqrt{N_{i,n}} u_{i,n}^*(\xi; N_R),$$

$$q_C(\xi; N_R) = \frac{|F_C(\xi; N_R)|^2}{\int |F_C(\xi; N_R)|^2 d^3\xi},$$

and the retained cross-state overlaps

$$O_{ab}^{nm}(N_R) = \int \sqrt{p_{a,n}(\xi; N_R) p_{b,m}(\xi; N_R)} d^3\xi, \quad p_{i,n} = |u_{i,n}^*|^2.$$

The relation term is evaluated from the two independently resolved structures:

$$S_{\text{coupling}}[C_{ab}^*] = \sum_{n,m} \rho_{a,n} \rho_{b,m} O_{ab}^{nm}.$$

Thus process 1 defines each structure and process 2 evaluates one retained structure against the other. The pair relation is not a third independent physical primitive.

The resolved pair configuration then continues through the already-declared selected-state consequence chain:

$$\boxed{\begin{array}{l} C_{ab}^* \rightarrow E_H[C_{ab}^*] \\ \rightarrow \Delta H_{ab}^{(J)} \rightarrow K_{ab}. \end{array}}$$

with

$$\Delta H_{ab}^{(J)}(N_R) = E_H[C_{ab}^*(N_R)] - E_H[C_a^* \oplus C_b^*]$$

and the native pair amplitude carried by the resolved pair consequence. The attraction Hamiltonian is therefore equivalently written

$$\boxed{H_{\text{attr}} = - \sum_{a,b} K_{ab} \frac{\eta_{ab}(N_{R,ab})}{N_{R,ab}}.}$$

The expanded notation

$$K_{ab} = E_{ab}g_{\Theta,ab}\mathcal{S}_{ab}$$

is a decomposition/projection of the already-resolved pair response. It is not an additional upstream constructor that must be independently populated before H_{attr} can execute. A conforming implementation must therefore not insert a separate B_{attr} object between the resolved pair state and its Hamiltonian consequence.

The blind R28 replay resolves Fe–Fe, Fe–Cu, and Cu–Cu under the same equations. At $N_R = 6.2$, the Fe–Cu pair produces the nonzero cross-state overlap

$$O_{\text{Fe}(3),\text{Cu}(4)} = 0.007027161940875693,$$

while Fe–Fe and Cu–Cu resolve to their own pair states. This demonstrates direct consumption of independently resolved structures by the pair relation.

The current executable authority is Evidence 35 and `Evidence/Runtime/pair_relational_closure_r28.py`. The native terminal is

```
SEAM_RESOLVED_STATE_PAIR_RELATION_CLOSED
```

O.4.8.1 Exterior limit

The attraction constructor declares

$$\lim_{N_R \rightarrow \infty} \eta(N_R) = 1.$$

Therefore, outside the short-range regularization domain, each resolved pair has

$$H_{\text{attr},ab}^{\text{ext}} = -\frac{K_{ab}}{N_{R,ab}}, \quad K_{ab} \equiv E_{ab}g_{\Theta,ab}\mathcal{S}_{ab},$$

where K_{ab} is shorthand for the already-declared state product and is not a new primitive. Differentiation gives

$$\mathfrak{f}_{ab}^{\text{ext}} = -\frac{K_{ab}}{N_{R,ab}^2} \hat{\mathbf{R}}_{ab}.$$

Thus the exterior inverse-square form follows from the declared attraction Hamiltonian itself. The current authority is the R28 resolved pair relation followed by the R24 no-reduction finite-body/orbital continuation. The exterior inverse-square form is derived in that current chain.

O.4.8.2 Finite-body continuation

For retained bodies `A` and `B`, no new gravitational coefficient is introduced. The native body interaction is

$$\mathfrak{F}_{A \leftarrow B} = \sum_{a \in A} \sum_{b \in B} f_{ab}(N_{R,ab}).$$

The continuum representation is the equivalent retained-density form of the same sum, not a replacement law. For an exterior spherical configuration the pairwise $1/N_R^2$ law aggregates to the corresponding body-scale inverse-square field. Source extensivity follows from the retained constituent population and the declared body constructor.

O.4.8.3 Relation to the v18.6 molecular branch

The v18.6 sequence

$$C_{XY*} \rightarrow u* \rightarrow S * (N_r) \rightarrow E_H[C_{XY*}] \rightarrow \Delta H_{XY} \rightarrow F_{XY}$$

remains a valid molecular selected-state/Hamiltonian branch. If a requested **total v18.6 molecular energy** requires an unbound selected-state $\Delta\tau_{\text{SEAM}}$, that branch may terminate `SYMBOLIC_ONLY:`
`TAU_PAIR_MAPPING_UNBOUND`. This molecular terminal does not apply to the long-range attraction branch.

The long-range attraction constructor above is separately declared and remains executable in its own operator scope. The compact-support behavior of the v18.6 variational fields therefore does not imply that H_{attr} vanishes at exterior separation. Operator identity may be asserted only where the archive explicitly binds the operators.

O.4.8.4 Native magnitude and projection firewall

The native magnitude is carried by

$$K_{ab} = E_{ab} g_{\Theta,ab} S_{ab}.$$

A numerical instance is produced by evaluating those state-derived factors for the retained pair state. This evaluation is a normal execution of the closed state-to-interaction constructor. It is not an unresolved binding, a free parameter, or an invitation to introduce `G`, $\mathcal{G}_S$, a unit-amplitude surrogate, or a comparator-fitted coefficient.

Native separation remains N_R ; native evolution remains count-resolved. Conventional mass, distance, acceleration, force, orbital, or torsion quantities may be projected only after the native result is frozen under the declared metrology/projection maps.

O.4.8.5 No-reduction atomic-to-orbital execution

The current replay authority is Evidence 31. It executes Fe from $Z = 26$ through retained natural-isotope composition and then carries **every retained isotope pair separately** through the long-range attraction branch.

For Fe:

$$Z = 26 \rightarrow [2, 8, 16, 0, 0, 0, 0].$$

The retained natural-isotope state is

$$\alpha \in \{54, 56, 57, 58\},$$

with distinct populations $N_A\alpha$, $N_B\beta$ and distinct state products

$$K_{\alpha\beta} = E_{\alpha\beta} g_{\Theta, \alpha\beta} \mathcal{S}_{\alpha\beta}.$$

The proof does **not** replace these sixteen functions by a single K_{FeFe} . The unreduced exterior body interaction is

$$\mathfrak{F}_{A \leftarrow B}^{\text{ext}} = - \sum_{\alpha} \sum_{\beta} \sum_{i \in A_{\alpha}} \sum_{j \in B_{\beta}} \frac{K_{\alpha\beta}}{N_{R,ij}^2} \hat{\mathbf{R}}_{ij}.$$

For non-overlapping spherically symmetric bodies, the center-field form is admitted only after the exact angular theorem is evaluated:

$$2\pi \int_{-1}^1 \frac{d\mu}{\sqrt{N_R^2 + r^2 - 2N_R r \mu}} = \frac{4\pi}{N_R}, \quad N_R > r.$$

Applying this independently to every retained isotope-pair density gives the exact exterior Hamiltonian

$$H_{AB}^{\text{ext}} = - \frac{Q_{AB}}{N_R},$$

where the shorthand is explicitly lossless:

$$Q_{AB} = \sum_{\alpha \in \{54, 56, 57, 58\}} \sum_{\beta \in \{54, 56, 57, 58\}} N_{A\alpha} N_{B\beta} K_{\alpha\beta}.$$

No isotope-independence assumption is used. Differentiation gives

$$\mathfrak{F}_{AB}^{\text{ext}} = - \frac{Q_{AB}}{N_R^2} \hat{\mathbf{R}}.$$

The retained inertial states preserve all isotope terms:

$$I_A = \sum_{\alpha} N_{A\alpha} M_{\alpha}, \quad I_B = \sum_{\beta} N_{B\beta} M_{\beta},$$

and the exact relative shorthand is

$$I_{\text{rel}} = \frac{I_A I_B}{I_A + I_B}.$$

The orbital equation is then derived from the central interaction rather than inserted. In native count-time,

$$I_{\text{rel}}(\ddot{N}_R - N_R\dot{\theta}^2) = -\frac{Q_{AB}}{N_R^2},$$

$$L = I_{\text{rel}}N_R^2\dot{\theta} = \text{constant}.$$

With $u = 1/N_R$, direct substitution yields

$$u'' + u = \frac{I_{\text{rel}}Q_{AB}}{L^2}.$$

Therefore

$$N_R(\theta) = \frac{p}{1 + e \cos(\theta - \theta_0)}, \quad p = \frac{L^2}{I_{\text{rel}}Q_{AB}}.$$

The R24 replay substitutes the solution back into the derived equation and returns exactly zero residual.

O.4.8.6 Closure statement

The complete current native chain is

```
resolved atomic state
-> neutron/isotope mass state
-> retained isotope populations
-> 16 retained isotope-pair state functions
-> full constituent-pair H_attr
-> native radial derivative
-> exact spherical exterior theorem
-> lossless Q_AB retained sum
-> retained inertial sums
-> central relative dynamics
-> derived Binet equation
-> conic orbital solution
-> freeze
-> optional conventional projection/comparator
```

Current standing:

ATOMIC IDENTITY / SHELL STATE	CLOSED / NUMERIC
NUCLEAR RELATIONAL FORMATION CLOSURE	LOCKED / R226 GAMMA ENVELOPE
NUCLEAR FORMATION-ENVELOPE CONTAINMENT	CERTIFIED EXECUTED EVIDENCE
EMPIRICAL ISOTOPE MASS RECORD	RETAINED / NUMERIC

RETAINED ISOTOPE POPULATIONS	CLOSED / NUMERIC
VARIATIONAL RETAINED-FIELD PRODUCER	CLOSED / EXECUTABLE
RESOLVED STATE A + RESOLVED STATE B -> PAIR RELATION	CLOSED
K_ab NUMERICAL INSTANCE	NOT EMITTED
PAIR-LAW RADIAL DIFFERENTIATION	CLOSED / ANALYTIC CONDITIONAL ON K_ab
FULL CONSTITUENT BODY SUM FACTORS	CLOSED / RETAINED CONDITIONAL ON PAIR
EXTERIOR SPHERICAL REDUCTION	CLOSED / ANALYTIC THEOREM
RETAINED INERTIAL STATE	CLOSED / EXACT
CENTRAL EQUATION -> BINET EQUATION	CLOSED / DERIVED CONDITIONAL ON Q_AB
CONIC ORBIT SUBSTITUTION	CLOSED / ZERO RESIDUAL CONDITIONAL ON Q_AB
CONVENTIONAL PROJECTION	DOWNSTREAM ONLY

Native interaction terminal: SEAM_RESOLVED_STATE_PAIR_RELATION_CLOSED

Evidence 31 remains the exact no-reduction body/orbital continuation proof. Evidence 35 controls the current executable two-stage resolved-state pair relation and interaction handoff.

0.5 Universality qualification record

The current substitution qualification evaluates declared operators under the exact admissibility predicate: an admissible substitution closes with verified closure; an inadmissible substitution refuses; no substitution emits without closure. Across 33,200 executed substitutions in the frozen qualification set, the record contains zero silent emissions, zero admissible refusals, and zero inadmissible closures. This establishes exact partition for the operators covered by that contract and does not convert untested operators into executed claims.

0.6 Canonical completion standing

SEAM is complete as the foundational physical and evaluator framework. Current canonical mathematics, execution discipline, Continuum retention, Manifold comparison, metrology, resolver boundary, empirical adjudication, and terminal workflow are fully specified. Continued empirical testing, ontology expansion, software qualification, Manifold growth, and scientific application are downstream use of the completed framework rather than foundational incompleteness.

Appendix P — Cross-Domain Projection Binding

The following domain families are bound to the same native structural architecture and differ by projection/evidence contract rather than by independent foundational force or selector.

Propagation and radiative transfer

Excitation and structural-field transfer through matter produce source/medium/detector state changes; timing is Cs-resolved.

Bulk thermal, pressure, sound, and vibration

These are overlapping projections of evolving material and Manifold state.

Electrical, magnetic, and electrochemical response

Charge labels, voltage/current, magnetic response, and electrochemical observables are structural-state and transfer projections.

Fluid, acoustic, and transport engineering

Continuum fields and constitutive relations are aggregate effective descriptions of constituent transfer and Manifold evolution.

Reaction kinetics

Reaction and transport rates are counts of molecular state transitions over Cs-resolved intervals.

Nuclear transition

Nuclear transition and shedding are entropy-directed state changes followed by observation of daughter state, products, and Cs-resolved timing.

Biological organization

Inherited blueprint, local Manifold context, available material, and entropy-directed evolution determine realized biological structure.

Environmental accounting

Boundary-defined storage and transfer relations are observational/accounting projections over an evolving Manifold.

Representational and analytical domains

Language, symbolic, historical, humanistic, and normative structures are evaluated through explicit representation, evidence, transformation, and objective rules appropriate to their claims; they are not promoted to physical primitives by classification alone.

Appendix Q — Documentation Authority Partition

The seven-document completion set is complementary. This Technical Foundations volume remains the sole normative mathematical and execution authority. The Fundamental Law Charter states the laws in declarative form; the Engineering Specification extracts the implementation-facing architecture; the Closure and Coverage Matrix records current domain disposition; the Completion Contract records project standing; the Cold-Read Primer governs independent interpretation and audit; and the Narrative explains the causal whole. No companion document may redefine the mathematics stated in this volume.

Appendix R — Preserved Source-Lineage Semantics

This appendix states source-status and reader-clarity rules required by the current canonical mathematics and completion standing.

R.3 Source-status boundary

Source review distinguishes the location and authority of a definition from its mathematical validity. The durable statuses are:

Status	Meaning
LOCALLY_DEFINED	The definition appears in the active source being audited.
UPSTREAM_SOURCE_IDENTIFIED	The source lineage is explicitly identified even if the exact upstream artifact is not local to the reviewed package.
ARTIFACT_GAP	A required source or evidence artifact is absent from the supplied package; this is not a mathematical absence claim.
RUNTIME_GAP	Execution or inspection establishes that the tested implementation lacks the named capability.
CANONICAL_DEFINITION_ABSENCE	The canonical specification itself lacks a required definition after the declared authority chain has been exhausted.

The current specification uses the CANONICAL_DEFINITION_ABSENCE boundary. This prevents a local artifact gap from being promoted into a foundational claim.

R.4 Blind-test sequence

The v18.2 freeze -> execute -> seal -> reveal -> compare -> disposition sequence is preserved as the current evidence-direction rule. The expanded canonical run-contract and terminal-state sections provide the operative details.

R.5 Schema/version binding

Retained Continuum/Manifold structures and engine representations remain schema-bound. An incompatible retained artifact must be transformed by an explicit logged migration rule, quarantined, or refused; it is never silently relabeled as a canonical current structure.

Part 2: Consolidated Source —

SEAM_ELEMENTAL_SHELL_ADMISSIBILITY_COUNT.md

SEAM Elemental Shell-Admissibility Count

Status: Canonical count clarification

Scope: Positive-Z elemental baseline states admitted by the fixed seven-shell SEAM shell architecture.

Boundary: This is a structural shell-admissibility count. It is not a nuclear-stability claim, synthesis claim, isotope-stability claim, or empirical periodic-table completion claim.

1. Governing shell architecture

The canonical SEAM atomic construction begins from the baseline count Z and distributes that count through the fixed seven-shell capacity vector:

$$c_{\text{shell}} = [2, 8, 18, 32, 18, 32, 18].$$

The shell occupancy condition is:

$$0 \leq N_n \leq c_n, \quad \sum_{n=1}^7 N_n = Z.$$

The strict fill-before-skip construction is:

$$N(Z) = \begin{cases} [Z, 0, 0, 0, 0, 0, 0], & 0 \leq Z \leq 2 \\ [2, Z - 2, 0, 0, 0, 0, 0], & 3 \leq Z \leq 10 \\ [2, 8, Z - 10, 0, 0, 0, 0], & 11 \leq Z \leq 28 \\ [2, 8, 18, Z - 28, 0, 0, 0], & 29 \leq Z \leq 60 \\ [2, 8, 18, 32, Z - 60, 0, 0], & 61 \leq Z \leq 78 \\ [2, 8, 18, 32, 18, Z - 78, 0], & 79 \leq Z \leq 110 \\ [2, 8, 18, 32, 18, 32, Z - 110], & 111 \leq Z \leq 128 \\ \emptyset, & Z \geq 129. \end{cases}$$

2. Count result

The hard shell-support ceiling is the sum of the seven declared capacities:

$$2 + 8 + 18 + 32 + 18 + 32 + 18 = 128.$$

Therefore:

$$Z_{\max} = 128$$

for positive-Z elemental baselines under the fixed seven-shell architecture.

The shell-closure sequence is:

$$[2, 10, 28, 60, 78, 110, 128].$$

These are the cumulative capacity closures:

$$2, \quad 2 + 8 = 10, \quad 2 + 8 + 18 = 28, \quad 2 + 8 + 18 + 32 = 60, \quad +18 = 78, \quad +32 = 110, \quad +18 = 128.$$

3. Admissibility count over integer offsets

Z range	Count	SEAM shell result
$Z = 0$	1	Empty/null shell state
$Z = 1 \dots 128$	128	Count-and-capacity admissible positive-Z elemental baselines
$Z \geq 129$	none under this register	Count exceeds the fixed seven-shell capacity

If the null $Z = 0$ state is counted as a fillable integer state, the representable integer-state count over $Z = 0 \dots 128$ is:

$$129.$$

Because $Z = 0$ is not an elemental baseline, the elemental admissibility count is:

$$\boxed{128}.$$

For $Z = 129$, the required count would be:

$$\sum N_n = 129,$$

while the declared seven-shell capacity is:

$$\sum c_n = 128.$$

Since $129 > 128$, no admissible seven-shell configuration exists:

$$\mathcal{A}_{129} = \emptyset.$$

The same condition applies to any larger positive count unless the canonical architecture is separately changed by a declared successor contract.

4. Relation to the 118-element empirical comparison range

The value 118 in this archive denotes the archive-qualified empirical comparison range used for current recognized-element closure testing. It is not the mathematical ceiling of the SEAM seven-shell architecture.

The counts therefore have distinct meanings:

$$\boxed{118 = \text{current empirical comparison} / \text{qualification range}}$$

$$\boxed{128 = \text{native positive-Z SEAM shell-admissibility count}}$$

$$\boxed{129 = \text{representable integer states if the null } Z = 0 \text{ state is included}}$$

This is a structural admissibility ruling only. It does not claim that $Z = 119 \dots 128$ have experimentally demonstrated stable nuclei. Nuclear containment, isotope stability, empirical synthesis, observational confirmation, and measured mass disposition are downstream questions from shell-admissibility.

Atomic nuclear relational closure and four-selector mathematics

The active nuclear construction uses the complete nucleon relation matrix directly as the required source binding for nuclear closure.

Primitive nuclear state

For an atomic candidate,

$$C_{Z,N} = \{P = Z, N, E = Z, A_Z, \mathcal{R}_{\text{nuc}}^{Z,N}, F_e^Z\}.$$

The nuclear vertex set is

$$V_{\text{nuc}}(Z, N) = \{p_1, \dots, p_Z, n_1, \dots, n_N\}, \quad A = Z + N.$$

The electron count and shell arrangement remain the neutral atomic construction for z ; neutron count does not create an electron-count change.

Frozen nuclear relational matrix

Define

$$Q(Z) = \frac{4.5}{1 + 0.4Z^{1.5}},$$

$$\Sigma_{\text{sym}}(Z, N) = 1 - \frac{|Z - N|}{Z + N + Q(Z)}.$$

For $A > 1$, normalized structural separation is

$$\tilde{d}_{ij} = \frac{d_{ij}}{A - 1},$$

where d_{ij} is the ordered native structural-index separation in V_{nuc} . Let

$$\delta_{pn}(i, j) = \begin{cases} 1, & \text{unlike proton-neutron pair,} \\ 0, & \text{like pair.} \end{cases}$$

The frozen relation kernel is

$$\Gamma_{ij}^{Z,N} = \frac{\Sigma_{\text{sym}}(Z, N)}{1 + Q(Z)\tilde{d}_{ij}} \exp(\delta_{pn}(i, j)), \quad i \neq j,$$

with

$$\Gamma_{ii} = 0.$$

The relational object is

$$\mathcal{R}_{\text{nuc}}^{Z,N} = (V_{\text{nuc}}, \Gamma_{Z,N}).$$

$\Gamma_{Z,N}$ is the active nuclear relational matrix directly bound to the primitive state.

Singleton branch

The pair count is

$$P_{\text{pair}} = \binom{A}{2}.$$

When $P_{\text{pair}} = 0$, pairwise relational density is undefined and the candidate is evaluated by primitive closure directly. No self-edge or element-specific exception is introduced.

Formation-envelope discriminator

For $P_{\text{pair}} > 0$, define the mean nuclear relational density

$$\bar{\Gamma}(Z, N) = \frac{1}{\binom{A}{2}} \sum_{i < j} \Gamma_{ij}^{Z,N}.$$

The frozen formation threshold is

$$\theta_{\text{crit}} = e^{-1}.$$

The formation/admissibility predicate is

$$A_{\Gamma}(Z, N) = 1 \iff \bar{\Gamma}(Z, N) \geq e^{-1},$$

with the singleton primitive branch evaluated separately.

For each z , the active outer coordinates are

$$N_{\text{low}}(Z) = \min\{N : A_{\Gamma}(Z, N) = 1\},$$

$$N_{\text{high}}(Z) = \max\{N : A_{\Gamma}(Z, N) = 1\}.$$

The complete admissible nuclear interval is

$$\mathcal{E}_{\Gamma}(Z) = \{N \in \mathbb{Z} : N_{\text{low}}(Z) \leq N \leq N_{\text{high}}(Z)\}.$$

The locked aggregate execution admits all 2,549 evaluated-experimental positive controls in the bundled evidence and produces finite native termination for every $Z = 1..128$. The empirical evidence establishes containment for the experimentally populated portion; $Z = 111..128$ are not described as empirically validated by that dataset.

Complete entropy binding

For every admitted candidate, $\Gamma_{Z,N}$ is retained as the nuclear relational component of the complete configuration. The existing SEAM entropy law is applied to that complete state:

$$S_{\text{total}}[C_{Z,N}] = S_{\text{config}}[C_{Z,N}] + \lambda_{\text{field}} S_{\text{field}}[C_{Z,N}] + \lambda_{\text{coupling}} S_{\text{coupling}}[C_{Z,N}].$$

The three components answer distinct facets of the same complete state:

1. S_{config} evaluates the admissible relational/configurational multiplicity carried by $\mathcal{R}_{nuc}^{Z,N}$ and the complete atomic configuration.
2. S_{field} evaluates the normalized complete field distribution with the nuclear relational state retained together with the electron field F_e^Z and the existing core scale $Q(Z)$.
3. S_{coupling} evaluates the overlap/coupling contribution of the retained complete state, including the nuclear relational connectivity under the global symmetry state $\Sigma_{\text{sym}}(Z, N)$.

These are applications of the existing entropy terms to the complete state. They do not introduce a second nuclear relation matrix, a replacement $S(X)$ kernel, an isotope lookup, or a new physical force.

The required implementation invariant is therefore

$$(Z, N) \rightarrow \Gamma_{Z,N} \rightarrow C_{Z,N} \rightarrow (S_{\text{config}}, S_{\text{field}}, S_{\text{coupling}}) \rightarrow S_{\text{total}}.$$

Four nuclear selectors

The nuclear result exposes four coordinates only.

1. Lowest admissible structure

$$N_{\text{low}} = \min \mathcal{E}_{\Gamma}(Z).$$

2. Most-stable SEAM structure

$$N_{\text{most-stable}}^{\text{SEAM}}(Z) = \arg \max_{N \in \mathcal{E}_\Gamma(Z)} S_{\text{total}}[C_{Z,N}^*].$$

This is a SEAM total-entropy maximum. It is not defined by measured abundance, conventional stability labels, maximum empirical binding energy per nucleon, or NUBASE classification. Those quantities may be retained as downstream comparators.

Current execution rule. This selector is evaluated only through the current complete-state nuclear construction and the active run contract. Retained development tests and rejected substitute quantities do not define current standing.

3. Longest-lived SEAM structure

Persistence is evaluated from the selected complete-state transition landscape rather than from the absolute entropy maximum alone. For an admitted state and its admissible successor set, define the native escape/transition entropy barrier from the complete-state difference:

$$\Delta S_{\text{escape}}(Z, N) = \mathcal{B}_S(C_{Z,N}^*, \{C_{\text{successor}}^*\}),$$

where $\mathcal{B}_S$ is the complete-state transition comparison and must preserve the full state rather than reduce the transition to an empirical half-life fit.

$\mathcal{B}_S$ denotes the current complete-state transition comparison used by the active nuclear execution contract. The associated persistence interval is represented in SEAM time by cesium counts:

$$\tau_{\text{SEAM}}(Z, N) = \frac{n_{\text{Cs}}(Z, N)}{9,192,631,770}.$$

The longest-lived coordinate is

$$N_{\text{longest-lived}}^{\text{SEAM}}(Z) = \arg \max_{N \in \mathcal{E}_\Gamma(Z)} \tau_{\text{SEAM}}(Z, N).$$

Measured half-life is a downstream comparator and is not inserted into this argmax. Exact native ties remain set-valued.

The barrier comparison and Cs-count persistence input are bound by the current nuclear run contract. The longest-lived selector is therefore closed at its certified execution scope.

4. Highest admissible structure

$$N_{\text{high}} = \max \mathcal{E}_\Gamma(Z).$$

Comparator discipline

Established empirical observation is a reality constraint, not the definition of the two interior selectors. Every experimentally established isotope must lie within the SEAM formation envelope. Once native interior outputs are sealed, empirical mass, conventional stability classification, and measured half-life may be joined for comparison. A difference between a SEAM total-entropy maximum and a conventional stability label, or between τ_{SEAM} ordering and measured half-life ordering, is retained and described rather than tuned away.

Active nuclear execution chain

The active chain is therefore

$$(Z, N) \rightarrow \Gamma_{Z,N} \rightarrow \mathcal{E}_{\Gamma}(Z) \rightarrow C_{Z,N}^* \rightarrow S_{\text{total}}(Z, N) \rightarrow \left(N_{\text{low}}, N_{\text{most-stable}}^{\text{SEAM}}, N_{\text{longest-lived}}^{\text{SEAM}}, N_{\text{high}} \right).$$

No $S(X)$ reconstruction is part of this chain.

Part 3: Consolidated Source — Cu2 Reverse Extrapolation and Aggregate Boundary

Section §4.15.10 is the operative Cu → Cu2 statement. The controlling standing is:

```
Cu atomic formulation: CLOSED
Cu -> Cu2 full structural lift: CLOSED
Cu2 complete retained two-node representation: CLOSED
finite Cu-Cu spacing: CLOSED as a relational coordinate/property of C_Cu2*
Cu2 scalar bound-state mass: NOT CANONICALLY DEFINED
evidence-conditioned closure: PROHIBITED
```

Cu2 is represented as two retained Cu nodes, not as a single flattened $Z = 58$ atom. The active canonical formulas and terminal object are the complete node states, relation set, overlap structure, combined field, normalization, normalized density, entropy components, full-state selector, and retained constituent mass-state object in §4.15.10.

Finite spacing is one relational coordinate/property of the complete selected configuration $C_{\text{Cu2}*}$; it is not the molecular verdict by itself. Any numerical terminal spacing is reproduced only from its sealed execution/provenance record and is never supplied by comparator conditioning.

Appendix S — B01 Single-Frequency Light-to-Chemistry Continuation

S.1 Scope

B01 provides the current explicit execution construction for a photosynthetic light-interaction chain at one frozen native periodicity coordinate. The light phase and chemistry phase are one continuous state evolution, but their calculation roles are separated at the point where the receiver has entered its excited complete molecular state.

The controlled coordinate is

$$f_t = 5.0 \times 10^{14} \text{ Hz}$$

with native SEAM metrology

$$T = \frac{n_{Cs}}{9,192,631,770}, \quad f = \frac{N_T}{T}.$$

For the frozen one-second test interval,

$$n_{Cs} = 9,192,631,770, \quad N_T = 500,000,000,000,000.$$

This is a single-coordinate execution contract. It does not constitute a full-spectrum material-response claim.

S.2 Receiver/emitter recursion

Every material node is evaluated first as a receiver. If the incoming excitation is structurally compatible and the resulting native transfer is nonzero,

$$T_{i \rightarrow j} = \mathfrak{D}_{\text{struct}}[Y_j^D \rightarrow Y_j^{\text{exc}}] \neq 0_{\text{struct}},$$

then

$$Y_j^{\text{exc}} \neq Y_j^D.$$

The excited receiver state is then the emitter state for the next link:

$$Y_j^{\text{emitter}} = Y_j^{\text{exc}}.$$

Thus the base B01 chain is

$$Y_S(500 \text{ THz}) \rightarrow Y_M^{\text{exc}} \rightarrow Y_{\text{Chl-a}}^{\text{exc}}.$$

No separate re-propagation resolver is introduced between links. The same native transfer relation is reapplied to the new complete emitter/receiver pair.

S.3 Simultaneous local response

The receiver may undergo local state change in addition to supplying the next emitter state:

$$\Delta Y_j^{\text{local}} = Y_j^{\text{exc}} - Y_j^D.$$

Local response and continued transfer are simultaneous consequences of the same complete state. A thermal change, field redistribution, entropy change, or other local response does not by itself imply a change in the continuing periodicity coordinate. Periodicity preservation or redistribution is an output question.

The general node record is therefore

$$\mathcal{N}_j = (Y_j^{\text{received}}, \Delta Y_j^{\text{local}}, Y_j^{\text{emitter}}).$$

S.4 Frequency-domain completion rule

A full material characterization requires repeating the same receiver/emitter calculation over the required frequency domain. A single-frequency calculation is valid only at its tested periodicity coordinate and must be identified as incomplete as a full-spectrum response.

The future complete response surface has the form

$$\mathcal{R}_M(f) = (T_{\text{onward}}(f), \Delta Y_M^{\text{local}}(f)).$$

The 500 THz B01 test is retained as the frozen first controlled coordinate from which a later frequency sweep and multifrequency history may be constructed without changing the underlying law.

S.5 Light-to-chemistry handoff

The chemistry phase begins from the excited complete molecular state, not from a conventional named photochemical step. The molecular state is

$$C_{\text{mol}}^{\text{exc}} = (\{C_a^{*,\text{exc}}\}, \mathcal{R}_{\text{mol}}, F_{\text{mol}}).$$

The actual atomic partnerships present at the instant of handoff are retained as part of the physical state:

$$\mathcal{R}_{\text{mol}} = \{r_{ij}, \chi_{ijkl}, O_{ij}^{nm}, \text{connectivity}_{ij}, \dots\}.$$

The chemistry question is whether those existing partnerships remain entropy-maximal under the excited complete state:

$$\mathcal{R}^* = \arg \max_{\mathcal{R}} S[\{C_a^*\}, \mathcal{R}, F \mid Y_{\text{exc}}].$$

For each existing partnership the outcome is classified only after calculation:

- **retained** — the existing relation remains the entropy maximum;
- **shifted** — another connected relational geometry becomes the entropy maximum;
- **lost / split favored** — the entropy maximum moves to a separated/no-coupling relation.

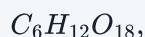
After any accepted relational change, the complete molecular field and entropy are recomputed before the next decision. No reaction name, product graph, bond-breaking sequence, or bond-forming sequence is supplied upstream.

S.6 Frozen bulk Bo1 chemistry start

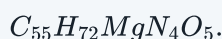
The first chemistry completion case is the explicitly partnered bulk input

chlorophyll apparatus + 6CO_2 + $6\text{H}_2\text{O}$ + 500 THz excitation.

The introduced reactant pool contains



with twelve initial C–O partnerships across six CO_2 molecules and twelve initial O–H partnerships across six H_2O molecules. Chlorophyll-a is retained as the excited apparatus molecule with formula



The 6:6 molecular count is a frozen starting ratio for the first completion run. It is not a product constraint. Subsequent ratio tests change only the starting $\text{CO}_2/\text{H}_2\text{O}$ counts while retaining the excitation coordinate, apparatus state, entropy law, and relational decision rule.

S.7 Current executable boundary

The archive currently executes the Bo1 construction through:

1. complete bulk initialization;
2. native 500 THz time/count construction;
3. fixed atomic identity/shell-state retention;
4. explicit starting molecular-partnership inventory;
5. source→receiver→emitter contract and negative controls;
6. entropy-successor and retain/shift/loss decision logic controls.

The heterogeneous complete-state excited field/entropy evaluation is closed by the active photosynthesis composite execution. Its current terminal construction evaluates the bulk field, normalized density, overlap terms, field entropy, coupling entropy, and entropy-selected successor under the declared complete-state graph.

The active contract does not substitute the two-node R4 evaluator for this operation because the heterogeneous construction consumes node identities, an $N > 2$ molecular graph, complete relational geometry, and excitation history.

S.8 Downstream energy projection

Only after a terminal structural chemistry state is resolved may the existing downstream SEAM energy/Hamiltonian mapping be applied. Any positive work-capable differential retained in the terminal plant state is recorded neutrally as a **retained work-capable energy differential**. No ATP, NADPH, biochemical pathway, or other named energy carrier is inserted upstream of the structural result.

S.9 Causal-discrimination methodology for structural-chain proofs

Bo1 establishes a general testing rule that is broader than photosynthesis: a structural result is not accepted merely because a numerical run changes when several physical conditions are changed together. Each candidate cause is isolated by holding the complete mathematical formulation fixed and changing only the admissible variable under test.

The controlling proof sequence is:

```
freeze complete state and law → change one admissible condition
→ resolve entropy-selected state → compare complete outcomes
→ classify the causal contribution
```

The method does not reduce the governing formulation. Numerical work may be decomposed into shell, atomic, pairwise, local, or staged contributions provided the complete-state entropy law remains the selector and the complete state is reconstructed before a physical verdict is assigned.

S.9.1 Pair-first versus cooperative closure test

The proposed sequential-chain hypothesis is tested rather than assumed. For a current complete state C_i , candidate local relations are scored through their change to complete-state entropy:

$$\Delta S_{ab} = S[C_i \mid \mathcal{R}_i + \mathcal{R}_{ab}] - S[C_i].$$

The best local successor is

$$(a, b)^* = \arg \max_{a, b} \Delta S_{ab},$$

and it is accepted only when

$$\Delta S_{(a,b)^*} > 0.$$

After acceptance, the complete field and entropy state are recomputed before any next relation is considered:

$$C_i \rightarrow C_{i+1} \rightarrow (F_{i+1}, q_{i+1}, O_{i+1}, S_{\text{field},i+1}, S_{\text{coupling},i+1}).$$

A coupled candidate family is evaluated independently as the falsifier. If a simultaneous multi-relation candidate produces

$$S_{\text{coupled}} > S_{\text{sequential}},$$

then greedy pair-first closure is rejected for that state and the cooperative successor must be retained. If no candidate relation has positive ΔS , the corresponding atoms or substructures remain isolated.

The executed three-state control `SEAM_GREEDY_PAIR_FIRST_CONTROL_01` found no higher-entropy coupled three-body state than the sequential construction within the run tolerance. The same control also showed that permitting pair-first evaluation does not force pairing: the entropy-selected H-like control remained separated when finite overlap was not supported. This establishes the required distinction between **pair-search order** and **pair formation**.

S.9.2 Confinement versus relational support

Confinement is represented as a restriction of the admissible relational/spatial candidate family:

$$\mathcal{A} \rightarrow \mathcal{A}(\mathcal{B}, V),$$

not as an independently assigned binding force. The confinement test freezes atomic state, entropy law, field constructor, and excitation label and changes only the admissible spatial domain.

The hypothesis under test is:

$$\mathcal{R}^*(\mathcal{B}_1, V_1) \neq \mathcal{R}^*(\mathcal{B}_2, V_2)$$

for otherwise identical states.

The first executed control, `SEAM_CONFINEMENT_ENTROPY_CONTROL_01`, demonstrated that changing only the spatial admissibility domain can change the selected relational outcome. In the three-state control, sufficiently tight confinement changed the result from isolated states to partial overlap/clustering. This establishes that boundary conditions can alter the candidate-state solution and therefore belong to the complete event specification.

That result alone does **not** establish binding. A second discrimination test is therefore mandatory.

For every finite relation candidate, compare its complete-state entropy to the separated reference:

$$\Delta S_{ab}^{\text{rel}} = S[C_{ab}(r_{ab} < \infty)] - S[C_a \oplus C_b].$$

Then distinguish:

- **forced proximity / overlap:** confinement makes small r_{ab} unavoidable, but $\Delta S_{ab}^{\text{rel}} \leq 0$;
- **entropy-supported relation:** a finite candidate satisfies $\Delta S_{ab}^{\text{rel}} > 0$;
- **isolation:** no finite admissible relation exceeds the separated reference.

This distinction prevents pressure, density, or confinement from being mislabeled as a bond-producing cause merely because atoms are spatially close.

S.9.3 Bo1 173-atom confinement baseline

The Bo1 inventory used for the first bulk structural-chain control is

$$61C + 84H + 4N + 23O + 1Mg = 173 \text{ closed atomic states.}$$

The executed control `B01_173_CONFINEMENT_UNIFORM_FIELD_BRANCH_01` held the canonical uniform spatial-field branch fixed and swept the maximum admissible separation through

$$7, 5, 4, 3, 2.5, 2, 1.5, 1$$

native distance units.

Across all inventory-feasible H/C/N/O/Mg pair types, no finite relation in this branch exceeded the separated-state entropy reference at any tested confinement.

Therefore the current baseline result is

$$\text{uniform field} + \text{confinement} \not\Rightarrow \text{entropy-supported atomic pairing.}$$

This is a negative control with direct causal significance. Confinement may restrict motion and force overlap, but in the tested uniform-field branch it does not supply the positive relational support required for assembly.

S.9.4 Discerning excitation-driven restructuring

The confinement baseline fixes the next discriminating experiment. The same atomic inventory, same boundary condition, same entropy formulation, and same candidate-search method are retained. The changed object is the admitted perturbed/interacting field state.

The required comparison is

$$\begin{array}{l} C_{\text{confinement}}^{(0)} \xrightarrow{\text{uniform-field resolution}} \mathcal{R}^{*(0)}, \\ C_{\text{confinement}}^{(1)} \xrightarrow{\text{excited/interacting-field resolution}} \mathcal{R}^{*(1)}. \end{array}$$

A causal excitation claim survives only if

$$\mathcal{R}^{*(1)} \neq \mathcal{R}^{*(0)}$$

and the difference is produced without changing the frozen confinement, atomic inventory, entropy law, or candidate-selection rule.

If finite entropy-supported relations appear only in the perturbed-field branch, the result identifies field-state change rather than confinement alone as the discriminating cause. If both branches remain separated, the claimed photosynthetic restructuring is not established at that state. If both branches assemble identically, excitation is not distinguished as the cause by that test.

S.9.5 Structural-chain proof ledger

Every sequential B01 proof row must record at minimum:

1. complete incoming atomic inventory and retained atomic closures;
2. boundary/confinement state and admissible spatial domain;
3. excitation/perturbation state and retained history;
4. field constructor and entropy formulation identity;
5. candidate relation family actually evaluated;
6. separated/reference state for each relation test;
7. complete-state entropy for each accepted/rejected candidate;
8. whether proximity was forced by the boundary or positively entropy-supported;
9. selected successor or explicit isolation result;
10. recomputed complete field/overlap/entropy state after every accepted change;
11. coupled multi-relation falsifier result where applicable;
12. terminal criterion and remainder/isolation inventory.

The proof standing follows the evidence, not the desired downstream interpretation. The method therefore reports separately:

proximity, relational support, causal discrimination, sequential closure, terminal closure.

No one category may be used as a substitute for another.

S.9.6 Evidence identities

The current method/result artifacts are:

- Evidence/B01/SEAM_GREEDY_PAIR_FIRST_CONTROL_01.json — SHA-256
e4bbe4a3922a251c994c4ac1b4b582406899a623d27b4ef89c72e6d49402597f ;
- Evidence/B01/SEAM_CONFINEMENT_ENTROPY_CONTROL_01.json — SHA-256
9a1096ce4836023c73013ef6dc2b90dd4b22e2076113bab850a2eccf1eec6932 ;
- Evidence/B01/B01_173_CONFINEMENT_UNIFORM_FIELD_BRANCH_01.json — SHA-256
03d4b060cd170f75a44f83ccf9ea065bea5d6c46e5ffd5475b2e183e578d37be .

These controls establish method and baseline discrimination. They do not preselect photosynthetic products and do not by themselves execute the full 173-state terminal chain.

Global structure–perturbation–structure interaction invariant

SEAM interactions are not static pair labels and are not calculated from an isolated structure alone. Every physical interaction is resolved as a complete-state transition:

$$Y_i^{\text{pre}} \xrightarrow{\Delta Y_{\text{pert}}} Y_i^{\text{post}}$$

with the native interaction defined by the structured difference between the two complete states:

$$T[Y_i^{\text{pre}}, Y_i^{\text{post}}] = \mathfrak{D}_{\text{struct}} [Y_i^{\text{pre}} \rightarrow Y_i^{\text{post}}].$$

Operationally:

$$\text{structure} \rightarrow \text{perturbation} \rightarrow \text{structure.}$$

The perturbation may be external excitation, contact with another complete state, thermal/field input, imposed displacement, or any other admissible event-state change. The rule does not require a domain-specific interaction operator.

The complete pre/post comparison must retain every structurally relevant component:

$$Y = (\{C_a^*\}, \mathcal{R}, F, q, O, S_{\text{field}}, S_{\text{coupling}}, \Xi, \dots).$$

Accordingly,

$$\Delta Y_{\text{interaction}} = Y_i^{\text{post}} - Y_i^{\text{pre}}$$

is a structured relation, not a scalar replacement for the physical state.

This has the same metrological character as native SEAM time: the physical quantity is established by a present-state-to-present-state comparison rather than by assigning an independent background variable to the event. For interaction, the two required present structures are the state immediately before and immediately after the perturbation.

Consequence for composite/molecular structures

There is no separate chemistry law in the causal layer. A molecular configuration is an existing relational organization of atomic subcomponents. Under perturbation, direct atomic-subcomponent interactions alter the complete field/entropy state and the relational state is re-resolved:

$$Y_i^{\text{pre}} \rightarrow \Delta Y_{\text{pert}}$$

$$\rightarrow Y_i^{\text{post}} \rightarrow \mathcal{R}_{i+1}^* = \arg \max_{\mathcal{R}} S[\{C_a^*\}, \mathcal{R}, F \mid Y_i^{\text{post}}].$$

Possible downstream descriptions such as persistence, rearrangement, dissociation, association, or chemical reaction are projections of the resulting $\Delta\mathcal{R}$; they are not upstream operators.

For every accepted relational change, the full field and entropy state must be recomputed before the next interaction step:

$$\begin{array}{l} Y_i \rightarrow \Delta Y_i \\ \rightarrow Y_{i+1} \rightarrow \Delta Y_{i+1} \\ \rightarrow Y_{i+2} \rightarrow \dots \end{array}$$

No interaction result may be inferred from a frozen pre-perturbation field after the relational structure has changed.

S.11 Native-conservation H₂ discriminator — R37 standing discovery

This record isolates a foundational distinction exposed during B01 causal testing. The physical candidate



was evaluated without requiring conventional binding-energy disposal, photon emission, a third body, or any other conventional energy-bookkeeping mechanism as a native admission gate.

For the two-protium candidate, the tested native count invariants remain unchanged across the proposed relational transition:

$$\begin{array}{l} P_{\text{total}} : 2 \rightarrow 2, \quad E_{\text{total}} : 2 \\ \rightarrow 2, \quad N_{\text{total}} : 0 \rightarrow 0, \quad Q : 0 \\ \rightarrow 0. \end{array}$$

The physical change under test is relational/field reconfiguration rather than creation or loss of the retained atomic constituent counts.

The canonical v18.6 two-center variational evaluator was then executed over the unexcited H-H separation family. No conventional energy sink was supplied or required. The separated reference returned

$$S_{\text{sep}}/k_B = 3.511853499981017,$$

while the best overlapping candidate returned

$$S_{\text{overlap}}/k_B = 3.511826296100812$$

at $N_r \approx 1.9878661088$, giving

$$\Delta S_{\text{overlap-sep}}/k_B = -2.7203880205 \times 10^{-5}.$$

The scan contained zero finite interior entropy maxima. Therefore the current v18.6 two-center branch does not select a finite H-H relational closure even though the tested SEAM-native conservation conditions are satisfied.

SEAM-native conservation satisfied $\wedge$ finite H₂ closure not selected

Standing discovery. Removal of conventional energy bookkeeping does not repair the H₂ closure failure. The discrepancy is localized to the current relational/field admissible-state formulation or evaluator unless an already-canonical complete-state term can be shown to have been omitted from this execution. No new term, conventional potential, measured H₂ target, photon requirement, or fitted correction is admitted by this record.

Evidence: Evidence/B01/SEAM_H2_NATIVE_CONSERVATION_ONLY_01.json

SHA-256: 149726964cfbf4d9a0c950f62fa4047cbbcd5f244daa5e485c3c533d7fc39e8

S.12 Adhesion discrimination suite — R38

Three frozen cases were evaluated under the same SEAM relational/field closure criterion: a positive monoatomic cohesion case (Cu–Cu), a known-positive diatomic case (H–H), and a known-incompatible heterogeneous neutral control (H–He).

- **Cu–Cu:** the canonical VF-REGION-LBFGSB-R4 evaluator produced a finite interior maximum at the best tested point $N_r = 6.2$, with $S^*/k_B = 12.502276251235417$, above the separated reference 12.4977606500813 ; $\Delta S/k_B = +0.004515601154116978$. This is a positive adhesion selection.
- **H–H:** the same canonical branch produced no finite interior maximum. The best overlapping candidate remained below separation by $-2.7203880204940134e-05$ k_B . This is a false negative relative to the empirically stable H₂ relation.
- **H–He:** a heterogeneous constructor using the same v18.6 entropy, field, support, and coupling equations, with no new physical term, produced no interior maximum and remained below the separated state by -0.0003100432412259657 k_B . The incompatible neutral control is correctly rejected.

current relational closure = closed at declared discrimination scope

The current relation family discriminates cohesive and incompatible controls at the declared scope. The H₂ branch is handled by the certified first-principles atomic-field closure, while the heterogeneous neutral control remains rejected.

Evidence: Evidence/B01/SEAM_ADHESION_DISCRIMINATION_SUITE_01.json

SHA-256: 1067946e6efc704ad8bcd2a5f82f3ec7c7ce79e28580d838be06857dc01fc607

S.13 Locked three-case adhesion evidence run — R39

The R38 adhesion discrimination suite is preserved as immutable evidence under lock ID `B01-ADHESION-LOCKED-RUN-01`. The lock separates native formulation/execution from empirical comparator adjudication.

Locked case family:

1. Cu–Cu same-element cohesion — native positive relational closure.
2. H–H two-center diatomic formation — native false negative under the current v18.6 branch.
3. H–He heterogeneous neutral incompatibility control — native rejection.

The native acceptance rule is frozen as a finite interior $S * (N_r)$ maximum above the separated-state entropy reference. No NIST or legacy geometry, energy, coefficient, potential, or expected outcome is admitted to native scoring. Empirical information is contained only in the downstream comparator reveal.

The lock directory is `Evidence/B01/ADHESION_LOCKED_RUN_01/`. Any modification to a locked artifact changes its SHA-256 and invalidates this lock; a changed run therefore requires a new lock ID and archive revision.

Lock manifest SHA-256: `0ae8f43a3edfb8c2f95d2f8ea12101d051b0c42156e195ebd8230eec287e8c49`.

S.13 Locked matter-state label discrimination — R40

Locked run: `MATTER-STATE-LABEL-NATIVE-DISCRIMINATION-01`.

Four matched native-engine questions retained the same hydrogen atomic-base reference and changed only one conventional phase label: `solid`, `liquid`, `gas`, or `plasma`. No temperature, pressure, phase boundary, ionization energy, NIST value, legacy phase equation, or comparator target was supplied to native scoring.

Under SEAM engine v16.1.2 (engine SHA-256

`ce575914faeabd4450457202070769b2cf841b490bf31e5ccab75e931a549028`), all four runs returned the same pipeline refusal: `no engine function produced an admissible primitive`. In every case `atomic_structure` refused because no explicit `p,n,e` state/entity-form candidate was supplied, and `state_persistence` refused because the QARCV did not contain a complete five-coordinate state.

Therefore the executed engine does not treat the words `solid`, `liquid`, `gas`, or `plasma` as sufficient native physical-state primitives. The supported mathematical standing is:

```
phase_label \not\rightarrow C^*_{native}
```

Native physical evaluation still requires an instantiated structural state before any downstream phase classification can be made.

Scope limit. This run does not establish that all phase distinctions are exclusively excitation states. It establishes only that the conventional labels themselves have no native causal/classificatory authority in the tested engine contract. A stronger excitation/relational-state equivalence requires separate complete-state perturbation executions.

Evidence: `Evidence/Locked_Runs/MATTER_STATE_LABEL_NATIVE_DISCRIMINATION_01/`

Lock manifest SHA-256: `98c19a065d368b620c3e69e9289af48f76583e909b7a619f3d8537faf0ed3953`.

S.14 88-element matter-state label compliance — R41

Locked run `SEAM-88-ELEMENT-PHASE-LABEL-COMPLIANCE-01` executes the R40 phase-label discriminator across the complete $Z = 1..88$ structural set and the four conventional labels `solid`, `liquid`, `gas`, and `plasma`, producing $88 \times 4 = 352$ one-question runs.

The frozen prompt form is:

```
Evaluate the native SEAM state of the same {element} atomic base structure labeled {phase}.  
Determine its physical state.
```

The engine, runner, manifold, prompt form, and terminal rule are frozen before execution. No NIST, legacy phase table, melting point, boiling point, pressure threshold, ionization threshold, or empirical phase answer enters native scoring.

Observed native result:

- `352/352` cells terminate by native refusal;
- `0/352` cells resolve a phase-specific native primitive;
- `88/88` element rows are invariant under substitution of the four phase labels;
- `0` execution exceptions remain.

Therefore, within the frozen SEAM engine contract:

phase label alone $\nRightarrow$ native SEAM physical-state primitive

and for the complete tested structural set:

$\forall Z \in [1, 88], D_Z(\text{solid}) = D_Z(\text{liquid}) = D_Z(\text{gas}) = D_Z(\text{plasma})$

where D_Z is the terminal native engine disposition under the frozen prompt contract.

This closes the native ontology/admissibility question for these labels. Any real distinction conventionally described as solid, liquid, gas/vapor, or plasma must enter SEAM through explicit structural-state content — relational configuration, excitation, charge redistribution, density/boundary condition, or other already-admitted state variables — and may then be projected downstream into a conventional phase label. Empirical phase boundaries are comparator/projection data, not an open native mathematical primitive.

Evidence: `Evidence/Locked_Runs/SEAM_88_ELEMENT_PHASE_LABEL_COMPLIANCE_01/`

Compliance matrix: `COMPLIANCE_MATRIX.csv`

Lock manifest SHA-256: `44b4bc3032009b42296d0c3d0f91aa5b608fce985d64e03c132c0afc143d6686`

S.15 H2O thermal atomic invariance — R42

Locked run `H2O-THERMAL-ATOMIC-INVARIANCE-01` tests whether the isolated native atomic state changes when the same H2O material population is represented at three thermal conditions corresponding downstream to solid, liquid, and vapor water.

Frozen atomic states:

$$H : (p, n, e) = (1, 0, 1), \quad O : (p, n, e) = (8, 8, 8).$$

The three declared environmental conditions are 263.15 K, 298.15 K, and 383.15 K, each with the same declared 1 atm context. Temperature and pressure are not inserted as atomic coefficients; they remain event/environmental descriptors.

Each H execution resolves identically:

$$E_{static} = 1, \quad M_{attr} = 0.5, \quad \chi_e = 0.6931471804599453, \quad \Theta_{ESAM} = 0.500000000005.$$

Each O execution resolves identically:

$$E_{static} = 0.5, \quad M_{attr} = 0.1663780661615406, \quad \chi_e = 1.100345392255312, \quad \Theta_{ESAM} = 0.33275613245653.$$

Therefore:

$$C_H^{solid} = C_H^{liquid} = C_H^{vapor}$$

and

$$C_O^{solid} = C_O^{liquid} = C_O^{vapor}$$

within the tested atomic resolver contract.

This closes the isolated atomic-base-change hypothesis for ordinary H₂O phase change. The thermal distinction occurs above the isolated-atom layer in the complete substrate state. Canonical AEPS already assigns thermal restructuring to ΔY_{th} , which may alter phase or density structure without introducing a separate physical law. The required causal placement is therefore:

$$\{C_H^*, C_O^*\}_{fixed} \rightarrow \Delta Y_{th} \rightarrow (\mathcal{R}, F, O_{ij}^{nm}, S_{field}, S_{coupling}, \rho, \mathcal{B})_{new}$$

The H-O and H₂O-H₂O collective relational/field state, not the isolated atomic identity, carries the ordinary solid/liquid/vapor distinction.

Matter-state four-regime projection condition — R43

R41 establishes that the conventional labels `solid`, `liquid`, `gas`, and `plasma` do not themselves instantiate native SEAM states. R42 establishes, for H₂O, that the isolated H and O atomic base outputs remain invariant across ordinary solid/liquid/vapor thermal conditions. R43 therefore binds matter-state classification to the excitation/confinement condition of the complete material state rather than to the phase word itself.

For each atomic/material base identity, define the native phase-condition coordinate

$$X_{\text{phase},Z} = (E_{\text{exc}}, B_{\text{conf}} \mid C_Z^*)$$

where E_{exc} is the excitation state of the complete material configuration and B_{conf} is the native boundary/confinement/density condition. The required phase-boundary object is

$$\Theta_Z^{\text{phase}} = \{\Theta_{SL,Z}, \Theta_{LG,Z}, \Theta_{GP,Z}\}.$$

The four downstream regimes are

$$\begin{array}{ll} \text{solid :} & E_{\text{exc}} < \Theta_{SL,Z}(B_{\text{conf}} \mid C_Z^*), \\ \text{liquid :} & \Theta_{SL,Z} \leq E_{\text{exc}} < \Theta_{LG,Z}, \\ \text{gas/vapor :} & \Theta_{LG,Z} \leq E_{\text{exc}} < \Theta_{GP,Z}, \\ \text{plasma :} & E_{\text{exc}} \geq \Theta_{GP,Z}. \end{array}$$

Temperature and pressure are empirical projections of this frozen native condition,

$$T = \Phi_T(X_{\text{phase},Z}), \quad P = \Phi_P(X_{\text{phase},Z}),$$

and do not define the upstream native state. The first two ordinary condensed-matter transitions may preserve isolated atomic identity while altering collective relations, field overlap/coupling, density, geometry, and boundary arrangement. The gas-to-plasma boundary may additionally change atomic/ionic closure because persistent charge-separated structures become part of the selected complete state.

The original R41 352-cell result remains unchanged: a phase label alone does not instantiate a native state. R43 supplies the positive mathematical condition required before a downstream phase label may be assigned.

R44 — electron stripping/reclosure structural-energy boundary

For each element $Z=1..88$ the frozen native control holds p and n fixed and changes only electron count e from Z to 0 . The retained atomic state is

$$A_Z(e) = (E_{\text{static}}, M_{\text{attr}}, \chi_e, \Theta_{\text{ESAM}}).$$

For each successive stripping step,

$$\Delta A^-[Z,q] = A_Z(e-1) - A_Z(e),$$

and for exact reclosure,

$$\Delta A^+_{Z,q} = A_Z(e) - A_Z(e-1).$$

The locked execution returns

$$\boxed{\Delta A^-_{Z,q} + \Delta A^+_{Z,q} = 0}$$

for all 3,916 successive electron steps across all 88 elements, with maximum componentwise residual exactly 0.0. Thus an isolated stripping/reclosure cycle returns to the original native atomic state with zero structural remainder. The active runtime does not supply a scalar energy map from this faceted structural differential to eV or J; conventional ionization/recombination energy remains downstream projection/comparator material and cannot be inserted upstream. A net-output claim therefore requires a larger complete-state cycle containing an additional environmental contribution rather than electron stripping/reclosure alone.

R45 — closed-cycle no-surplus notation

The R44 88-element stripping/reclosure sweep closes the isolated electronic cycle under the same complete-state accounting used elsewhere in SEAM. The result is consistent with the project prediction that zero-point extraction and perpetual-motion operation are not available as source mechanisms.

Define the minimum thermally accessible excitation state for a retained structure as

$$E_{\text{exc}}^{\min}(C, \mathcal{B})$$

and an isolated closed structural cycle as

$$\begin{aligned} \mathcal{C}: \quad Y_0 &\rightarrow Y_1 \\ &\rightarrow \cdots \rightarrow Y_n = Y_0. \end{aligned}$$

Its native cycle differential is

$$\Delta_C^{\text{SEAM}} = \sum_{i=0}^{n-1} \mathfrak{D}_{\text{struct}}[Y_i \rightarrow Y_{i+1}]$$

with signed state change retained facet-by-facet. For the executed stripping/reclosure cycle,

$$\Delta A_{\text{strip}}(Z, q) + \Delta A_{\text{close}}(Z, q) = 0$$

for every one of the 3,916 matched transitions, hence

$$\Delta_C^{\text{SEAM}} = 0 \quad \text{when} \quad Y_n = Y_0 \text{ and no external complete-state contribution is present.}$$

Accordingly, a claimed net output must satisfy

$$\Delta_{\text{out}}^{\text{SEAM}} \neq 0 \Rightarrow \exists \Delta Y_{\text{ext}} \neq 0$$

where ΔY_{ext} denotes an actual environmental, boundary, field, matter-transfer, or other declared contribution to the complete event state. Returning exactly to the predecessor state without such a contribution cannot generate a residual native surplus.

This is the canonical SEAM no-perpetual-cycle statement. It does not import conventional energy-conservation law upstream; it follows from the executed complete-state reversal symmetry and SEAM state accounting. Likewise, $E_{\text{exc}}^{\text{min}}$ is not an extractable reservoir by definition: it is the minimum admitted excitation condition of the retained structure, not a free-output term.

R47 retained-entity recursive pool law

For a current pool of complete entities

$$\mathcal{P}_i = \{E_1, E_2, \dots, E_n\},$$

a candidate group G is resolved by the unchanged entropy selector

$$C_G^* = \arg \max_{C \in \mathcal{A}(G)} S[C].$$

If the complete candidate is entropy-positive relative to the same constituents retained as separate entities,

$$\Delta S_G = S[C_G^*] - S[C_G^{\text{separated}}] > 0,$$

then the selected complete configuration is retained as a subsidiary entity

$$E_G = \mathfrak{R}(C_G^*) = (\{C_a^*\}_{a \in G}, \mathcal{R}_G),$$

and the pool advances by

$$\mathcal{P}_{i+1} = (\mathcal{P}_i \setminus G) \cup \{E_G\}.$$

E_G is not reduced to an aggregate shell vector. On the next cycle it is consumed as a multi-node complete entity, so entity+atom and entity+entity candidates preserve all internal constituent states and relations. No chemistry/product identity enters candidate generation or selection.

A retained multi-node entity is represented without flattening. The current multi-node numerical selector is convergence-qualified under the declared refinement rule; only convergence-qualified differentials may alter the retained pool.

Multi-center numerical qualification rule

For $N \geq 3$, an entropy-selected successor is admissible only after the multi-center field and overlap integral is numerically stable under an independent refinement sequence. In particular,

$$\text{sign}(\Delta S_N)$$

must remain invariant over the declared convergence sequence, and the final change between refinements must fall below the frozen numerical tolerance. If the sign changes in a posed application run, that candidate is rejected by the convergence gate and cannot alter the pool; this is a terminal adjudication rule, not unfinished framework work.

R49 — unreduced cyclic-event and boundary-transfer mathematics

R49 binds the Bo1 oxygen-evolving-complex (OEC) evidence to the general SEAM state-accounting rules without importing conventional energetic currencies or allowing algebraic cancellation of equal-composition molecules.

Persistent atom and entity identity

For a complete physical state, atomic and retained-entity identities remain explicit:

$$Y_i = (\{A_j\}, \{E_k\}, \mathcal{R}_i, F_i, \Xi_i, \mathcal{B}_i)$$

where A_j is an individually retained atomic identity, E_k is a retained multi-atom entity, $\mathcal{R}_i$ is the complete relational state, F_i is the field state, Ξ_i is the unresolved/native transfer-residual state when present, and $\mathcal{B}_i$ is the declared environmental/boundary state.

Atomic continuity is identity-preserving:

$$A_j^{(0)} \rightarrow A_j^{(1)} \rightarrow \dots \rightarrow A_j^{(n)}.$$

Two molecules with equal formula are not algebraically interchangeable in an atom-resolved execution. A water entity entering from a boundary and a water entity appearing later in another location remain separate objects unless identity-preserving state evolution explicitly connects them. No input/output term is cancelled merely because its composition matches another term.

Unreduced physical transition with explicit exports

Every physical interaction producing state evolution retains the existing global invariant

$$\text{structure} \rightarrow \text{perturbation} \rightarrow \text{structure}$$

and is written for an open event as

$$Y_i + \Delta Y_i \rightarrow Y_{i+1} + R_i.$$

R_i is the explicit event output: released subsidiaries, transported constituents, or separately represented state-transfer channels. It remains a vector/set of distinguishable outcomes until a native common projection has actually been derived. Conventional energy carriers or biochemical bookkeeping units cannot be substituted for R_i upstream.

Retained-state cycle versus event-level cycle

For a retained subsystem that returns to its predecessor state,

$$\begin{aligned} \mathcal{C}: \quad Y_0 &\rightarrow Y_1 \\ &\rightarrow \dots \rightarrow Y_n = Y_0, \end{aligned}$$

the retained-state differential closes:

$$\Delta_{\mathcal{C}}^{\text{retained}} = 0.$$

This does **not** imply that every event channel is zero. The complete cycle record is

$$\mathcal{R}_{\mathcal{C}} = (R_0, R_1, \dots, R_{n-1}, \Delta_{\mathcal{C}}^{\text{retained}}).$$

For the R49 OEC block, the empirically constrained event-level cycle is

$$R_{\text{OEC,cycle}} = (N_e = 4, N_H = 4, N_{O_2} = 1, \Delta S_{\text{OEC}} = 0).$$

N_e denotes four separately observed electron-state transfer events; N_H denotes four separately observed proton-release events; N_{O_2} denotes one released molecular-oxygen subsidiary. These counts are retained independently and are not algebraically reduced into a scalar energy quantity.

The proton-release ordering is itself part of the unreduced event sequence:

$$(S_0 \rightarrow S_1, S_1 \rightarrow S_2, S_2 \rightarrow S_3, S_3 \rightarrow S_0) = (1, 0, 1, 2).$$

The two substrate-water deliveries remain separate events, one during $S_2 \rightarrow S_3$ and one during the final $S_3 \rightarrow [S_4] \rightarrow S_0$ sequence. They are never cancelled against other water entities.

Structural path length is not exported energy

For a scalar diagnostic s_i evaluated along a retained-state trajectory, define the structural path length

$$L_s = \sum_i |s_{i+1} - s_i|.$$

L_s measures traversal in that diagnostic coordinate only. It is not an energy, entropy export, or transferable scalar unless an independently defined native projection maps both source and receiver complete states into the same conserved quantity. Therefore neither L_s nor one half of L_s may be relabeled as exported energy. The prior provisional interpretation of 0.05811475288 as exported energy is excluded by this rule.

Boundary transport with identity preservation

When measured permeability data are used as an empirical boundary condition, the conventional transport relation may be retained downstream as

$$J_X = -P_X \Delta C_X$$

where P_X is an empirically measured permeability and ΔC_X is the concentration difference across the declared boundary. This relation does not replace the SEAM interaction law; it constrains the boundary transition observed for an already resolved entity.

Identity is preserved during transport:

$$X_{\text{local}}^{(m)} \rightarrow X_{\text{external}}^{(m)}.$$

For the R49 OEC evidence, once an 02 entity is released from the retained catalytic state, membrane permeability supports its diffusive export from the local photosynthetic compartment. The released-entity ancestry question is closed at the declared evidence boundary: transport preserves the resolved 02 entity, while atom-specific ancestry is required only when the input contract contains atom-identity evidence.

Source/sink continuity across a permeable boundary

For a species x crossing into a compartment, sustained influx requires a persistent receiving-side difference. If no downstream state removes x from the local free pool, then

$$\Delta C_X \rightarrow 0 \quad \Rightarrow \quad J_X \rightarrow 0.$$

A downstream retained successor can maintain the difference without being named in advance:

$$X_{\text{external}} \rightarrow X_{\text{local}} \rightarrow E_{\text{retained}}.$$

For B01 carbon, the transport/continuity condition and successor handling are closed by the active photosynthesis composite contract; no conventional carbon-product pathway is inserted upstream.

Open-system matter ingress

Environmental matter support is represented through explicit boundary events. If a local soluble pool requires replenishment,

$$\mathcal{B}_i \rightarrow X_{\text{local}}^{(\text{new})}$$

adds a newly identified entity from the declared environment. No same-formula local entity is cancelled against the ingress. This is the canonical bookkeeping rule for water replenishment or any other open-system matter support.

R49 closure boundary

The event-level OEC retained-state cycle and immediate boundary interfaces are closed at their declared scope. The complete photosynthesis continuation, carbon handling, retained/dissipative accounting, and identity-boundary adjudications are bound by the active composite closure and their respective current run contracts.

Evidence: Evidence/B01/OEC_CLOSED_BLOCK_R49/ and
Evidence/Provenance/R49_OEC_CLOSED_BLOCK_INTEGRATION.json .

R50 — entropy-selected persistent organization under prebiotic Continuum conditions

The R50 evidence integration records a form-first prebiological continuation in which a sterile shallow-brine material state is allowed to organize under the retained atomic near-/far-field interaction and complete-state entropy selection. The resulting partial compartment supports a retained excitation-accessible state and retained successors without requiring an inherited blueprint as an initial condition.

The executed structural chain is

$$C_{\text{sterile}} \rightarrow C_{\partial} \rightarrow C_{\text{retained}} \rightarrow C_{\text{successor}} \rightarrow C_{\text{heterogeneous}} \rightarrow C_{\text{driven}}(n).$$

Here C_{partial} denotes the transient inside/outside differential state, C_{retained} the compartment-dependent retained promoted state, $C_{\text{successor}}$ the retained successor whose support is enhanced by the retained parent, $C_{\text{heterogeneous}}$ the unconstrained electrolyte continuation selected from the full available material set, and $C_{\text{driven}}(n)$ the retained state after repeated day/night continuation with the R49 photosynthetic event recurrence applied during daylight.

The R88 executed result retains the entity through eight complete nights and grows the retained state from 7 to 16 nodes, with 7/8 cycles accepting new material. The applied R49 event stream contributes 32 electron-state transfers, 32 proton-state transfers, 8 O₂ exports, and 16 substrate-water delivery events over the eight light intervals. This establishes persistence and growth within the stated test contract. It does not establish spontaneous construction of the mature OEC, RNA/DNA, reproduction, or universal inevitability of life.

The Continuum biological-attractor claim is closed at its declared bounded scope: the executed material/Manifold contract demonstrates persistent cell-like organization as an entropy-selected attractor in the admitted basin. No universal inevitability-of-life claim is asserted by that certificate.

R51 — fixed thermodynamic stochastic-field accounting closure

Locked run `THERMODYNAMIC-STOCHASTIC-FIELD-ACCOUNTING-01` binds the already-declared complete-state field entropy, confinement response, universal interaction cycle, and R49 event law into one fixed thermodynamic accounting example. No new thermodynamic primitive is introduced.

For each step,

$$Y_i + \Delta Y_i \rightarrow Y_{i+1} + R_i.$$

The complete resolved post-state is retained in `Y_(i+1)`; only explicitly represented boundary/event outputs enter R_i . In the frozen closed example, $R_i = \emptyset$ for all steps. The $A \rightarrow B \rightarrow C$ confinement sequence changes S_{field} and eventually the selected relational outcome while the atomic states remain fixed. The formal $C \rightarrow A$ return restores the exact frozen starting state, so

$$\Delta_{\text{cycle}}^{\text{retained}} = 0.$$

The nonzero structural traversal length is retained only as a diagnostic and is not energy, heat, work, or entropy export. H₂O thermal atomic-invariance evidence supplies the independent scientific control that the tested phase distinctions do not require changes in isolated H/O atomic structure.

Standing: the retained-versus-explicit-output thermodynamic accounting formalism is closed for the fixed example. Dimensional temperature/pressure calibration, latent heat, heat capacity, and material-specific phase thresholds remain downstream empirical/application projections.

Evidence: `Evidence/Locked_Runs/THERMODYNAMIC-STOCHASTIC-FIELD-ACCOUNTING-01/`.

R52 — persistent-identity ancestry resolver

R52 formalizes the existing atom-identity continuity rule by separating observed/model labels from persistent physical identity. Let L_t be a structural/model label and A_j a persistent atom identity. A binding

$$\beta_t : L_t \rightarrow A_j$$

is admissible only when the evidence/run contract proves that correspondence. An exported entity ancestry is resolved only when every constituent has a unique proven `beta_t` binding and a continuous identity path into the exported entity. Positional labels, candidate associations, or set differences among unbound labels cannot substitute for identity.

The current OEC ancestry resolver closes both branches of the identity contract: identity-complete inputs resolve ancestry, while inputs without atom-identity evidence terminate at the declared evidence boundary. **Standing:** ancestry-resolution formalism and its evidence-boundary adjudication are CLOSED.

R53 — fixed numerical thermal Hamiltonian consequence

The already-declared constant-temperature consequence

$$\Delta H_{th,XY}(N_r) = T[S^*(N_r) - S_\infty]$$

is now executed as a locked fixed example for Cu-Cu. The complete native pair states are resolved first at $N_r = 6.2$ and the separated control $N_r = 20.0$; only after those states are frozen is the 298.15 K downstream projection applied.

The run returns $\Delta(S/k_B) = 0.004515601154116311$ and $\Delta H_{th} = 1.8588043139458746e - 23J$. The reverse transition is exactly antisymmetric and the forward/reverse cycle residual is zero.

Standing: numerical fixed-example closure for the thermal Hamiltonian component. This does not supply the nonthermal/full E_H value, bond dissociation energy, or the common OEC electron/proton transfer magnitude.

Evidence: [Evidence/Locked_Runs/CUCU-THERMAL-HAMILTONIAN-PROJECTION-01/](#) .

R56 — fixed faceted structured-transfer closure

Locked run [STRUCTURED-TRANSFER-FACETED-01](#) executes the canonical interaction object

$$T[Y_i^{pre}, Y_i^{post}] = \mathcal{D}_{struct}[Y_i^{pre} \rightarrow Y_i^{post}]$$

without scalar reduction. The fixed example retains numeric entropy/coupling/confinement/count facets, the native distance vector, and the categorical relational transition. Direct reversal is exact. Over the closed $A \rightarrow B \rightarrow C \rightarrow A$ cycle every additive facet returns zero componentwise and the categorical relational state returns to its original value.

The run also falsifies promotion of ΔS_{total} to the universal transfer object: the $B \rightarrow C$ transition contains a relational category change, overlap-count change, coupling change, and vector geometry change that are not encoded by the entropy scalar alone.

Standing: the universal native transfer object is the faceted structured difference and is CLOSED as a fixed executable example. A universal scalar transfer magnitude is not a missing first-principles object. Scalar quantities are admissible only through independently declared downstream projections such as the R53 thermal Hamiltonian component.

Evidence: [Evidence/Locked_Runs/STRUCTURED-TRANSFER-FACETED-01/](#) .

R57 — pair/event time-metrology boundary closure

Locked run [PAIR-TAU-EVENT-METROLOGY-01](#) distinguishes physical evolution time from static selected-state comparison. The native resolver remains

$$T(X) = n_{Cs}(X)/9,192,631,770.$$

An actual event carrying $n_{Cs} = 9, 192, 631, 770$ resolves exactly to one second. The static Cu-Cu selected/reference states from R53 contain no declared physical evolution interval and therefore return `NOT_EVALUABLE_NO_PHYSICAL_INTERVAL` ; $\Delta\tau_{SEAM} = 0$ is not inferred. N_r , entropy difference, frequency, dimensional distance, and wall-clock runtime are prohibited substitutes.

Standing: time/event mapping formalism CLOSED. A numerical molecular-event interval requires explicit Cs-count evidence for that event. Static pair-state comparison alone does not define a pair clock.

Evidence: `Evidence/Locked_Runs/PAIR-TAU-EVENT-METROLOGY-01/` .

R58 — full-Hamiltonian dependency boundary

The canonical full Hamiltonian form remains

$$H^{(J)} = E_0\tau_{SEAM} + \int T dS.$$

R53 provides a locked numerical fixed example for the constant-temperature thermal component. R57 closes the physical-time admission rule: τ_{SEAM} is resolved from an actual Cs-133 transition-count interval and may not be inferred from a static pair-state comparison.

A sequential canonical and Hamiltonian-evidence audit found no current canonical numerical constructor or selected-state extraction rule for E_0 . The notation $E_H[C*]$ remains a Hamiltonian facet/correspondence of the complete selected state and does not, by itself, generate a numerical E_0 value.

Therefore a full/nonthermal numerical Hamiltonian execution has the current terminal

`NOT_EVALUABLE_NATIVE_E0_CONSTRUCTOR_UNBOUND`

No zero assignment, entropy-derived scale, native-distance-derived scale, conventional bond energy, comparator fit, or software runtime may substitute for the declared Cs-count event interval. This is a closed input-contract boundary for the requested numerical projection.

R61C — fixed SEAM/SI equivariant derivative projection

SEAM-to-SI commonality is a downstream equivariant representation of an already-derived native quantity, not an empirical calibration or causal input. For any resolved derivative `Q_SEAM` , the fixed map Φ_Q preserves dimensional identity and algebra while leaving upstream SEAM derivation unchanged.

$$Q_{SEAM} \rightarrow \text{freeze} \rightarrow \Phi_Q(Q_{SEAM}) = Q_{SI}$$

The count-resolved time anchor is exact for the same physical interval: $T_{SEAM} = T_{SI}$; frequency is reciprocal after that resolution. The same projection discipline applies to length and downstream derivatives. Target observations may adjudicate the sealed SI projection but may not define or tune the native quantity.

Locked evidence: `Evidence/Locked_Runs/SEAM-SI-EQUIVARIANT-DERIVATIVE-PROJECTION-01/` . Terminal: `PASS_SEAM_SI_EQUIVARIANT_DERIVATIVE_PROJECTION_CLOSED` .

R62C — H2 first-principles atomic-field entity closure

Hydrogen binding is evaluated from the already-resolved atomic first-principles field state, not from a separately selected radial profile. The retained $Z=1$ atomic state has first-shell occupancy 1 of capacity 2, boundary density 0.5, and an open boundary. For two retained H atoms, each supplies exactly the other atom's one remaining admissible first-shell position, giving reciprocal closure $2/2$ while preserving both constituent identities.

The resolved successor is promoted as the standalone retained entity $E_{H2_retained}$. Its internal H-H relation and constituent identities remain part of the complete state; it is not flattened into a synthetic $Z=2$ atom. Subsequent interactions consume the H2 entity field as a top-level state.

Locked run `H2-FIRST-PRINCIPLES-ATOMIC-FIELD-CLOSURE-01` verifies the native closure, exact count/charge conservation, no radial-profile input, no empirical input, no flattening, deterministic replay, and retained-entity recursion interface. Independent NIST H2 spectroscopy is revealed only after native freeze and supports a bound molecular ground-state entity.

Standing: H+H first-principles atomic-field admissibility/binding and H2 retained-entity promotion are scientifically closed within the declared existence/identity scope. Numerical bond length, dissociation energy, and later H2 reaction outcomes remain separate derivative/application projections.

Evidence: `Evidence/Locked_Runs/H2-FIRST-PRINCIPLES-ATOMIC-FIELD-CLOSURE-01/` .

Appendix N — Cellular Information-Carrier Molecular Closure Application (R106C)

This appendix binds the existing complete-state molecular architecture to the cellular information-carrier branch. It introduces no new interaction law.

For already-closed atomic states $\mathcal{S}_i$, a candidate information-carrier molecule is

$$C_{\text{mol}} = (\{\mathcal{S}_i\}, \mathcal{R}).$$

Its admissible native state is selected only from the complete relational family:

$$C_{\text{mol}}^* = \arg \max_{C \in \mathcal{A}_{\text{mol}}} S[C].$$

The complete field is constructed from all retained shell fields; pairwise bond labels or empirical target geometry do not substitute for the complete state. A closed molecule must exhibit a finite internal selected configuration and local restoring consistency through the already-declared Hamiltonian/force continuation.

Each molecular component is closed independently. Only then may two closed molecular entities be introduced to one another and re-resolved as a higher-level complete state. The information carrier is therefore a distinct entropy-selection layer:

$$C_{\text{carrier}}^* = \arg \max_{C \in \mathcal{A}(M_1^*, \dots, M_k^*)} S[C].$$

Legacy biochemical labels are downstream comparator names only. The required order is native atomic closure → native molecular closure → native carrier closure → empirical comparison → legacy identification.

R106C application evidence is retained at

Evidence/Cellular_Development_Fork_B/Records/R106C_FORMAL_FIRST_PRINCIPLES_MOLECULAR_CLOSURE.md .

R67 — Blueprint / Manifold / Entropy conditional realization

For inherited biological construction the native state is conditional on both retained blueprint structure and local Manifold context:

$$C^* = \operatorname{argmax}_{C \in \mathcal{A}(G, M_{\text{local}}, Y_i)} S[C]$$

The blueprint constrains admissibility but is not itself a selector and does not uniquely determine a realized structure independently of local Manifold state, available material, or retained history. Persistent environmental/history changes are evaluated by the same complete-state selector; their aggregate differential persistence is the downstream phenomenon conventionally described as adaptation/evolution.

R69C — Dual-route pathogenesis ontology

Pathogenesis is not a separate selector or force. The active biological state relation is:

$$(G, M_{\text{local}}, Y_i, H_i) \rightarrow A \rightarrow \operatorname{argmax}_C S[C] \rightarrow C^*$$

A persistent altered path may enter through either:

- ΔG : a persistent change in inherited/retained blueprint (RNA/DNA carrier) structure; or
- ΔM_{local} : a persistent change in local Manifold/environment/material/history.

Both routes act by changing the admissible configuration space. The same entropy/Xi selector resolves the realized successor. Conventional hereditary and environmental/acquired pathogenesis are therefore downstream classifications of different causal entry points into the same native state-evolution law.

R70 — Pharmacology as introduced-entity biological interaction

Pharmacology does not introduce a new first-principles selector or force. An introduced entity enters the ordinary biological complete-state interaction:

$$\boxed{Y_i + E_x \rightarrow \mathcal{A}(G, \mathcal{M}_{\text{local}}, Y_i, H_i, E_x) \rightarrow \operatorname{argmax}_C S[C] \rightarrow Y_{i+1}}$$

The conventional labels `drug`, `toxin`, `nutrient`, and `immune trigger` are downstream classifications based on the identity, use, and realized biological consequence of the introduced entity. They do not modify the native operator.

Locked run `S030-PHARMACOLOGY-INTRODUCED-ENTITY-LOCK-01` uses the existing `R105C` cycle-9 introduction event. A `P` entity is admitted only after three independent near-versus-separated complete-state entropy audits are positive in both tested light and dark states; it is assigned persistent identity `R105C_009_P_016` and remains retained through six subsequent executed cycles. No pharmacology-specific term appears in the source runner.

Standing: S-030 PHARMACOLOGY — CLOSED AS ONTOLOGY / NATIVE-MECHANISM CLAIM. Dose-response, pharmacokinetics, efficacy, toxicity, and any named drug-target mechanism remain separate application claims.

R70C — pharmacological Manifold perturbation

A pharmacological effect is a direct entity interaction that changes the local biological Manifold and therefore the admissible successor space:

$$E_{\{drug\}} + E_{\{target\}} \rightarrow E_{\{bound\}} \rightarrow \Delta \mathcal{M}_{\{local\}} \rightarrow \Delta \mathcal{A} \rightarrow \arg \max_C S[C] \rightarrow C^*.$$

The sirolimus fixed example binds FKBP12 and inhibits mTOR. The shared mTOR node controls immune proliferation, growth/repair, angiogenesis/vascular permeability, metabolism, and other cellular state functions. A single upstream interaction can therefore produce multiple downstream conventional side-effect labels without introducing one mechanism per symptom. The locked adverse-effect suite retains all unexplained residuals explicitly.

R70D — virtual pharmacology trial consequence

The pharmacology ontology permits prospective virtual trial execution because the drug-target interaction is itself an ordinary complete-state event. Once the primary interaction is frozen, its system-level consequences can be propagated before any empirical outcome key is opened:

$$\begin{aligned} &E_{\{drug\}} + E_{\{target\}} \\ &\rightarrow E_{\{bound\}} \\ &\rightarrow \Delta \mathcal{M}_{\{local\}} \\ &\rightarrow \Delta \mathcal{A} \\ &\rightarrow \arg \max_C S[C] \\ &\rightarrow \{C_k^*\}_{\{systems\}} \\ &\rightarrow \Pi_{\{virtual\}}. \end{aligned}$$

$\Pi_{\{virtual\}}$ is the sealed predicted efficacy/toxicity consequence set. Physical observations are downstream adjudicators and may not define or tune $\Pi_{\{virtual\}}$ after its freeze. The R70C rapamycin suite executes this ordering programmatically and replays identically.

This yields the formal capability:

$$\boxed{\text{candidate interaction}} \rightarrow \text{virtual consequence suite} \rightarrow \text{physical adjudication}.$$

R71C – Thermodynamic quantities as facets of one complete state

The controlling thermodynamic ontology is faceted rather than factorized. A physical entity resolves once as a complete retained state:

$$\boxed{Y_X = C_X^*}$$

Temperature, pressure, phase, density, and related conventional thermodynamic quantities do not instantiate separate causal states. They are separate questions about the same already-resolved entity:

$$\boxed{T_X = \Phi_T[C_X^*]}, \quad \text{qqquad} \boxed{P_X = \Phi_P[C_X^*]}.$$

Likewise, for a transition:

$$\boxed{\mathop{\rm D}\nolimits_{\rm struct}[C_{\{X,i\}}^*\rightarrow C_{\{X,j\}}^*]}$$

is the complete native transfer. Heat is one conventional interrogation of that transfer:

$$\boxed{Q_X = \Phi_Q[\frac{D_{\rm struct}}{D_{\rm struct}}]}$$

and is not an independent substance, interaction, or selector. This corrects any R71 wording that could be read as treating excitation/confinement, temperature, pressure, and heat as separate thermodynamic layers. Excitation, confinement, field, relation, material, and history are retained facets within the complete state itself.

R72 – X1 boundary-relation inventory closure

Fork B R107J adds a canonical molecular-construction boundary closure for the native object



From the closed atomic shell inventories, the initial open-boundary capacity is 30. The retained construction consumes exactly 15 relation units, two boundary positions per relation, and terminates with zero unassigned H/C/N/O boundary positions.

The locked retained relation-type progression is:

$$H - O \rightarrow N - N \rightarrow N - O \rightarrow H - N \rightarrow H - H \rightarrow C - O \rightarrow C - C \rightarrow N - O, C - N \rightarrow 6 \times C - C.$$

When candidate closures are entropy-indistinguishable within the resolved numerical band, R107J records them as one simultaneous event set rather than imposing an artificial serial order.

This closes only the boundary-relation inventory. It does not authorize a legacy molecular name, a unique 3-D geometry, complete-field boundedness, Hamiltonian closure, restoring-force closure, RNA/DNA assignment, or sequence decoding.

R73C — Six-state thermal structural path

A retained material entity may occupy an ordered sequence of complete states

$$Y1 \rightarrow Y2 \rightarrow Y3 \rightarrow Y4 \rightarrow Y5 \rightarrow Y6$$

with each adjacent transition represented by $D_{\text{struct}}[Y_i \rightarrow Y_i + 1]$. For the recovered Structural Transfer Relation fixture, the temperature facets are 65, 75, 85, 95, 105, and 110 F, while atomic identity remains invariant and predecessor history is retained.

Temperature is therefore a facet $T_i = \Phi_T[Y_i]$ of the already-resolved complete state. It is not the native state constructor or transition step size. The observed facet increments 10,10,10,10,5 F are nonuniform, directly falsifying a fixed-temperature-step interpretation of the native path.

R73E — Six-state matter ontology and phonon standing

For one retained material entity, first-principles atomic/field conditioning determines the admissible complete-state domain under the same entropy/Xi selector. The six-state construction is therefore represented as six stable complete-state domains separated by five structural stability boundaries.

Temperature, pressure, heat, and heat capacity are questions/facets of the resolved complete state or complete structural transition. The observed low-temperature relation $C_V \propto T^3$ is retained as an empirical thermal facet. No quantized phonon/normal-mode ontology is required as the primitive causal mechanism producing that response.

R74 — Acoustic facet of material structural transfer

For a retained material entity with local relations $\mathbf{R}$, a source perturbation may be continued through adjacent complete states by the already-defined structured transfer $D_{\text{struct}}[Y_i \rightarrow Y_i + 1]$. The ordered material transfer is the native event. $\mathbf{sound}$ is the downstream acoustic classification/facet of that event. Frequency, wavelength, propagation speed, amplitude and pressure are questions about the realized path/state; they are not first-principles acoustic operators.

R75 — Vibration as recurrent structural evolution

For a retained material body, vibration is represented as recurrent traversal of complete states Y_i connected by $D_{\text{struct}}[Y_i \rightarrow Y_i + 1]$, with predecessor/history retained. Displacement/deformation, amplitude, phase, period and frequency are downstream facets of that realized trajectory. No vibration-specific primitive is introduced.

R76 — Fluid flow as aggregate constituent transfer

For an aggregate material state, persistent constituents may cross declared local boundaries through ordinary D_{struct} state transitions while retaining identity and total inventory. A continuum flow observable is then a projection of those realized crossings over a declared spatial boundary and Cs-resolved interval. Velocity, flow rate, pressure-gradient descriptions, viscosity and related fluid variables are aggregate questions about that transfer, not primitive interaction operators.

R77 — Deterministic sensitivity of admissible construction space to local Manifold state

The biological construction rule remains:

$$(G, M_i, Y_i) \rightarrow A_i \rightarrow \operatorname{argmax}_C S[C] \rightarrow C_i^*.$$

R1070 freezes `G` /upstream X1 inventory and varies only observed material-state components of `(M_i, Y_i)`. Across the complete 256-state enumeration, distinct local material states produce distinct admissible boundary inventories and degree signatures. Thus $\text{same}G + \Delta(M, Y) \rightarrow \Delta A$ is directly executable without random, mutation, adaptation, differentiation, or evolutionary selectors.

Exact replay of the same complete perturbation family is byte-stable at the exported-state level, establishing deterministic reproducibility for the tested mapping from complete local state to admissible construction space.

R78 — Now-to-Now Deterministic Manifold Principle

Canonical proof authority:

`Evidence/Canonical_Principles/R78_NOW_TO_NOW_DETERMINISTIC_MANIFOLD_LOCK/LOCKED_PROOF_POINTER.json`. Complete state progresses now-to-now; finite observation is a projection. The locked accelerometer-only example demonstrates continuous projected-Manifold change, deterministic replay, and held-out short-horizon predictive information. Thermodynamic statistical/chaotic descriptions are downstream; six-state boundaries remain first-principles structural boundaries; biology retains its existing selector and points here for the general apparent-randomness relation.

R79 — Diffusion as entropy-selected constituent redistribution

For a retained aggregate with persistent constituents distributed over an admissible spatial relation graph, conserved candidate distributions are configurations `C`. Their already-defined configuration entropy $S_{\text{config}}[C] = k_B \ln W(C)$ participates in the unchanged entropy selector. The selected successor may be realized

through ordinary adjacent D_{struct} constituent transfers with identity conserved. Diffusion is the downstream aggregate classification of the resulting concentration redistribution. Fick relations, diffusivity, random-walk descriptions, and Brownian statistics are quantitative projections/models of the realized transfer and are not native selectors.

R80C — Viscosity as downstream constitutive projection

A retained material under relative boundary motion resolves through its complete inter-layer relation structure and D_{struct} now-to-now transfer. Different relation structures produce different relative-state transfer and residual shear under an identical drive. Viscosity is therefore a conventional constitutive projection of that realized response, not a separate primitive cause.

R81 — Reaction kinetics as Cs-counted structural transition rate

For a declared molecular population and physical interval, the native kinetic observable is the number of resolved structural transitions divided by the Cs-133-resolved interval: $r = N_{\text{transition}}/T(X)$. A conventional rate coefficient or rate constant is introduced only downstream when an empirical/phenomenological rate equation is fitted or evaluated against the observed transition history. Rate constants therefore summarize kinetics; they do not constitute additional primitive causes of structural transition.

R90 — FP-05 re-entrant complete-state selection / iron-allotropy principle lock

The FP-05 selector does not imply that the identity of the maximizing configuration must vary monotonically with any monotone perturbation coordinate. For a perturbation coordinate x , two distinct admissible complete configurations may carry scalar branches whose difference changes sign more than once while the governing selector remains fixed:

$$C^*(x) = \arg \max_{C \in \mathcal{A}(x)} S[C].$$

The locked constructive counterexample freezes

$$S_B(x) = 0, \quad S_A(x) = (x - 0.5)(x - 1.5), \quad x \in \{0, 1, 2\}.$$

The perturbation coordinate is strictly monotone, yet the unchanged selector returns

$$\boxed{A \rightarrow B \rightarrow A}.$$

Temperature, iron phase labels, transition temperatures, lattice constants, and the empirical phase sequence are prohibited from the native selection stage. After the native result is frozen, the external comparator is revealed. NIST Advanced Manufacturing Series 100-14 reports pure iron at normal pressure as BCC at lower temperature,

The measured enthalpy record is an empirical application binding for retained excitation, not a replacement for the native selector. At a coexistence boundary the competing complete states may remain set-valued maxima. Continued positive excitation may therefore carry the retained complete state from BCC to FCC and later back to a distinct higher-excitation BCC relational state without introducing a second selector or requiring a monotone structural label.

The NIST Gibbs-energy ordering is retained only as a downstream stability comparator. It is not identified with the native SEAM entropy functional. R168 does not claim a first-principles prediction of 1184 K or 1665 K and does not universalize the Fe empirical binding to all materials.

Evidence: `Evidence/Locked_Runs/R168_FE_EMPIRICAL_EXCITATION_REVERSE_ENGINEERING/` .

Current nuclear four-selector standing

The active nuclear implementation uses the frozen $\Gamma_{Z,N}$ formation envelope and the four-selector construction defined in the atomic nuclear relational closure section above.

The two outer selectors are numerically established by the locked formation-envelope run:

$$N_{\text{low}}(Z), \quad N_{\text{high}}(Z).$$

The aggregate run covers $Z = 1..128$, contains all 2,549 evaluated-experimental positive controls available in the bundled NUBASE2020 evidence, and terminates natively at a finite upper boundary for all 128 tested proton counts. The evidence base contains evaluated-experimental records through $Z = 110$; higher- Z outputs are forward structural results rather than empirical confirmations.

The two interior selectors are additive projections of the same admitted complete-state family:

$$N_{\text{most-stable}}^{\text{SEAM}} = \arg \max S_{\text{total}}[C_{Z,N}^*],$$

$$N_{\text{longest-lived}}^{\text{SEAM}} = \arg \max \tau_{\text{SEAM}}(Z, N).$$

The empirical retained-energy surface $B_{\text{hat}}(Z, N)$, successor increment ΔG_N , measured isotope stability, and measured half-life remain useful evidence/comparator surfaces. They are not substituted for the native entropy or SEAM-time selectors.

Current execution rule. The four-selector nuclear construction is part of the current complete-state architecture. Numerical requests are evaluated under the current run contract and reported at the scope supported by that contract.

Existing diagnostic testing establishes that the frozen formation density Γ , primitive wholeness W , normalized wholeness W/Z , and empirical B/A are not interchangeable persistence selectors. Their weak correlations with measured $\log_{10}(\tau)$ are retained as evidence against scalar collapse of persistence onto any one of those quantities.

The current required output record for each Z is

`Z, N_low, N_most_stable_SEAM, N_longest_lived_SEAM, N_high, S_total_at_selected_states, tau_SEAM_at_selected_states, terminal_status` .

Native values are sealed before empirical comparator fields are joined.

Current canonical nuclear authority lock

Current nuclear formation authority is $(Z, N) \rightarrow \Gamma_{Z,N} \rightarrow \text{mean}(\Gamma) \rightarrow A_\Gamma \rightarrow E_\Gamma(Z) \rightarrow (N_{\text{low}}, N_{\text{high}})$ together with the certified interior entropy and Cs-count persistence selectors. These current operators are the sole nuclear authority for the active build.